STORAGE, RETRIEVAL, AND EVIDENCE ACCESS FOR LONG-TERM CONVERSATIONAL MEMORY

by

Wang Wenqi
A0329680W
e1554007@u.nus.edu

An internal capstone project report submitted for the degree of

Master of Computing

in the School of Computing

of the National University of Singapore

Supervisor: Zhang Jiaheng
Submission date: [Submission Month and Year]

Contents

1 Introduction **1**
- 1.1 Research Background . . . 1
- 1.2 Problem Statement . . . 1
- 1.3 Research Landscape and Gaps . . . 2
- 1.4 Research Objectives . . . 3
- 1.5 Research Questions . . . 3
- 1.6 Research Contributions . . . 4
- 1.7 Research Scope and Boundaries . . . 4

2 Background and Related Work **5**
- 2.1 Foundations of Long-Term Memory for LLM Agents . . . 5
- 2.2 Main Representations of External Memory . . . 5
- 2.3 Memory Retrieval Methods . . . 6
- 2.4 Filesystem-Based Memory . . . 6
- 2.5 ReFind and Iterative Retrieval over Raw Dialogue . . . 7
- 2.6 Long-Term Dialogue Evaluation and LoCoMo . . . 7
- 2.7 Evaluating Long-Term Memory Systems . . . 7
- 2.8 Positioning and Differences of This Study . . . 9

3 System Design and Methodology **10**
- 3.1 From Research Gaps to Methodological Requirements . . . 10
- 3.2 Task Formalisation . . . 11
- 3.3 Overall System Architecture . . . 12
- 3.4 Data Normalisation and Traceability . . . 12
- 3.5 Three Storage Representations . . . 13
- 3.6 Three Retrieval Systems . . . 14
- 3.7 Shared EvidenceBundle and Answerer . . . 15
- 3.8 Mapping of E1–E7 Conditions . . . 16
- 3.9 Alternative Evidence-Access Paths . . . 17

4 Experimental Design and Evaluation Methodology **19**
- 4.1 Research Questions, Variables, and Identification Strategy . . . 19
- 4.2 Dataset and the Frozen Main-60 Test Set . . . 20
- 4.3 Experimental Matrix and Core Comparisons . . . 20
- 4.4 Models and Evaluation Roles . . . 20
- 4.5 Controlled Experimental Conditions . . . 21
- 4.6 Evaluation Metrics and Cost Accounting . . . 21
- 4.7 Statistical Analysis Plan . . . 23
- 4.8 Error Analysis Method . . . 24

5 Results and Analysis **26**
- 5.1 Analysis Scope and Research Questions . . . 26
- 5.2 RQ1: Matching Memory Representation with Retrieval . . . 26
- 5.3 RQ2: Robustness across Backbone Models . . . 28
- 5.4 RQ3: Dual-Source Fusion . . . 28
- 5.5 RQ4: Full Context versus Selective Retrieval . . . 30
- 5.6 RQ5: Error Chain and Bottleneck Diagnosis . . . 31
- 5.7 RQ6: Quality, Cost, and Practical System Choice . . . 33
- 5.8 Chapter Summary and Consolidated Answers to RQ1–RQ6 . . . 34

6 General Discussion: Towards Evidence-Controlled Memory Systems **35**
- 6.1 The Evidence-Conversion Gap . . . 35
- 6.2 The Missing Architectural Layer: An Evidence Controller . . . 35
- 6.3 Adaptive Evidence Access as a Control Policy . . . 35

Chapter 1. Introduction

1.1 Research Background

1.1.1 From Single-Turn Question Answering to Long-Term Interactive Agents

Large language models (LLMs) are increasingly used as agents that perform tasks, call tools, and interact with users over time. In personal assistance, collaborative work, customer service, and similar settings, users reveal preferences, relationships, events, and plans across many sessions. An effective agent must therefore understand the current request and access information introduced much earlier. LoCoMo frames this challenge as long-term conversational understanding, while MemoryAgentBench emphasises incremental memory in multi-turn environments [1, 2].

Long-term interaction is difficult for more than one reason. Relevant facts may be scattered across sessions, expressed in different words, or updated by later events. A question may also require several pieces of evidence. For example, a user may propose a plan, revise its date, and later report the outcome. Retrieving only the first statement can produce an outdated answer. Retrieving all statements is still insufficient if the model cannot resolve entities, temporal order, and the current state. LoCoMo includes single-hop, multi-hop, and temporal questions, which capture these different requirements [1].

Long-term agents therefore need mechanisms to preserve, update, locate, and use historical information. Many systems provide these capabilities through external memory. Retrieval-augmented generation combines a parametric generator with retrievable non-parametric memory [3]. Filesystem-based memory instead treats files, memory-management agents, search agents, and tool interfaces as parts of a persistent memory system [4]. Reliable memory is important because an agent that cannot access past information may contradict earlier statements, ignore user preferences, or rely on outdated facts.

1.1.2 A Long Context Window Is Not a Long-Term Memory System

A direct approach is to place the entire conversation history in the model context. This preserves the raw source and avoids omissions caused by an explicit retrieval step. However, information that fits in the context window is not always used reliably. Long-context performance can vary with the position of relevant evidence, especially when it appears in the middle of the input [5]. Context capacity and effective memory are therefore not equivalent.

As the history grows, Full Context increases the amount of text that the Answerer must process on each query. This can raise token use, latency, and computation. Raw conversations contain repetition, irrelevant dialogue, similar entities, conflicts, and outdated states. A long-term memory system must address selection, organisation, updating, provenance, and auditability, not only input capacity. Filesystem-Based Memory and ReFind address these needs through maintainable file memory and controlled search over raw chat logs, respectively [4, 6].

This study therefore uses Full Context as a **diagnostic baseline without explicit retrieval selection**. It is not a storage representation, an additional experimental condition in the selective-retrieval matrix, or an oracle. It exposes the complete raw source to the answering model, referred to as the Answerer, but remains subject to long-context localisation, noise, conflict, and reasoning failures [1, 5].

1.1.3 External Memory as a Storage–Retrieval–Utilisation System

External memory has three interdependent stages. **Storage** determines how history is represented. The same conversations may be kept as flat raw records, organised into topic directories, or consolidated into entity-, event-, or topic-centred memory. These choices trade off source fidelity, navigation, compression, write cost, and maintenance. Their value also depends on the tools and agents used to access them [4].

Retrieval selects evidence for the current question. An agent may browse directories, search for terms, and inspect files. A lexical method such as BM25 may instead rank conversation units, supported by query reformulation, context expansion, temporal constraints, and iterative search. ReFind shows how an agent can use iterative lexical search over raw chat logs without first constructing a semantic memory structure [6]. Narrow retrieval may miss evidence, whereas broad retrieval may introduce noise or conflict.

Evidence utilisation occurs when the Answerer produces the final response. Relevant evidence does not guarantee a correct answer. The model may fail to resolve references, order events, combine facts, or identify the latest state. Recent diagnostic work distinguishes retrieval-stage failures from downstream utilisation failures, but groups missing stored detail and missed retrieval under the former category [7]. This study extends that separation by checking stored content, retrieved evidence, evidence packaging, and answer generation as distinct points in the end-to-end system.

1.2 Problem Statement

1.2.1 Storage: Fidelity, Organisation, and Compression

Raw logs preserve wording, temporal cues, and dialogue context, but distribute relevant facts across many files and turns. They reduce information loss at write time while increasing the search burden at query time. Topic directories can provide useful navigation cues. Organised stores may require more navigation steps while reducing the amount of text read, and they do not improve answer quality uniformly [4].

Curated memory can consolidate related facts and remove repetition. It can reduce the text read at query time, but curation is a lossy transformation. It may condense away details, flatten temporal updates, or leave stale states unreconciled [4]. The first problem is therefore whether storage organisation improves retrieval and answering when the source data are fixed, and whether the effect depends on the retrieval system.

1.2.2 Retrieval: Coverage, Noise, and Cost

Retrieval must collect sufficient evidence within a limited context and cost budget. Agent-led file search can adapt after each observation, but it requires repeated model calls and may follow an early wrong decision. Lexical retrieval is efficient and auditable, but depends on query formulation, index units, ranking, and context expansion. ReFind shows that these surrounding design choices affect retrieval from raw conversations [6].

More retrieved text may improve coverage but also add duplicates and conflicts. Less text may reduce noise but remove a critical condition. Retrieval should therefore be assessed through gold-source coverage and the evidence retained in the EvidenceBundle, not only through final answer quality. LoCoMo provides source turn identifiers that support this separation [1]. The broader question is whether retrieval-system effects remain stable across storage representations or form a storage × retrieval interaction.

1.2.3 Utilisation: Visible Evidence Does Not Guarantee a Correct Answer

An Answerer may still fail when the EvidenceBundle contains the required information. Common demands include entity resolution, temporal ordering, multi-hop composition, and separation of plans from completed events [1]. Retrieval sufficiency and evidence use must therefore be diagnosed separately [7].

Full Context supports the same distinction. If it repairs an error made by selective retrieval, the likely bottleneck may be access or evidence budget. If selective retrieval succeeds while Full Context fails, selection may have improved focus or reduced noise. If both fail, the cause may lie in reasoning, interpretation, or data ambiguity [5, 7]. Backbone models may differ in both search behaviour and evidence use, so an architecture observed with one backbone model may not transfer unchanged to another.

1.2.4 Evaluation: One Final Score Is Insufficient

LoCoMo evaluates question answering with normalised token-level partial-match F1 [1]. This study reports that official-style F1 together with project-defined, aligned token-overlap Precision and Recall as diagnostic measures. The aggregates are means of per-question values, so aggregate Precision and Recall should not be used to recompute aggregate F1. Exact normalisation and category-specific procedures are deferred to Chapter 4, where the evaluator implementation is documented. These metrics are reproducible, but remain sensitive to wording and answer length [1].

An LLM judge can complement lexical overlap with semantic assessment, but prior work documents position, verbosity, and self-enhancement biases, as well as limited reasoning ability [8]. This study therefore uses an independent LLM judge under a fixed rubric and reports Strict Accuracy, Partial Rate, and Soft Semantic Score alongside the deterministic metrics. None is treated as a complete measure of answer quality.

For retrieval, Evidence Recall is the fractional coverage of annotated gold source turns. Any-hit and All-hit capture binary source-access outcomes. These evidence measures apply only to the 58 evidence-eligible Main-60 questions and are not applicable to Full Context, which performs no retrieval. Process evaluation also includes model calls, input and output tokens, latency, construction cost, and maintenance burden.

The studies reviewed here do not provide one controlled comparison that jointly covers storage organisation, retrieval strategy, backbone model, dual-source fusion, and Full Context under shared data and evaluation [4, 6, 7]. This study addresses that combined gap and traces outcome differences to evidence access, evidence packaging, evidence utilisation, and cost.

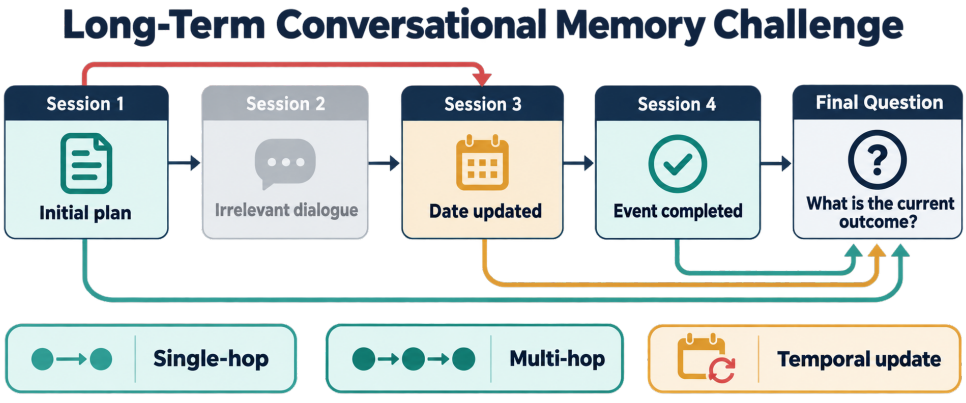


Figure 1.1. Long-term conversational memory challenge: relevant evidence may be distributed across sessions, revised over time, and surrounded by irrelevant dialogue.

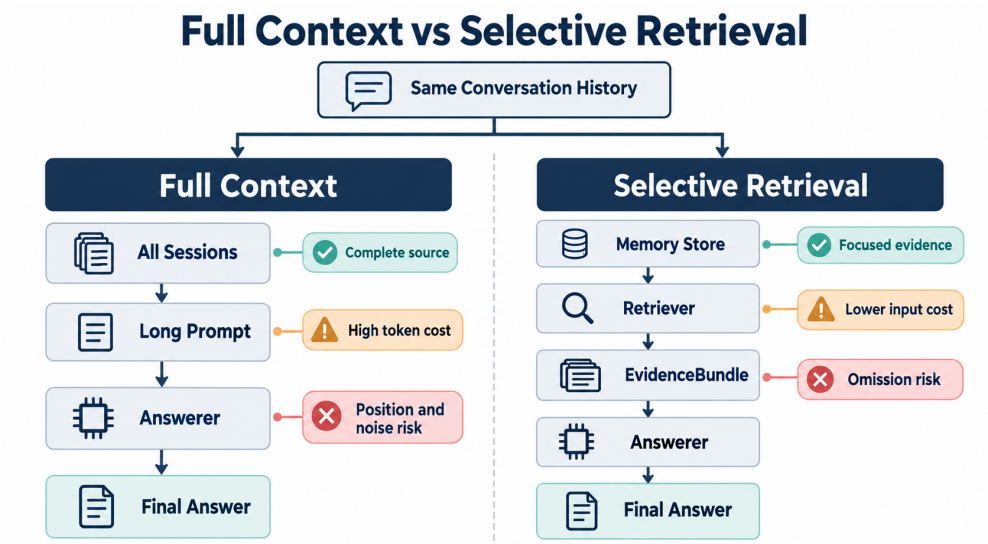


Figure 1.2. Full Context preserves complete source visibility but increases context length and noise, whereas selective retrieval provides focused evidence at the risk of omission.

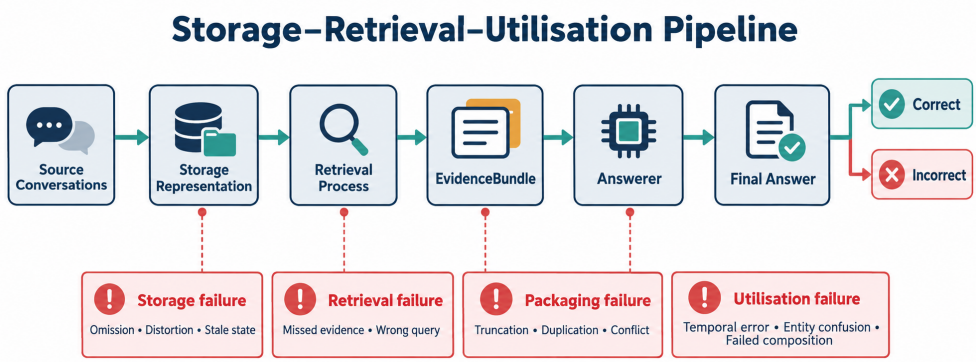


Figure 1.3. End-to-end storage–retrieval–utilisation pipeline and the principal failure points examined in this study.

1.3 Research Landscape and Gaps

1.3.1 Existing Memory Representations

Research on long-term agent memory has produced a wide range of representations. Retrieval-augmented generation introduced the general pattern of combining a parametric generator with externally retrievable, non-parametric memory [3]. Subsequent systems have represented memory as extracted facts, summaries, graph structures, linked notes, discourse units, embedding-organised indexes, raw interaction records, and hierarchical files [4]. Long-conversation benchmarks such as LoCoMo and incremental-agent evaluations further show why these representations matter: relevant information may be introduced across sessions, revised later, or require several pieces of evidence to answer one question [1, 2].

These representations make different trade-offs. Raw conversations preserve wording, provenance, temporal cues, and surrounding dialogue, but leave relevant facts distributed across many turns. Topic-based files add navigational structure without rewriting the source. Curated memory can consolidate related material and reduce repetition, but curation is a lossy operation that may omit qualifiers, weaken temporal relationships, or preserve an outdated state. Filesystem-Based Memory treats organisation, management agents, search agents, tool interfaces, cost, and store health as parts of one persistent-memory design space [4]. Its analysis also shows that a well-organised store should not automatically be assumed to produce better answers: the value of a representation depends on how it is searched and used.

1.3.2 Retrieval Paradigms

Memory representations are accessed through several retrieval paradigms. The simplest diagnostic is Full Context, in which the complete raw history is placed in the model input. This removes explicit retrieval selection but does not guarantee that the model will locate and use the relevant evidence. Long-context performance can depend on evidence position, with information in the middle of an input often used less reliably than information near its boundaries [5]. Full Context also increases input size and exposes the Answerer to irrelevant, repeated, or conflicting material.

Selective retrieval attempts to reduce this burden. Agent-led filesystem search allows a model to inspect directories, formulate search terms, open files, and revise its search after each observation [4]. ReFind instead uses an agent-controlled iterative lexical process over raw chat logs, combining query formulation, ranked retrieval, local context expansion, temporal constraints, and stopping decisions before a separate answering stage [6]. These approaches differ not only in search algorithm but also in the actions available to the agent, the units being ranked or viewed, the number of iterations, and the evidence finally passed to the Answerer. Retrieval quality must therefore be analysed as a process rather than treated as a single interchangeable component.

1.3.3 Evaluation Practices

Evaluation has likewise moved beyond one final accuracy number. LoCoMo uses normalised token-level partial-match F1 and provides answer-source turn identifiers for questions with annotated evidence [1]. Token overlap is transparent and reproducible, but it is sensitive to wording and answer length [1]. Semantic evaluation with an LLM judge can recognise meaning beyond exact token overlap, yet LLM judges may exhibit position, verbosity, and self-enhancement biases, as well as limited reasoning ability [8]. LLM judge scores should consequently complement rather than replace deterministic metrics.

A second distinction concerns where failure occurs. Recent diagnostic work distinguishes retrieval-stage failures from downstream utilisation failures and hallucinations, although its retrieval-stage category includes both missing stored detail and missed retrieval [7]. This study further separates storage fidelity, evidence access, evidence packaging, and answer utilisation. Filesystem-Based Memory additionally measures search effort, token use, content read, and store health, showing the importance of evaluating operational cost alongside answer quality [4]. Together, these practices motivate a multi-level evaluation comprising answer quality, gold-evidence coverage, process traces, and resource use.

1.3.4 Unresolved Gaps

Four gaps remain. First, emerging studies have begun to isolate memory shape [4] or cross write and retrieval strategies [7], but many end-to-end system comparisons still vary representation and access procedures jointly [6]. A result obtained with a curated store and a specialised retriever cannot reveal whether the gain came from representation, search procedure, their interaction, or the Answerer. Cross-paper comparisons are especially weak for this purpose because datasets, prompts, tools, budgets, and models also differ.

Second, the backbone model is itself a source of variation. Model capability can change query formulation, tool use, stopping behaviour, memory maintenance, and downstream outcomes [4]. An apparent property of a memory architecture may therefore depend on the Retrieval Agent or Answerer used to test it.

Third, the studies reviewed here do not establish the per-question paired diagnostic between selective retrieval and Full Context used in this study [1, 4, 6, 7]. Complete visibility may repair retrieval omissions, while selective evidence may outperform a longer context by reducing noise. If both fail despite access to the same source, the bottleneck may instead lie in evidence utilisation or answer generation [5, 7].

Fourth, final rankings provide limited causal explanation. Similar answer scores can conceal different evidence coverage, token cost, latency, and failure stages. Practical recommendations therefore require question-level comparisons, evidence measures, process records, and manual case analysis, not only aggregate F1 or LLM judge scores.

1.3.5 Response of This Study

This study addresses these gaps through one controlled framework built on a unified Main-60 question set. It compares three storage representations—flat raw conversations, topic-organised source files, and agent-curated memory—and three retrieval processes—agent-led filesystem search, ReFind-style iterative lexical retrieval over raw conversation units, and an adaptation of that iterative process to curated memory. All selective-retrieval paths produce a standardised EvidenceBundle for the Answerer. Matched raw-memory and topic-organised conditions support cleaner storage and retrieval contrasts; the curated-memory conditions are evaluated as complete systems and are not interpreted as pure single-component ablations.

The same selective-retrieval configurations are run with Qwen and DeepSeek Retrieval Agents and Answerers while the stored snapshots remain fixed, enabling matched analysis across backbone models. A DeepSeek Full Context run supplies a separate diagnostic comparison. Full Context performs no explicit retrieval selection and is therefore neither a storage representation, an eighth selective-retrieval condition, nor an oracle.

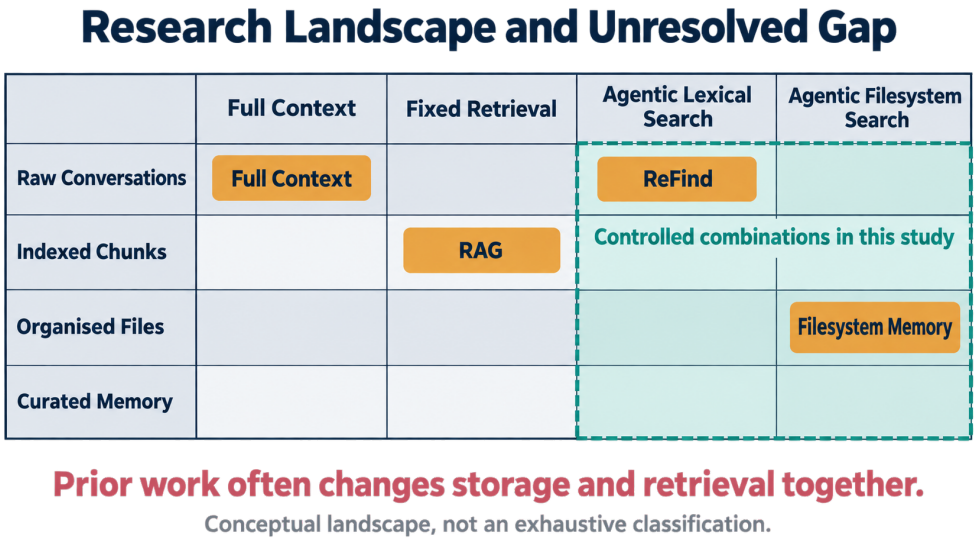


Figure 1.4. Conceptual synthesis of retrieval-augmented generation [3], organised and curated filesystem memory [4], ReFind-style raw-chat search [6], and controlled write–retrieval crossings [7]. The figure motivates cross-combination analysis rather than treating any cited system as one fixed storage–access pair.

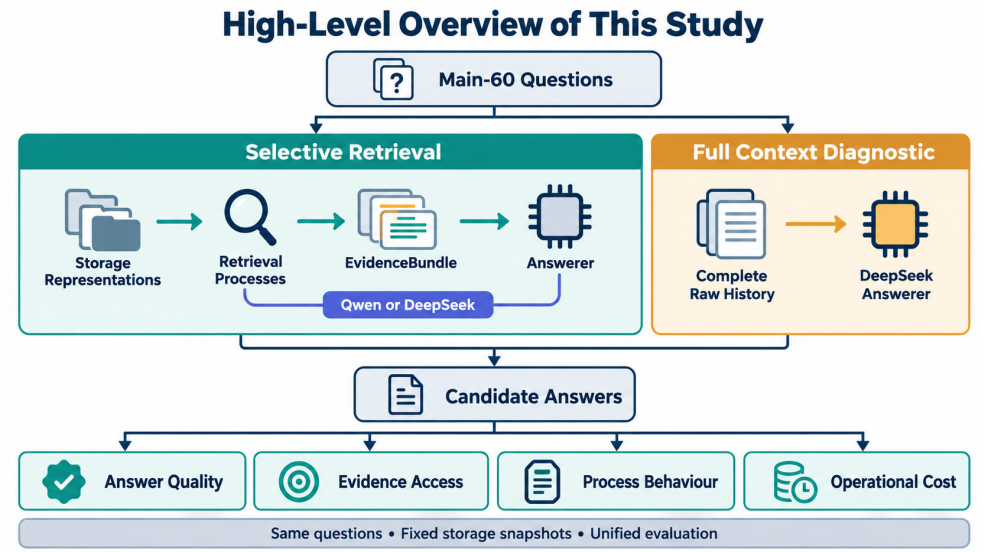


Figure 1.5. High-level overview of the study, including the selective-retrieval and Full Context paths and the four levels of evaluation.

Evaluation combines LoCoMo-style token F1 [1] with this study's aligned token Precision and Recall, semantic Strict Accuracy, Partial Rate, and Soft Semantic Score produced by an independent LLM judge under a fixed rubric [8], gold-source coverage for evidence-eligible questions, and process measures including model calls, tokens, latency, construction cost, and maintenance burden. Paired contrasts and manual review then trace likely failures to storage fidelity, retrieval access, evidence packaging, answer utilisation, or long-context processing. The purpose is not to identify one context-free winner, but to determine which conclusions are architecturally stable, which depend on the backbone model, and which system offers the most defensible quality–cost trade-off under different deployment priorities. Chapter 3 defines the system components and condition identifiers, while Chapter 4 specifies their implementation, controls, and evaluation protocol.

1.4 Research Objectives

1.4.1 Overall Objective

This study aims to evaluate long-term conversational memory under a controlled framework. It examines how storage representation, retrieval strategy, and backbone model capability affect answer quality, evidence coverage, and operational cost. Dual-source fusion and Full Context are used to diagnose system bottlenecks. The goal is to determine when each design is useful and whether its gains justify its cost.

1.4.2 Specific Objectives

The study has six objectives:

1. Construct three traceable storage representations from the same source data.
2. Connect agent-led file search, iterative lexical retrieval, and curated-memory retrieval to a common EvidenceBundle interface.
3. Compare the same selective-retrieval configurations with Qwen and DeepSeek Retrieval Agents and Answerers.
4. Use a DeepSeek Full Context diagnostic baseline to examine access and long-context utilisation.
5. Evaluate lexical quality, semantic quality, gold-evidence coverage, model calls, tokens, latency, and maintenance burden.
6. Develop a stage-based error analysis and derive bounded practical implications for different quality, cost, scalability, and auditability needs.

1.5 Research Questions

1.5.1 Architecture: RQ1

RQ1: What effects do storage representation and retrieval strategy have, and do they interact?

The experiment uses matched conditions that vary storage organisation while holding the retrieval approach fixed, and vary retrieval while holding storage fixed. These comparisons estimate storage effects, retrieval effects, and their interaction. Complete curated-memory systems are analysed separately because their outcomes cannot be attributed to one isolated component.

1.5.2 Robustness: RQ2

RQ2: Are these architectural effects stable across Qwen and DeepSeek?

The analysis compares the direction and magnitude of matched contrasts across the two backbone models. Consistency supports a more general architectural interpretation; a direct difference between model-specific effects indicates dependence on the backbone model.

1.5.3 Mechanisms: RQ3–RQ5

RQ3: When does dual-source fusion repair errors, and when does it introduce interference?

The dual-source condition reuses evidence from two existing retrieval paths and adds no new retrieval process. Its repair, preserve, harm, and unresolved outcomes are compared with those of the two single-source parent conditions.

RQ4: Does Full Context outperform selective retrieval, and what bottlenecks does each approach reveal?

The within-DeepSeek paired descriptive comparison tests whether complete source visibility repairs selective-retrieval failures and whether selected evidence can reduce noise. Full Context is not a storage representation, an additional selective-retrieval condition, or an oracle.

RQ5: Do errors mainly arise from storage fidelity, retrieval coverage, evidence packaging, answer utilisation, or long-context processing?

Stored content, retrieval traces, EvidenceBundles, gold evidence, candidate answers, and evaluation outputs are combined with manual review to locate the likely failure stage.

1.5.4 Decision-Making: RQ6

RQ6: Considering quality, cost, stability, and maintenance complexity, which conditions are most suitable for practical use?

RQ6 examines quality–cost trade-offs and stability across backbone models. It develops conditional guidance for quality-first, cost-sensitive, and audit-focused settings, while treating implications for continuously growing memory as tentative because sustained memory updates are outside the present experiment.

Research question	Main experimental material	Main evidence
RQ1	Controlled storage–retrieval comparisons and complete curated-memory systems	Paired quality differences, Evidence Recall, storage–retrieval interaction
RQ2	The same selective-retrieval configurations with two backbone models	Effect direction and magnitude across backbone models
RQ3	Two single-source conditions and their dual-source fusion condition	Repair, preserve, harm, unresolved, evidence complementarity and conflict
RQ4	DeepSeek selective retrieval and Full Context	Paired outcomes, answer quality, context size, and cost
RQ5	Storage, retrieval traces, EvidenceBundles, and answers	Stage-based errors, evidence conversion, and manual cases
RQ6	All formal conditions	Quality–cost Pareto analysis, cost amortisation, and decision matrix

Chapter 3 formally defines the system components and experimental condition identifiers. Chapter 4 presents the execution protocol, metric definitions, and statistical tests.

1.6 Research Contributions

1.6.1 Controlled Experimental Design

The study provides one storage–retrieval–answering framework with shared data, questions, storage snapshots, and evidence interfaces. Matched conditions support architecture contrasts, while separate integrated conditions evaluate complete curated-memory systems. This separation limits unsupported single-component claims and supports explicit analysis of storage–retrieval interaction.

1.6.2 Evidence Across Backbone Models and Full Context

The same selective-retrieval configurations are evaluated with Qwen and DeepSeek Retrieval Agents and Answerers while storage snapshots remain fixed. A DeepSeek-family Full Context diagnostic adds evidence about selective retrieval, complete source visibility, and long-context utilisation.

1.6.3 Multi-Level Evaluation and Error Diagnosis

The study combines LoCoMo-style token F1 [1] with this study's aligned token Precision and Recall, plus three semantic metrics defined for this study and produced by an independent LLM judge. The LLM judge follows the general semantic-evaluation paradigm described in prior work [8]. Gold-evidence measures and process and cost records provide further diagnostic evidence. Paired comparisons and manual case review distinguish failures to retrieve evidence from failures to use it correctly.

1.6.4 Practical Decision Support

Answer quality is considered together with model calls, tokens, latency, construction cost, maintenance burden, and stability across backbone models. The study provides bounded, conditional guidance rather than one unconditional system ranking.

1.7 Research Scope and Boundaries

1.7.1 Included Scope

The formal study uses the frozen Main-60 question set from LoCoMo `conv-50`. All 60 questions form one unified test set and cover the C1–C4 categories used in this work. The study uses only the question-answering material and relevant source annotations [1].

The experiments include three storage representations, three retrieval processes, seven selective-retrieval configurations, two backbone model configurations for the Retrieval Agent and Answerer, and one DeepSeek-family Full Context diagnostic. The evaluation covers token-level metrics and answer metrics produced by an LLM judge, evidence measures for the 58 eligible questions, model calls, tokens, latency, and maintenance cost. Retrieval traces, EvidenceBundles, and run audits support stage-based analysis.

1.7.2 Excluded Scope

This is not a full reproduction across all LoCoMo conversations. Other self-organising, fact-based, graph-based, and vector-RAG memory systems are not formal baselines. The study does not simulate months of continuous writing, active forgetting, conflict accumulation, or automatic reorganisation.

The backbone model comparison is not a full Retrieval-Agent × Answerer factorial design. The Retrieval Agent and Answerer change together, while the topic-organised and curated-memory snapshots remain fixed. It therefore measures dependence on the backbone model across the query-time pipeline, not the separate effect of each model role or the writing model. Full Context is evaluated only with DeepSeek, so it cannot support a conclusion across backbone models about Full Context prompting.

1.7.3 Limits on Interpretation

The matched raw-memory and topic-organised conditions provide the cleaner architecture contrasts, but retrieval-system comparisons still include their full search procedures. The integrated curated-memory conditions jointly change the memory representation and its corresponding retrieval process. They cannot support a claim about changing only storage or only retrieval.

Full Context performs no retrieval, but it is neither a new storage method nor an accuracy ceiling. Its comparison with selective retrieval is diagnostic rather than a strict causal ablation because the input mechanism and context scale differ. All conclusions are limited to the present question set, storage snapshots, models, and protocols. Broader generalisation requires further validation.

The remainder of this report is organised as follows. Chapter 2 reviews the relevant literature. Chapters 3 and 4 define the system and experimental methodology. Chapters 5 and 6 present and discuss the empirical findings. Chapters 7 and 8 address validity and future research directions, followed by the conclusion in Chapter 9.

Chapter 2. Background and Related Work

This chapter defines the concepts and prior work needed to study long-term conversational memory. It reviews memory representations, access methods, the two core system papers, the LoCoMo benchmark, and relevant evaluation methods. Exact system definitions, parameters, and formulas are deferred to Chapters 3 and 4.

2.1 Foundations of Long-Term Memory for LLM Agents

2.1.1 Parametric, Contextual, and External Memory

An LLM agent can obtain past information from three broad forms of memory. **Parametric memory** is encoded in model weights and is difficult to inspect or update with one local fact after deployment. **Contextual memory** is placed in the current input. It is immediately available, but bounded by the context window and the cost of long prompts. Models may also use different positions within a long input with unequal reliability [5].

External memory stores information outside the model as logs, files, summaries, indexes, vectors, or graphs. Relevant content can then be selected and returned to the model. This follows the retrieval-augmented principle of combining a parametric generator with an updateable, non-parametric source [3]. External memory can support provenance, local editing, and auditability, depending on its representation and tools [4, 9]. This study concerns text-centred external memory rather than weight editing or memory-policy training. Full Context uses the same external source but places it directly in the current input.

2.1.2 The Long-Term Memory Lifecycle

Long-term memory is a process rather than a static database. A conversation produces events, preferences, plans, corrections, and personal facts. A system must decide what to retain, how to represent it, and how to organise it by session, time, topic, entity, or relation. At query time, retrieval locates candidate evidence and an answering component uses it. Later maintenance may add entries, resolve conflicts, remove stale material, or reorganise the store. Incremental evaluation matters because memory agents receive information over time rather than as one fixed document [2].

The lifecycle therefore contains a feedback loop:

interaction → writing → organisation → retrieval → evidence use → answer → maintenance

An early error can propagate, while maintenance can change what later searches find. Chapter 3 maps this lifecycle to the study's concrete components.

2.1.3 Storage, Retrieval, and Utilisation as Distinct Problems

Three questions should be separated. **Storage fidelity** asks whether memory retains the required fact and its qualifiers. **Retrieval coverage** asks whether relevant material reaches the Answerer. **Utilisation** asks whether the Answerer correctly interprets and combines that material. A final score combines all three.

Consider an abstract question about the latest date of an event. The updated date may have been omitted during summarisation. This is a storage failure. It may be present in the store but absent from the retrieved evidence. This is a retrieval failure. It may also appear in the evidence while the Answerer still selects an older date. This is a utilisation failure. These cases require different remedies.

Recent diagnostic work crosses write strategies with retrieval methods and separately probes retrieval relevance and downstream memory utilisation [7]. Its failure taxonomy groups missing stored detail and missed retrieval under one retrieval-stage category; it does not independently measure write-time information loss. This study therefore uses a finer project-defined distinction among storage, retrieval, and utilisation while preserving the paper's retrieval-to-generation separation.

2.1.4 Principal Design Axes of Long-Term Memory Systems

Long-term memory systems can be organised along five design axes.

1. **Fidelity versus compression.** Raw records preserve detail, whereas summaries and extracted facts reduce size but may lose context.
2. **Flat versus hierarchical organisation.** A flat store is simple. A hierarchy may support navigation, but only if the search process can use it.
3. **Passive versus agent-controlled retrieval.** A fixed ranker returns one result list. An agent may reformulate queries, inspect paths, and stop when evidence is sufficient.
4. **Selective evidence versus Full Context.** Selection reduces noise and input size. Full Context avoids explicit search misses but creates a long-context utilisation problem.
5. **Quality versus total cost.** A system may move cost between memory construction, query-time search, answering, maintenance, and recovery.

These axes are not independent. Compression may lower query cost while increasing construction and fidelity risks; hierarchy helps only when access tools expose it; and Full Context removes explicit retrieval but not selection inside the model. Evaluation must therefore compare complete data paths as well as isolated components. The later chapters use these axes to distinguish clean contrasts from pipeline-level comparisons. This distinction also prevents a tidy store from being treated as evidence of better memory.

2.2 Main Representations of External Memory

2.2.1 Raw Dialogue and Log-Based Storage

Raw dialogue retains wording, speaker labels, session boundaries, dates, and turn order. It therefore preserves provenance and allows later reinterpretation without relying on an earlier summary. This matters when a qualifier, correction, or temporal cue becomes important.

Its weakness is dispersion. An answer may depend on turns from several sessions, while repetition, pronouns, and omitted context make isolated matches difficult to interpret. Raw logs still have minimal structure through session, turn, time, and speaker metadata, but they require effective search or full-archive processing.

2.2.2 Filesystem-Based Memory

A filesystem adds addressable structure to textual memory. Paths, filenames, folders, headings, and descriptions can act as retrieval cues. An agent can list directories, search text, inspect files, and follow new leads. Files are also editable and compatible with standard tools.

Filesystem-Based Memory defines a memory store as files under a rooted hierarchy and a tool harness as the operations through which agents access that store [4]. Store shape and access interface can therefore jointly influence what the agent finds.

Hierarchy is not automatically useful. A taxonomy can add maintenance work without improving answers if the searcher ignores it. Filesystem quality must be evaluated with navigation behaviour, tool constraints, and model capability.

2.2.3 Agent-Curated or Summarised Memory

Agent-curated memory rewrites interaction history into a compact form. A manager may extract facts, group events, merge repetition, or maintain topic notes. Mem0 uses extraction and consolidation, while A-MEM adds linked, evolving notes [10, 11].

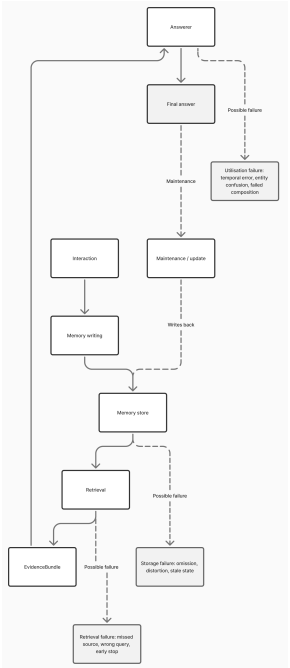


Figure 2.1. Long-term memory lifecycle and the project-defined separation of storage, retrieval, and utilisation failures. The retrieval-to-generation distinction is informed by [7], which does not independently measure write-time information loss.

Curation introduces another interpretation layer. It can surface stable facts, but may omit qualifiers, flatten temporal order, merge entities, or preserve stale states. Its value depends on the writing model, prompt, update policy, and construction and maintenance costs.

The study's curated store belongs to this broad family but remains a fixed experimental snapshot. Chapter 3 defines its construction and audit rules.

2.2.4 Vector, Graph, and Other Structured Memories

Vector memory encodes source units as embeddings and retrieves by semantic similarity. This extends the retrieval-augmented pattern of selecting information from non-parametric memory [3]. In practice, semantic similarity is not identical to task relevance, and a vector system also depends on choices such as source units, embedding model, query, and return depth. Graph memory represents entities, events, or notes as connected nodes. It supports relation traversal, but construction and conflict maintenance add assumptions and cost. Mem0 includes a graph-enhanced variant, while A-MEM links evolving notes [10, 11].

Hybrid systems combine several forms. MemGPT, for example, manages information across memory tiers [9]. Such systems define the wider design space but are not formal baselines here because their writing, indexing, and update protocols introduce further variables.

2.2.5 Comparison of Representations and Scope of This Study

Table 2.1 compares the main representation families through observable design properties rather than an ordinal ranking.

Table 2.1. Conceptual comparison of external-memory representations.

Representation	Primary stored unit	Source-text preservation	Added organisation or typical access	Typical update	Principal risk
Raw dialogue logs	Turns or exchanges with session metadata	Verbatim	Session and turn chronology; lexical or long-context access	Append new turns	Relevant facts are dispersed in long histories
Foldered raw files	Verbatim turns grouped into session or topic files	Verbatim, subject to grouping	Paths, filenames, and headings; file navigation and text search	Append or reorganise files	The taxonomy may not help the searcher
Curated textual memory	Extracted facts, summaries, or topic notes	Selective and rewritten; size may decrease or increase	Explicit thematic or entity structure; file or ranked access	Rewrite, merge, and reconcile notes	Omission, distortion, or stale consolidation
Vector index	Embeddings linked to source chunks	Depends on whether the source chunks are retained; the index is additive	Similarity ranking over indexed units	Encode, delete, or re-index chunks	Semantic similarity may miss task relevance
Graph or linked memory	Entities, events, or notes plus relations, often linked to source text	Depends on node construction and provenance	Relation traversal or hybrid access	Add, merge, or revise nodes and edges	Construction, linkage, and conflict-maintenance errors

Note. This table is a project-defined conceptual synthesis, not an empirical ranking. The representation families are not mutually exclusive; vector and graph indexes can be layered over textual sources. Sources: [3, 4], [9–11].

This study selects the first three representations because they remain human-readable and auditable while spanning different levels of organisation and compression. Adding vector or graph systems would also change the retrieval infrastructure.

2.3 Memory Retrieval Methods

2.3.1 Agentic File Navigation and Text Search

In agentic filesystem retrieval, an LLM controls a sequence of tool actions. It may inspect directories, formulate keywords, search text, open a matching region, and continue when evidence is incomplete. `grep` is an access tool, not a retrieval strategy; the agent determines the query, scope, inspection sequence, and stopping point.

This approach can use both lexical matches and filesystem structure. Its visible actions support audit and query-specific adjustment, but add model rounds, tool calls, context growth, and behavioural variation. *Filesystem-Based Memory* treats the harness as part of the system rather than a neutral interface [4].

This study adopts the paradigm but separates evidence collection from final answering. Exact controls appear later.

2.3.2 Lexical Ranking and Iterative Retrieval

Lexical rankers score stored units against a query. BM25 rewards informative term overlap while accounting for frequency and document length [12]. The query may be the original question or an LLM reformulation, and ranked units may be aggregated at a higher level. Reciprocal Rank Fusion combines multiple ranked views by favouring items that appear near the top of those lists [13].

ReFind applies lexical ranking to dialogue turns and adds session-level signals, local expansion, temporal filtering, and duplicate control [6]. An LLM controller can search again, retain evidence, or stop. Iteration may repair a poor first query, but adds state and cost. The ranker, controller, actions, and final evidence should therefore be distinguished.

2.3.3 Multi-Source Retrieval and Evidence Fusion

A system may retrieve from raw logs, curated notes, vector indexes, or other sources. Sources can be complementary: one may preserve exact wording and time, while another offers a cross-session summary. Fusion may also add repetition, conflicting states, or excessive context.

Multi-source evidence fusion is not necessarily a new joint retriever. It may combine outputs from existing paths without another search. The study's dual-source condition follows this design: it reuses two fixed evidence bundles and applies deterministic fusion rules before answering. Chapter 3 defines its provenance and budget handling.

2.3.4 Full Context and Long-Context Answering

Full Context places the complete available conversation in the Answerer input. It avoids explicit search misses, but does not guarantee correct localisation or use. Long-context models can be sensitive to evidence position [5], while repetition, similar speakers, and changing facts add interference. Search calls disappear, but input cost and attention burden grow with the history.

Here, Full Context is a no-retrieval diagnostic. It is neither a storage representation, an eighth selective condition, nor an oracle. It tests whether complete visibility repairs omissions or whether selected evidence improves focus.

2.3.5 Framework for Comparing Retrieval Methods

Table 2.2 separates four access paradigms. It focuses on control flow rather than implementation details.

Table 2.2. Conceptual comparison of memory-access paradigms.

Access paradigm	Control and selection mechanism	Main unit	Iteration	Use of hierarchy	Principal cost or risk
Agentic filesystem retrieval	An LLM selects paths, queries, tool actions, and the stopping point	Files, sections, and text regions	Multi-round	When hierarchy is exposed and used	Model rounds, tool calls, and behavioural variation
Ranked retrieval	A user or LLM formulates a query; a ranker scores indexed units	Exchange, passage, or fact	Optional controller loop	Not required; grouping or reranking may use metadata	Query mismatch and ranker error
Multi-source evidence fusion	Existing retrieval outputs are combined by fixed or learned fusion rules	Multiple evidence bundles	Inherits the parent paths; no additional retrieval in this study	Inherits the parent paths	Duplication, conflict, and context expansion
Full Context	No external selector; the Answerer receives the complete history	Complete history	No explicit retrieval loop in this study	Not applicable	Long input, attention dispersion, and scaling cost

Note. These access patterns can be composed; the rows identify where explicit selection occurs rather than mutually exclusive algorithms. The filesystem and ReFind-style rows draw on [4] and [6]. The fusion and Full Context operationalisations are project-defined.

2.4 Filesystem-Based Memory

2.4.1 Memory Store, Tool Harness, and Agent Roles

Filesystem-Based Memory for LLM Agents treats the filesystem as an external memory medium [4]. Its **memory store** is a rooted hierarchy of files and folders, while the **tool harness** defines the role-specific operations through which agents access it. Paths, descriptions, content, and headings expose structure.

The paper distinguishes three roles. A management agent writes and maintains memory. In the conversational setting, a search agent answers queries with cited sources. In the skill setting, an execution agent completes tasks using retrieved skills. The same store may therefore behave differently under different tools, prompts, or models.

The filesystem should therefore not be treated as one indivisible algorithm: store shape, access behaviour, and downstream use remain distinct.

2.4.2 Experimental Axes and Control Logic

The paper varies memory builder or shape, stream scale, tool harness, and model capability, while identifying what is fixed within each comparison [4, Sec. 2.4]. It organises the analysis around questions of organisation, retrieval, growth, cost, and sustainability. This controlled logic is more informative than a simple system ranking. The present study follows it where possible, while treating inseparable configurations as complete pipelines rather than false single-component ablations.

2.4.3 Main Findings and Open Questions

The paper reports that no memory shape gives the best answer quality in every setting [4]. At larger scales, organised variants can reduce the material read and search cost relative to verbatim storage, while heavier curation is not universally superior. Effects depend on the task, scale, model, and harness.

The evaluation also covers cost, growth, and store health, exposing trade-offs between management work and later search. Open questions include horizons beyond those measured, whether useful hierarchy can be maintained, and longer-horizon store health under continued updates.

These findings are conditional on the paper's tasks and systems. Its numerical results are not directly comparable with this study.

2.4.4 What This Study Inherits, Changes, and Extends

The relationship is summarised in Table 2.3.

Table 2.3. Relationship of this study to *Filesystem-Based Memory for LLM Agents* [4].

Relationship	Elements in this study
Adapted from [4]	Filesystem as an external store; raw, foldered, and curated textual memory as broad comparison families; role separation as a design principle; agentic navigation through tools
Project-specific modifications	A common evidence interface before final answering; the composite frozen Main-60 set from one LoCoMo conversation; audited memory snapshots; independently logged retrieval and answer stages
Additional components beyond [4]	ReFind-style retrieval; two query-time backbone models; fixed dual-source fusion; a DeepSeek-family Full Context diagnostic; LoCoMo category-aware F1 following the pinned official evaluator [14]; project-defined Precision/Recall, semantic, evidence-coverage, and stage-diagnostic measures

Note. Only the first row describes elements adapted from [4]. The remaining rows document this study's own design choices and evaluation scope; they do not imply that these additions were evaluated in [4].

This is not a line-by-line reproduction. The paper supplies a framework for representation, roles, controlled variation, and operational evaluation; the next section introduces iterative search over unmodified dialogue.

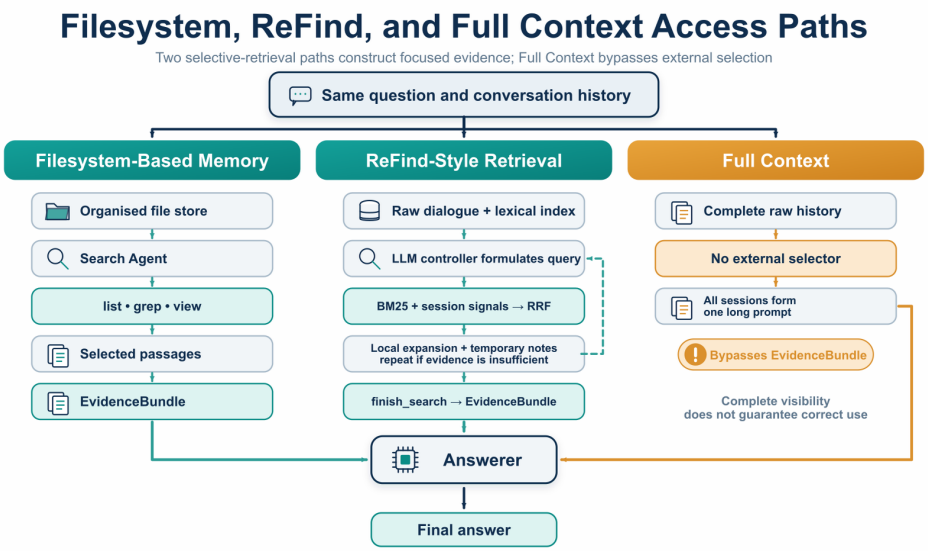


Figure 2.2. Conceptual access paths for filesystem-based memory, ReFind-style retrieval, and Full Context. The two selective paths construct an EvidenceBundle before answering, whereas Full Context passes the complete history directly to the Answerer. Adapted conceptually from [4] and [6]; the common interface and Full Context path are project-defined.

2.5 ReFind and Iterative Retrieval over Raw Dialogue

2.5.1 Motivation for Searching Raw Dialogue

Preprocessing can ease search but change evidence by removing a name, negation, or event order. Raw chat avoids this write-time transformation and retains speaker and local context, while leaving facts dispersed across the archive. ReFind therefore shifts effort from LLM-based memory construction to LLM-controlled query-time lexical search [6]. The controller must formulate effective searches and recognise sufficient evidence.

2.5.2 Two-Level Ranking and Iterative Search

ReFind indexes and ranks dialogue turns with BM25 and derives session-level signals [6, 12]. Reciprocal Rank Fusion combines the turn and session views [13]. High-ranking turns are expanded with local context. Temporal filtering narrows candidates by date, while seen-session filtering prevents later rounds from revisiting returned sessions.

Search and final answering are separate. The controller can query, inspect results, and save question-local notes without rewriting long-term memory. `finish_search` ends evidence collection; a later answering stage uses the notes. Stopping search is therefore not answering the question.

ReFind limits search iterations. This study's exact parameters appear in Chapters 3 and 4.

2.5.3 Empirical Findings and Cost Perspective

ReFind reports that agent-controlled lexical search can rival structured-memory methods in its evaluated settings [6]. Its controls and ablations show that single-shot BM25, or BM25 without the iterative and chat-native controls, is insufficient to explain the gains. Local expansion, session-level fusion, temporal filtering, seen-session control, and multi-round control interact to shape the evidence.

The design avoids LLM-based memory construction but uses an LLM controller at query time. Its value therefore requires joint analysis of answer quality and resource use.

The claim remains setting-specific. Relative performance may change with implicit matches, language, benchmark, or controller model.

2.5.4 Boundaries of the ReFind-Style Adaptation in This Study

This study adapts ReFind's turn-level idea to exchange-level lexical units while retaining higher-level grouping, rank fusion, local context, iterative control, temporary notes, and separate final answering. The topic-organised raw store adds project-specific path handling.

The study also applies the iterative principle to curated memory, where retrieval units and grouping differ from ReFind's turns and sessions. The common evidence interface, LoCoMo mapping, and curated adapter are project designs.

The report therefore uses **ReFind-style** or **ReFind-inspired**, not exact reproduction, and does not attribute the curated variant to ReFind.

2.5.5 Complementarity of Filesystem Memory and ReFind

The papers address complementary questions. *Filesystem-Based Memory* studies store shape, maintenance, and tool access [4], whereas ReFind studies iterative retrieval over raw chat [6]. This study combines their ideas to examine storage–retrieval interaction under shared data, questions, evidence interfaces, and Answerer protocols, not to rank the papers globally.

2.6 Long-Term Dialogue Evaluation and LoCoMo

2.6.1 LoCoMo Data Structure

LoCoMo is a benchmark for very long-term conversational memory [1]. Its conversations span multiple sessions and retain speakers, order, dates, and image captions where relevant. Human verification and editing were used to reduce long-range inconsistencies in the generated conversations.

Each question has a reference answer, and samples include turn identifiers for supporting dialogue evidence. This supports both answer evaluation and source-retrieval analysis.

LoCoMo also contains event-summarisation and dialogue-generation tasks. This study uses question answering; Chapter 4 describes its frozen sample.

2.6.2 Question Categories and Required Capabilities

LoCoMo defines five question categories [1]. **Single-hop** questions use a fact from one session; **multi-hop** questions combine sessions; **temporal** questions require ordering or tracking change; **open-domain** questions combine dialogue with commonsense or world knowledge; and **adversarial** questions use misleading, unanswerable premises.

This study uses one unified Main-60 set from the four non-adversarial categories. Category analysis is secondary to paired overall comparisons; Chapter 4 gives the selection rationale and distribution.

2.6.3 Gold Answers, Gold Evidence, and Evaluation Boundaries

LoCoMo favours reference wording drawn directly from conversations where feasible and uses partial-match F1 after normalisation [1]. Lexical scoring is reproducible, but can penalise valid paraphrases.

Evidence annotations identify source turns containing the answer and enable retrieval measures. They are not a perfect account of every valid reasoning path: some questions use background knowledge, and alternative passages may support the answer. Evidence metrics should therefore be limited to eligible questions and treated as diagnostics.

Question answering also does not test the full lifecycle, including prolonged updates, active forgetting, conflict repair, or autonomous reorganisation.

2.6.4 Why LoCoMo Is Used

LoCoMo offers dispersed multi-session history, varied reasoning demands, and source-turn annotations. It also provides a common reference point for long-context and memory-system research [1, 7].

The experiment nevertheless uses one frozen conversation and 60 questions. It supports controlled, traceable comparison rather than universal coverage.

2.7 Evaluating Long-Term Memory Systems

This section reviews general evaluation perspectives and explains why they are adopted in this study. It does not define the final experimental protocol. The exact metrics, formulas, prompts, eligibility rules, and aggregation procedures are specified in Chapter 4.

2.7.1 Token Precision, Recall, and F1

Token metrics compare a candidate with the reference after normalisation. **Precision** is the fraction of candidate tokens overlapping the reference; **Recall** is the fraction of reference tokens recovered by the candidate; and **F1** balances the two for each question. Low precision often reflects extra or non-matching text, while low recall often reflects omission.

The F1 measure aligns with LoCoMo's paper-described QA protocol [1], and its exact category-aware implementation follows the pinned official evaluator [14]. Precision and Recall are project-defined diagnostics that use the same matching basis. These token metrics are deterministic but sensitive to wording and answer length. A valid paraphrase may share few tokens, while a wrong answer may repeat key terms.

Chapter 4 defines normalisation and aggregation. Aggregate F1 is the mean of per-question values, not a recomputation from aggregate Precision and Recall.

2.7.2 Semantic Evaluation with an LLM judge

An LLM judge compares meaning rather than exact wording. Under a fixed rubric, **Strict Accuracy** counts fully correct answers, **Partial Rate** records partly correct responses, and **Soft Semantic Score** assigns partial credit.

Judging can recognise meaning beyond exact token overlap, but remains sensitive to position, verbosity, self-enhancement bias, prompt wording, and limited reasoning ability [8]. Accordingly, this study keeps the LLM judge independent of the answering backbone model, blind to condition names, and fixed to one rubric and prompt. It complements rather than replaces token metrics.

The rubric and label mapping are project-specific and specified in Chapter 4.

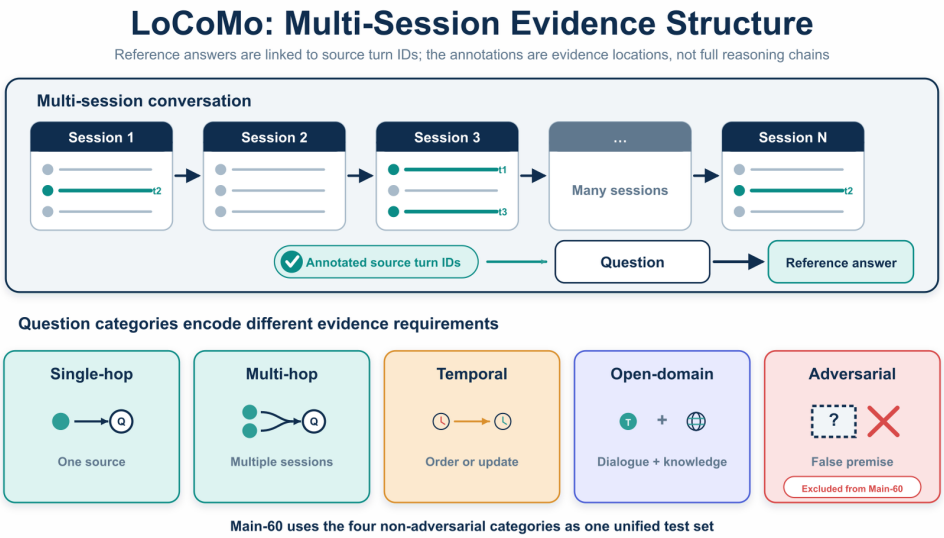


Figure 2.3. LoCoMo's multi-session question-answering structure, question categories, and source-turn annotations. Source-turn identifiers locate annotated evidence but do not constitute complete reasoning chains. The Main-60 set uses the four non-adversarial categories as one unified test set. Adapted conceptually from [1].

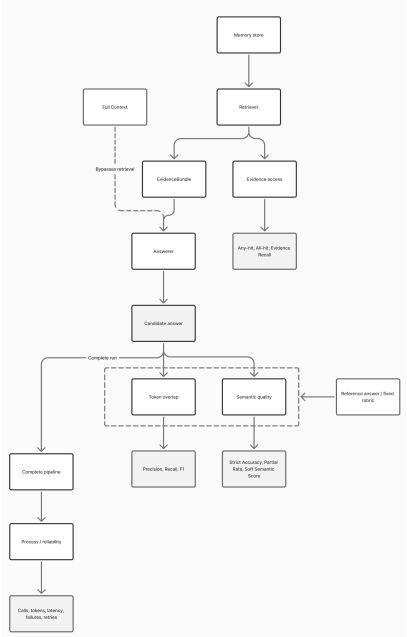


Figure 2.4. Observation points of answer-level, evidence-level, and process-level evaluation. Full Context bypasses explicit retrieval, so evidence-retrieval metrics are not applicable to that path.

2.7.3 Evidence-Level Evaluation

Evidence evaluation asks whether retrieval surfaced annotated sources. **Any-hit** records at least one match, **All-hit** requires all annotated sources, and **Evidence Recall** is the fraction covered.

These measures help locate failure, but are not causal. Retrieved text may be noisy, annotations incomplete, and the Answerer may use unannotated support. High recall therefore does not imply precise or correctly used evidence.

They apply only to selective retrieval and eligible questions. Full Context performs no retrieval selection, so complete visibility should not be labelled perfect retrieval recall.

2.7.4 Process, Efficiency, and Reliability Metrics

Quality must be interpreted with process cost. Query-time measures include model and tool calls, tokens, latency, failures, retries, and recovery. Store-level measures include size, hierarchy, and construction cost. Failed attempts remain part of the record. A filesystem round and an iterative lexical action expose different tools, so their counts are not equivalent. Chapter 4 separates construction, retrieval, answering, judging, and recovery costs.

2.7.5 Why Multi-Perspective Evaluation Is Needed

No single metric explains answer quality and system behaviour. Table 2.4 summarises the four complementary views.

Table 2.4. Evaluation perspectives, scope, and limits.

Metric family	Main question answered	What it cannot answer alone
Official LoCoMo F1 and project-defined token Precision/Recall diagnostics	How much normalised token or stem overlap does the candidate share with the reference?	Whether a valid paraphrase is semantically complete or factually correct
LLM judge measures: Strict Accuracy, Partial Rate, and Soft Semantic Score	Is the answer fully correct, partly correct, or incorrect in meaning under the fixed rubric?	Why the system failed or whether required evidence was retrieved
Gold-source coverage: Any-hit, All-hit, and Evidence Recall	Did selective retrieval cover the eligible annotated sources?	Evidence precision, source uniqueness, or whether the Answerer used the evidence correctly
Process, efficiency, and reliability measures	How many resources, actions, failures, retries, and recovery steps were observed?	Whether output quality is acceptable, or monetary cost without a stated pricing rule

Note. LoCoMo describes normalised partial-match F1 in the paper [1]; the exact category-aware implementation, including its normalisation and stemming rules, follows the pinned official evaluator [14]. Precision and Recall are project-defined diagnostics computed on the same matching basis. The judge-derived and evidence-coverage measures, including label mapping, source matching, and eligibility rules, are also project-defined. LoCoMo supplies the reference answers and source-turn identifiers [1], while [8] motivates the complementary use of an LLM judge.

Together, the four views support mechanism analysis and system choice rather than only a leaderboard.

2.8 Positioning and Differences of This Study

2.8.1 Related-Work Comparison Matrix

Table 2.5 locates the study relative to its three main foundations. Two panels separate system design from evaluation so that the comparison remains readable in a standard report layout.

Table 2.5. Panel (a). System-design comparison of the principal related work and this study.

Work	Primary focus	Memory representation or benchmark source	Access method	Model scope
Filesystem-Based Memory [4]	Effects of organisation, tools, scale, and model capability on persistent filesystem memory	Chunk-retrieval control; verbatim dump; foldered sessions; reorganised and agent-curated stores	Agentic filesystem search; BM25 chunk-retrieval control; targeted harness variants	Model capability is an experimental axis
ReFind [6]	Whether agent-controlled search over raw chat can rival structured memory	Raw chat logs	Turn-level BM25; session-rank RRF; ±2-turn local context expansion; temporal and seen-session filtering; iterative ReAct controller	GPT-4o-mini in the main suite; GPT-5-mini on LongMemEval subsets rather than a full cross-backbone replication
LoCoMo [1]	Evaluation of very long-term conversational memory	Multimodal multi-session conversations plus QA and source-turn annotations; no prescribed system memory store	The paper evaluates base or truncated context, long-context, and RAG variants; the benchmark itself is method-agnostic	Multiple models and methods are compared; backbone is not isolated under one shared pipeline
This study	Interaction among storage, retrieval, backbone model, fusion, and context scope under shared controls	Three auditable textual representations built from one frozen conversation source	Agentic file search; ReFind-style iterative retrieval; fixed evidence fusion; Full Context	Two selective-retrieval backbone pipelines under one scientific protocol; one DeepSeek-family Full Context diagnostic

Table 2.5 (continued). Panel (b). Evaluation and diagnostic comparison of the principal related work and this study.

Work	Long-context or complete-history comparator	Outcome evaluation	Process or diagnostic analysis
Filesystem-Based Memory [4]	No matched complete-history, no-retrieval diagnostic equivalent to this study's Full Context condition	Task quality, cost, and store properties	Growth, search effort, cost, and store health
ReFind [6]	Long-context comparator baselines are reported or reused, but not as a matched complete-history diagnostic	Answer quality, retrieval behaviour, ablations, and cost	Component ablations and search-process analysis
LoCoMo [1]	Includes long-context baselines	QA, event summarisation, dialogue generation, and RAG context-retrieval recall@k	Category-level QA analysis and RAG recall@k; manual error taxonomy for event summarisation
This study	Completed DeepSeek-family, no-explicit-retrieval Full Context diagnostic over the same Main-60 source	Official LoCoMo F1 plus project-defined token, semantic, evidence, process, and cost measures	Stage-based diagnosis, paired contrasts, and manual review

Note. The two panels compare scope and method, not reported scores. Direct numerical comparison would be misleading because the works use different tasks, models, prompts, and system boundaries. RAG = retrieval-augmented generation; RRF = reciprocal rank fusion.

2.8.2 From Literature Gaps to the Study Design

The methodological limitations identified in prior work and in this project's synthesis lead to the six design responses in Table 2.6.

Table 2.6. Methodological limitations motivating this study and project-defined responses.

Methodological limitation	Evidence basis	Response in this study
Storage and retrieval effects are not always isolated	Store shape and tool harness interact in [4]; [7] explicitly crosses write and retrieval strategies	Use clean storage–retrieval contrasts where possible; treat inseparable configurations as complete pipelines rather than pure ablations
Evidence from one backbone model has limited generalisability	[7] identifies single-backbone evaluation as a limitation	Repeat the selective-retrieval conditions with Qwen and DeepSeek backbone models
Retrieval-to-answer interfaces differ across systems	The tool-mediated filesystem pipeline in [4] and iterative retrieval pipeline in [6] expose different intermediate states	Use a common EvidenceBundle interface and a common Answerer prompt within each backbone pipeline
Final-answer scores can hide retrieval and utilisation failures	[7] separates retrieval relevance, memory utilisation, and downstream correctness	Add gold-source coverage, retrieval traces, stage-based diagnosis, and manual review
Complete visibility does not guarantee reliable use of long context	[5] shows that long-context utilisation can depend on evidence position	Add a DeepSeek-family Full Context condition to contrast explicit selection with complete visibility
Accuracy alone does not describe operational cost	[4] and [6] both report process or cost dimensions alongside answer quality	Record model calls, tokens, latency, construction cost, failures, and recovery

Note. The limitations are a project-defined synthesis rather than claims from a systematic literature review. The response column records this study's design choices; it does not attribute those choices to the cited papers.

These responses define the conceptual design rather than its implementation. Chapter 3 introduces the formal components and conditions; Chapter 4 defines the exact controls, statistical contrasts, and error rules.

2.8.3 Boundaries Relative to the Two Core Papers

The study inherits the filesystem store, role separation, and harness perspective from *Filesystem-Based Memory* [4], but requires auditable evidence before final answering through a common interface. From ReFind it inherits raw-chat, lexical, session-aware, iterative search [6], while adapting turn-level retrieval to project units. The adapters, evidence schema, source mapping, curated variant, prompts, and action limits are project choices. The objective is a controlled synthesis, not reproduction of either full leaderboard.

2.8.4 Transition to Chapter 3

Prior work shows that representation, search control, tools, model capability, context scale, and evaluation perspective all affect memory performance. A final answer score cannot identify the failing stage. Chapter 3 therefore defines the study's stores, retrieval processes, selective conditions, common EvidenceBundle and Answerer interfaces, fusion path, and Full Context diagnostic; Chapter 4 specifies execution and evaluation.

Chapter 3. System Design and Methodology

Chapter 2 identified several unresolved problems in long-term conversational memory. Storage and retrieval are often changed together, which makes their effects difficult to separate. Curated memory can improve organisation but may also omit or alter information. Selective retrieval can reduce context length, but it can fail to surface required evidence. Complete context avoids explicit retrieval omission, yet it may introduce noise and long-context utilisation problems. A final answer score alone does not reveal which stage caused an error.

This chapter translates these general problems into a concrete system design. The study separates memory representation, evidence retrieval, evidence packaging, and answer generation. It then derives a set of conditions that supports controlled component comparisons, complete-system comparisons, dual-source fusion, and comparison with a no-retrieval Full Context path.

The chapter explains why the system has this structure and how its components operate. Method-defining constants are introduced where their mechanisms are specified. Chapter 4 consolidates the experimental settings and defines the frozen question set, backbone models, remaining run controls, execution protocol, evaluation metrics, and statistical analyses. Chapter 5 presents and interprets the results. The purpose of this chapter is therefore to define the system and justify the condition structure, rather than to determine which condition performs best.

3.1 From Research Gaps to Methodological Requirements

3.1.1 Separating Storage, Retrieval, and Answering

A long-term memory system should not be treated as one indivisible component. Its end-to-end process can be divided into four stages:

$$\begin{aligned}\text{Conversation history} &\rightarrow \text{Memory representation}\\ &\rightarrow \text{Retrieval and evidence selection}\\ &\rightarrow \text{Answer generation}.\end{aligned}$$

Each stage can produce a different type of failure. The required information may not be retained when the memory is constructed. It may be present in the memory but remain unretrieved. Relevant material may appear in search results but not be selected for the final evidence input. Evidence may also be removed by a fixed budget. Finally, complete evidence may reach the Answerer but still be interpreted incorrectly.

These cases require different remedies. A write omission requires a better memory-construction process. A retrieval omission requires a better search strategy. Evidence loss may require a different packaging or budget policy. A utilisation error instead concerns the Answerer's ability to interpret the available evidence. One end-to-end answer score cannot distinguish among these mechanisms.

Building on the retrieval-to-generation separation in [7], the system therefore makes storage, retrieval, evidence packaging, and answering explicit. This four-stage distinction is a project-specific extension: [7] groups missing stored detail and missed retrieval under one retrieval-stage category. Each retriever produces an inspectable retrieval trace and a standardised EvidenceBundle. The Answerer operates only after retrieval has ended. This separation supports later analysis of where relevant information was preserved, found, retained, and used.

3.1.2 Requirement 1: Isolating Storage Organisation

Prior studies compare flat or raw archives with structured, curated, or summarised memory variants [4, 7]. Such a comparison can change both content and organisation. Any observed difference may result from rewritten text, removed detail, new semantic labels, or a different path structure.

The present study introduces two raw memory representations to isolate directory organisation. S1 stores the original sessions in one flat directory. S2 places the same complete session files into topic directories. The intended content relationship is:

$$\text{content}(S1) = \text{content}(S2),$$

while their path structures differ:

$$\text{paths}(S1) \neq \text{paths}(S2).$$

S2 does not summarise, split, merge, or rewrite a session. It only adds a directory hierarchy. This creates a controlled way to ask whether navigational structure helps memory access when the underlying dialogue remains unchanged.

The comparison does not measure semantic compression or fact extraction. It is restricted to the incremental effect of directory organisation and the way a retriever responds to that organisation.

3.1.3 Requirement 2: Comparing Retrieval Strategies on the Same Memory

A storage representation cannot be assessed independently if every store is paired with a different retriever. The study therefore applies two retrieval families to both raw stores. On S1, R1 defines E1 and R2 defines E2. On S2, the corresponding conditions are E3 and E4. These four conditions form the controlled raw-memory core.

Holding the memory representation constant supports two retrieval comparisons. E2 versus E1 compares R2 with R1 on flat raw sessions. E4 versus E3 compares the same retrieval families on foldered raw sessions. Holding the retrieval family constant supports two storage comparisons. E3 versus E1 examines how directory organisation affects R1, while E4 versus E2 examines whether the same organisation also affects R2.

The four conditions also support an interaction question: does directory organisation help one retrieval family more than the other? R1 can inspect and navigate paths directly. R2 instead begins from lexical ranking over raw conversation units. The R2 index excludes directory names, so the same query and date filter produce the same first-stage ranking over S1 and S2. However, result provenance can expose an S2 path to the Retrieval Agent. That path may influence a later query. Directory structure can therefore affect adaptive retrieval even when it is excluded from lexical scoring.

This is a controlled partial-factorial core rather than a claim that all retrieval interfaces are identical. Each retrieval family requires representation-appropriate observations and instructions. Chapter 4 defines the paired contrasts and controls used when interpreting them.

3.1.4 Requirement 3: Evaluating Curated Memory as a Complete System

S3 differs fundamentally from S1 and S2. It is an agent-curated memory rather than a relocated copy of the raw dialogue. Its construction process can rewrite statements, combine evidence from several sessions, group facts by entity and topic, reconcile later updates, and remove repetition. It can also omit qualifiers, preserve an outdated state, or join facts incorrectly.

S3 must therefore be evaluated as a complete representation rather than as a third directory layout. The study provides two ways to access the same frozen S3 store:

- E5 combines S3 with R1, the general filesystem retriever.
- E6 combines S3 with R3, a retrieval adapter designed for factual units and topic groups.

This pair addresses two questions. First, can a general filesystem Retrieval Agent use a curated store effectively? Second, does a retriever adapted to the internal structure of curated memory access the same store differently?

The conditions do not form an exhaustive 3×3 storage–retrieval matrix. R2 and R3 are not two independent generic retrievers. R2 operates over exchanges grouped by raw sessions. R3 adapts the same broad iterative retrieval principle to curated facts grouped by topic sections. R3 is therefore representation-specific. By contrast, R1 can browse S1, S2, and S3 through a common filesystem interface.

The condition set includes theoretically meaningful and operationally valid pairings rather than every nominal combination. Comparisons involving S3 must also remain system-level comparisons. Since S3 changes content, granularity, organisation, and provenance presentation together, a difference between S3 and a raw store cannot be attributed to one isolated storage feature.

3.1.5 Requirement 4: Testing Raw–Curated Complementarity

Raw dialogue and curated memory provide different forms of evidence. Raw dialogue retains exact wording, local context, temporal order, and complete source history. Its information may nevertheless be dispersed across many sessions and surrounded by irrelevant material. Curated memory can consolidate related facts and organise them by entity and topic. However, curation may omit detail, alter wording, or combine inconsistent states.

The two sources may therefore be complementary. Raw evidence may repair a write omission or distortion in curated memory. Curated evidence may connect facts that remain separated in raw sessions. Conversely, combining the sources may introduce duplication, conflicting descriptions, or additional distraction.

E7 tests this relationship by combining the EvidenceBundles already produced by E2 and E6. E2 represents raw dialogue accessed through the raw-chat retrieval path. E6 represents curated memory accessed through its adapted retrieval path. This pairing is fixed by the system design and is not selected after observing performance.

E7 does not formulate a new query or execute a third retriever. It performs deterministic post-retrieval fusion and then invokes the shared Answerer once. The condition therefore examines whether already retrieved raw and curated evidence are complementary or interfering.

3.1.6 Requirement 5: Distinguishing Retrieval Failure from Long-Context Failure

A failed selective-retrieval answer does not show whether the Answerer lacked the required information or failed to use it. To support this distinction, the study adds a Full Context diagnostic path. Full Context sends the complete ordered raw conversation directly to the Answerer without an external retrieval stage.

This path supports several diagnostic questions. If Full Context repairs an error made by selective retrieval, the selective path may have omitted necessary evidence. If selective retrieval succeeds while Full Context fails, evidence selection may have reduced distracting material. If both paths receive the required material but still fail, the result indicates a downstream utilisation or reasoning problem [7]. Subsequent analysis can then test question-specific mechanisms such as temporal ordering, entity resolution, multi-source composition, and long-context position effects [5].

Full Context is not a fourth memory representation, a fourth retriever, or an eighth selective condition. It is also not an oracle. Complete source visibility does not guarantee correct use. It is a separate no-retrieval path used to examine the consequences of removing explicit evidence selection.

3.1.7 Requirement 6: Controlling the Answering Stage

A comparison of memory access becomes difficult to interpret if every retriever uses a different evidence format or answering prompt. The study therefore requires all selective conditions to follow the same downstream structure:

$$\begin{aligned}\text{Condition-specific retrieval} &\rightarrow \text{Common EvidenceBundle}\\ &\rightarrow \text{Shared Answerer}\\ &\rightarrow \text{Common output schema}.\end{aligned}$$

The Answerer receives the question, an ordered EvidenceBundle, and shared answering instructions. It does not receive the condition identifier, retriever name, retrieval score, retrieval trace, reference answer, gold evidence, or evaluation label. It also has no retrieval tools. Its task begins only after the evidence input has been finalised.

The same component graph and interface definitions are used when the experiment is instantiated with different backbone models. Within each backbone pipeline, the selective conditions use the same Answerer protocol. Chapter 4 describes the backbone deployments and identifies which factors remain fixed across them.

3.1.8 Derivation of the Final Condition Set

The preceding requirements lead to four related parts of the study. E1–E4 form the controlled raw-memory core. E5–E6 extend the design to agent-curated memory. E7 tests dual-source evidence fusion. Full Context removes explicit retrieval as a diagnostic comparison.

Table 3.1. Derivation of the system conditions from the research problems.

Research problem	Design response	Relevant conditions or path
Storage organisation is confounded with content transformation	Compare flat and foldered stores with identical raw session contents	E1/E3 and E2/E4
Retrieval effects are confounded with storage effects	Apply R1 and R2 to the same raw stores	E1–E4
Curated memory changes both content and structure	Evaluate one S3 store through general and adapted retrieval	E5 and E6
Raw and curated evidence may be complementary	Fuse the existing E2 and E6 EvidenceBundles	E7
Retrieval omission is difficult to separate from utilisation failure	Remove selective retrieval and expose the complete raw history	Full Context
Answer generation may confound memory comparisons	Use a shared EvidenceBundle and Answerer protocol	E1–E7

The resulting design is theory-driven rather than an exhaustive Cartesian product. Each condition has a defined role in isolating raw-store organisation, comparing retrieval behaviour, examining curated-memory access, testing evidence fusion, or removing explicit retrieval.

3.2 Task Formalisation

3.2.1 Conversation, Session, and Source Turn

Let a conversation be an ordered sequence of sessions:

$$\mathcal{C} = (\sigma_1, \sigma_2, \ldots, \sigma_M).$$

Each session σ_i is an ordered sequence of source turns:

$$\sigma_i = (u_{i,1}, u_{i,2}, \ldots, u_{i,n_i}).$$

LoCoMo supplies ordered sessions, speaker contributions, session dates, dialogue text, optional image captions, and answer-source turn identifiers [1]. In this study, a **source turn** is one such speaker contribution. The local canonicaliser preserves the source fields and adds a stable project locator and content identity.

The report uses *source turn* rather than the unqualified term *turn*. This avoids confusion with the larger exchange units used by R2 and with the reasoning rounds used by R1.

3.2.2 Question and Evaluation-Only Information

Let q denote the question presented to the retrieval and answering pipeline. During online execution, the system may access q and the permitted memory representation. It cannot access the reference answer or the benchmark's annotated evidence.

Let a^* denote the reference answer. Let G_q denote the set of annotated gold source identifiers when such evidence is available. The question category and evaluation scores are also evaluation-only information. These fields are introduced only after the candidate answer has been generated.

This boundary applies to every online component. The memory-construction process, Retrieval Agent, deterministic retrieval backend, fusion procedure, and Answerer cannot use a^*, G_q, or any evaluation label.

3.2.3 Memory Representation

A memory representation is a transformation of the same canonical conversation:

$$\mathcal{M}_s = T_s(\mathcal{C}), \qquad s \in \{1, 2, 3\}.$$

Here, $\mathcal{M}_1$, $\mathcal{M}_2$, and $\mathcal{M}_3$ correspond to S1, S2, and S3. A representation includes both its text and its access-relevant structure. This structure may include filenames, directory paths, headings, topic groups, session boundaries, and source locators.

The transformation T_s does not imply that every store rewrites the source. S1 and S2 serialise the raw sessions without rewriting their contents. S3 instead produces curated factual memory through an agent-managed construction process.

3.2.4 Retrieval Trace and Evidence Selection

For a compatible memory representation and retrieval system, retrieval is defined as:

$$\mathcal{R}_r(q, \mathcal{M}_s) \rightarrow (\tau_{s,r,q}, \mathcal{B}_{s,r,q}),$$

where $\mathcal{R}_r$ is the retrieval system, $\tau_{s,r,q}$ is the retrieval trace, and $\mathcal{B}_{s,r,q}$ is the resulting EvidenceBundle.

A retrieval trace is an ordered sequence of observations and actions:

$$\tau = ((o_1, a_1), (o_2, a_2), \ldots, (o_L, a_L)).$$

Depending on the retriever, a trace may include directory observations, file reads, search expressions, generated lexical queries, ranked results, evidence-selection decisions, seen-state, and stopping decisions. R1 measures interaction through provider rounds, whereas R2 and R3 use bounded controller actions. These units are not assumed to be equivalent and are defined separately in Section 3.6.

3.2.5 EvidenceBundle

An EvidenceBundle is an ordered sequence of validated EvidenceItems:

$$\mathcal{B} = (e_1, e_2, \ldots, e_K).$$

Each item contains exact evidence text and the metadata required to identify its origin. Conceptually, this includes an evidence identifier, source kind, file path or structural group, source locators, canonical source identifiers, dates, speakers where applicable, content identity, and selection order.

The model does not provide arbitrary text as formal evidence. It selects material that has appeared in a valid observation. The host then re-reads the declared source and resolves the final text and provenance. This distinction separates model-directed selection from evidence authenticity.

3.2.6 Answer Generation

Let A_θ denote the Answerer instantiated with backbone model parameters θ. For a selective-retrieval condition, answer generation is:

$$\hat{a} = A_\theta(q, \mathcal{B}),$$

where $\hat{a}$ is the candidate answer.

The Answerer receives only the question, the final ordered evidence, and the shared answering instructions. It does not inspect the retrieval trace or access the memory store directly.

For Full Context, the evidence-access path changes:

$$\hat{a}_{FC} = A_\theta(q, \mathcal{C}).$$

Conceptually, the complete ordered raw conversation replaces a selectively constructed EvidenceBundle. In the shared Answerer interface, the complete session bodies are still wrapped as non-selective EvidenceItems so that provenance and citation validation remain available. The Answerer remains responsible for producing the candidate response, but no external retriever selects the preceding context.

3.2.7 RunRecord and Offline Evaluation Record

An online RunRecord can be represented as:

$$r = (q, c, \tau, \mathcal{B}, \hat{a}, \rho),$$

where c identifies the system condition and ρ contains operational metadata such as model-call and token measurements. For Full Context, the retrieval trace and selective EvidenceBundle are absent. The record instead contains the non-selective full-context bundle and identifies its complete source input.

An offline evaluation record is created after answer generation:

$$z = (r, a^*, G_q, \mathbf{s}),$$

where $\mathbf{S}$ is the vector of answer, evidence, and process measures defined in Chapter 4. Separating r from z makes the evaluation boundary explicit. Gold information is attached to an existing answer record rather than supplied to the online pipeline.

Table 3.2. Principal notation used in the system definition.

Symbol	Meaning
$\mathcal{C}$	Complete ordered conversation
σ_i	Session i
$u_{i,j}$	Source turn j in session i
q	Online question

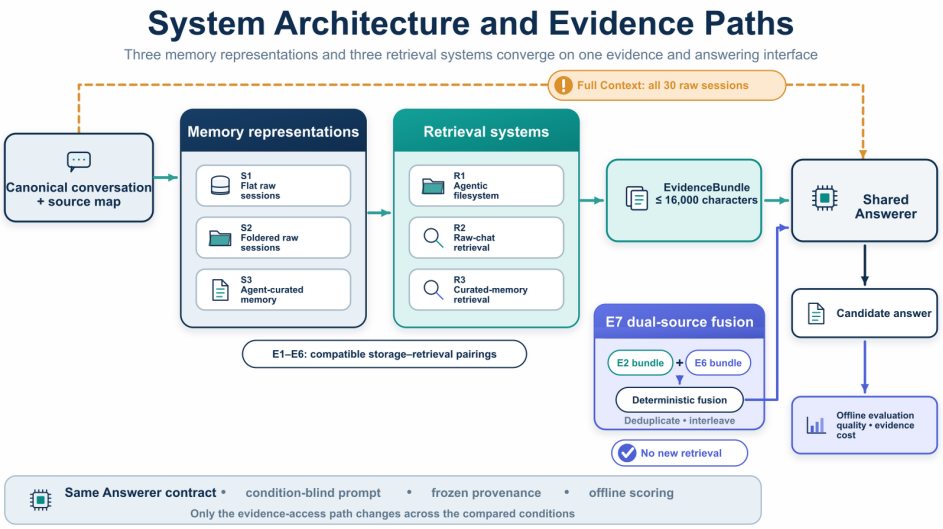


Figure 3.1. Overall system architecture and evidence-access paths. The canonical conversation is transformed into three memory representations and accessed through compatible retrieval systems. Selective conditions converge on a common EvidenceBundle and shared Answerer; E7 fuses the completed E2 and E6 bundles, whereas Full Context bypasses question-dependent retrieval.

Continued from the previous page

Symbol	Meaning
a^*	Reference answer used only for evaluation
G_q	Annotated gold source set, where applicable
T_s	Transformation that constructs memory representation s
$\mathcal{M}_s$	Memory representation S1, S2, or S3
$\mathcal{R}_r$	Retrieval system R1, R2, or R3
τ	Retrieval trace
e_k	One validated EvidenceItem
$\mathcal{B}$	Ordered EvidenceBundle
A_θ	Answerer using backbone model parameters θ
$\hat{a}$	Candidate answer
r	Online RunRecord
z	Offline evaluation record
$\mathbf{s}$	Evaluation metric vector

3.3 Overall System Architecture

3.3.1 Layered System Structure

The system contains six logical layers. The first layer is the canonical source conversation. It preserves the original session order, source turns, dates, speakers, dialogue identifiers, and available captions.

The second layer performs normalisation and establishes provenance. It converts the source into canonical records, raw session files, exchange units, and a common source map. These artefacts provide the shared foundation for all later paths.

The third layer contains the three memory representations. S1 preserves raw sessions in a flat directory. S2 preserves the same session contents under a topic hierarchy. S3 stores agent-curated facts organised by entity and topic.

The fourth layer performs evidence access. R1 navigates the filesystem directly. R2 performs iterative lexical retrieval over raw exchanges and sessions. R3 adapts this iterative approach to curated facts and topic groups. The Full Context path bypasses this layer.

The fifth layer standardises the retrieved information. All selective paths produce the same conceptual EvidenceBundle before answer generation. E7 operates at this layer by fusing the completed bundles from E2 and E6.

The sixth layer contains the shared Answerer and the offline evaluation boundary. The Answerer produces a candidate answer from either a selective EvidenceBundle or the complete raw history. Reference answers, gold evidence, and evaluation procedures are introduced only after this stage.

3.3.2 Selective-Retrieval Paths

The six single-source selective conditions follow the same high-level process:

$$\begin{aligned}\text{Canonical conversation} &\longrightarrow \text{Memory representation}\\ &\longrightarrow \text{Retriever}\\ &\longrightarrow \text{EvidenceBundle}\\ &\longrightarrow \text{Answerer}\\ &\longrightarrow \text{Candidate answer}.\end{aligned}$$

The internal retrieval process differs by condition, but the boundary before answering remains common. R1 produces evidence from observed file ranges. R2 produces evidence from raw exchange windows. R3 produces evidence from curated fact windows. These outputs are converted into the same evidence interface before they reach the Answerer.

This structure prevents a retriever from answering the benchmark question directly. Search decisions and answer generation remain separate model roles, even when they use the same backbone model within one pipeline.

3.3.3 Fusion Path

E7 begins only after E2 and E6 have completed retrieval for the same question. Their validated EvidenceBundles become the two parent inputs to a deterministic fusion procedure:

$$(\mathcal{B}_{E2}, \mathcal{B}_{E6}) \rightarrow F \rightarrow \mathcal{B}_{E7},$$

where F is the fixed fusion function.

The fused bundle is then passed to the shared Answerer. Fusion does not access S1 or S3 directly and does not produce a new retrieval trace. Its role is limited to validating, ordering, and deduplicating the existing parent evidence.

3.3.4 Full Context Path

Full Context follows a separate route. The canonical raw sessions are assembled into one complete, ordered context package. This package is supplied directly to the Answerer together with the question.

The path does not invoke S2, S3, R1, R2, R3, or E7. It does not convert the complete conversation into a selective EvidenceBundle; it only wraps the complete session bodies in the shared evidence interface. Its purpose is to remove external evidence selection while retaining the same source conversation.

3.3.5 Online and Offline Boundaries

The architecture maintains a strict distinction between online execution and offline evaluation. Online execution includes memory access, retrieval, evidence construction, fusion where applicable, and answer generation. Offline evaluation loads the reference answer and annotated sources only after the candidate answer has been finalised.

This boundary prevents gold information from shaping queries, stopping decisions, selected evidence, citations, or answer wording. It also permits one completed answer record to be assessed through several evaluation views without rerunning the online system.

Figure 3.1 summarises how the system layers and evidence-access paths converge on the shared Answerer.

3.4 Data Normalisation and Traceability

3.4.1 Canonical Source Records

The study uses one complete conversation selected from the released LoCoMo benchmark [1]. Local preprocessing verification identifies 30 sessions and converts their contents into one canonical sequence of 568 source-turn records before any memory representation is constructed.

Each canonical record contains:

- the conversation identifier;

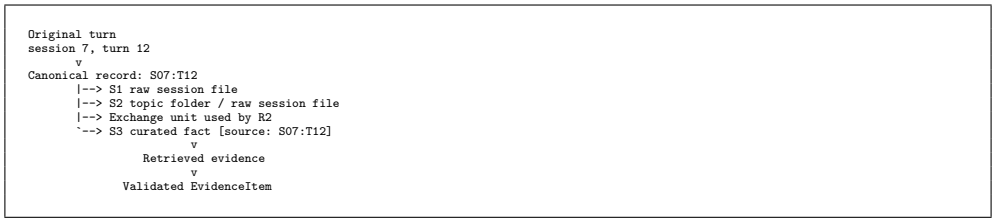


Figure 3.2. One canonical source turn remains traceable across raw, foldered, exchange-based, and curated representations.

- session identifier and session date;
- position within the session;
- global position within the conversation;
- speaker;
- original dialogue text;
- original LoCoMo dialogue identifier;
- a stable project locator;
- an available image caption;
- a content identity.

A locator follows the form $[SxTy]$, where x identifies the session and y identifies the source turn within that session. For example, $[S3T4]$ denotes the fourth source turn in the third session.

The canonical layer standardises identity and ordering without intentionally rewriting dialogue content. S1, S2, and S3 are derived from this shared layer. Differences among the stores therefore do not result from separate source parsing procedures.

The source contains 30 sessions and 568 source turns. Existing image captions are retained as fixed text when available. The system does not reopen image URLs or run a new vision model. The experiment is therefore based on the textual dialogue and the captions already present in the benchmark source.

3.4.2 Raw Session Serialisation

The canonical records are serialised into one Markdown file per natural session. Within each file, source turns remain in their original order. The serialisation preserves speaker labels, session date, source text, available captions, and stable locators.

These session files form the common textual basis of S1 and S2. S1 places them in one flat directory. S2 moves each complete file into a topic directory. Since the file bodies remain unchanged, any difference between the two raw representations arises from their navigational structure rather than rewritten dialogue.

Session serialisation also preserves natural temporal boundaries. No source turn is moved from one session body to another, and no source turn is duplicated or deleted during construction of the raw stores.

3.4.3 Exchange Construction

R2 requires retrieval units smaller than a full session but broader than one isolated source turn. The system therefore constructs deterministic exchange units within each session.

For session σ_i, each complete adjacent pair forms an exchange:

$$x_{i,j} = (u_{i,2j-1}, u_{i,2j}), \qquad j \leq \left\lfloor \frac{n_i}{2} \right\rfloor .$$

Pairing follows the original order:

$$(T_1, T_2),\ (T_3, T_4),\ (T_5, T_6), \ldots$$

If n_i is odd, the final exchange is the singleton:

$$x_{i,\lceil n_i/2 \rceil} = (u_{i,n_i}).$$

Exchanges never cross a session boundary. Every source turn appears in exactly one exchange.

The 568 source turns produce 292 exchange units across 30 sessions. An exchange is a retrieval unit rather than a summary. Its indexed content is derived directly from the corresponding speakers, source text, and available captions.

This unit provides more local conversational context than one source turn while remaining small enough for lexical ranking and neighbourhood expansion.

3.4.4 Source Map and Stable Provenance

The source map connects the project's canonical identifiers to the original benchmark records. For each locator, it resolves:

- the canonical source-turn record;
- the original LoCoMo dialogue identifier;
- session and date;
- speaker;
- source text;
- available caption.

This map creates a common provenance layer for all memory and retrieval paths. It is not supplied to a Retrieval Agent as hidden benchmark information. It is used by the host to verify selected sources and to support later evidence evaluation.

Stable provenance is important because the same source may appear in different forms. In S1 and S2, the source text appears directly in a session file. In S3, one curated fact may restate or combine several source turns. The attached locators preserve a link from that rewritten fact to the canonical source.

3.4.5 Traceability across S1, S2, and S3

Traceability is implemented differently for each representation.

S1 contains the original session text and locator directly. Evidence selected by R1 can therefore be resolved from the observed file and line range. Evidence selected by R2 is resolved from exchange windows whose source turns already have canonical identifiers.

S2 preserves the same file bodies. Its additional directory path changes navigation but not source identity. A source turn therefore retains the same locator and original dialogue identifier in S1 and S2.

S3 contains curated factual units rather than verbatim session records. Each formal fact retains one or more source locators. R1 may select a range of curated text, while R3 selects fact windows. In either case, the locators are parsed and resolved through the common source map.

This design does not imply that S3 contains every source turn. Curation may omit source material. It means that every formal S3 factual unit used as evidence must identify the source records on which it is based. Provenance validity and source coverage are separate properties.

Figure 3.2 illustrates how one original source turn remains traceable across the raw, foldered, exchange-based, and curated representations before becoming a validated EvidenceItem.

3.4.6 Conceptual Integrity Requirements

The normalised data and provenance layer must satisfy several requirements.

First, every raw source turn must resolve to one canonical record. Second, S2 must preserve the contents and source identities of the corresponding S1 session files. Third, every formal factual unit in S3 must contain valid provenance. Fourth, every EvidenceItem must resolve to the declared memory representation and its underlying canonical records. Fifth, missing or invalid provenance cannot be silently reconstructed from model output. Finally, reference answers, gold evidence, question categories, and evaluation labels remain outside the online memory pipeline.

These requirements make it possible to compare representations that differ substantially in structure while retaining a common source basis. They also support the later distinction between information that was absent from a store, information that was present but unretrieved, and information that reached the Answerer but was not used correctly.

3.5 Three Storage Representations

The three memory stores originate from the same canonical conversation, but they expose the information in different forms. S1 preserves the original sessions in a flat directory. S2 preserves the same session contents but introduces a topic-based directory hierarchy. S3 rewrites the source dialogue into entity- and topic-centred facts. The three stores therefore differ in both the amount and the type of transformation applied before retrieval.

3.5.1 S1: Flat Raw Sessions

S1 is the least transformed memory representation. It contains one Markdown file for each of the 30 conversation sessions. All session files are stored in a single directory. Within each file, source turns remain in their original chronological order. Speaker names, session dates, dialogue text, image captions, and source locators are preserved.

S1 does not add topic folders, summaries, extracted facts, or cross-session consolidation. Its main advantage is source fidelity. A retriever can inspect the original wording and surrounding dialogue without depending on an earlier summarisation decision. Direct source locators also make retrieved evidence straightforward to trace back to the canonical records.

However, source fidelity does not guarantee easy access. Information about the same person, event, or plan may be distributed across many sessions. Repeated descriptions and later corrections can also make the relevant state difficult to identify. S1 therefore provides a raw-filesystem reference representation rather than an assumption that unmodified dialogue is the best memory format.

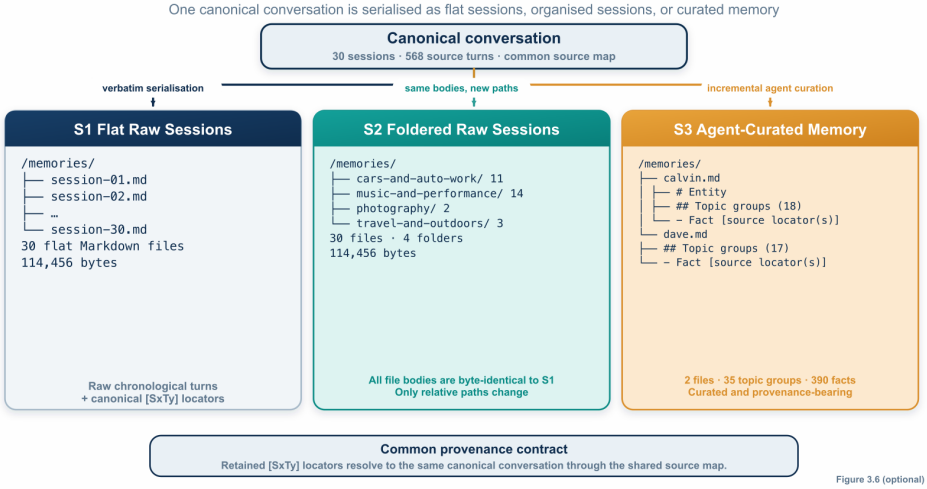


Figure 3.3. Organisation of the three memory representations. S1 stores 30 raw session files in a flat directory; S2 places byte-identical session bodies under four topic folders; S3 reorganises the conversation into two entity files containing 35 topic groups and 390 provenance-bearing facts.

3.5.2 S2: Foldered Raw Sessions

S2 is constructed from a verified copy of S1. It contains the same 30 session files, but places each complete file under one of four topic folders: `music-and-performance`, `photography`, `travel-and-outdoors`, and `cars-and-auto-work`. The body of every session file remains byte-identical to its S1 counterpart. Only the relative path changes.

This design isolates the navigation structure as closely as possible. S2 does not summarise a session, split it into new semantic units, or rewrite its dialogue. It instead adds topic labels that a filesystem Retrieval Agent may use when deciding where to search. The contrast between S1 and S2 therefore asks whether directory organisation adds value when the underlying source text remains fixed.

The path information affects the retrievers in different ways. R1 can inspect and navigate the folder hierarchy directly. R2 does not include folder names in its first-stage BM25 text, so the initial lexical ranking remains based on the same dialogue content as in S1. However, relative paths are visible in the returned observations. They may therefore influence a later query produced by the Retrieval Agent.

S2 also has a limitation. A complete session may contain several themes even though it must occupy one directory. Its folder assignment is therefore a navigation cue rather than a complete semantic description of every source turn.

3.5.3 S3: Agent-Curated Memory

S3 is an agent-curated filesystem. Unlike S2, it changes both the organisation and the wording of the stored information. A Management Agent processes the conversation incrementally and maintains persistent Markdown files. It may create files, add or revise facts, group related information, reconcile later updates, and introduce cross-references. This design follows the broad management-agent paradigm of *Filesystem-Based Memory* [4], while the local provenance and validation rules are project-specific.

The frozen S3 store contains two entity-oriented files, one for each main speaker. Each file uses an H1 heading for the entity and H2 headings for topic groups. Factual list items appear under these topics. The store contains 390 locator-bearing fact units under 35 H2 groups. Every formal fact contains one or more source locators that link it back to the canonical conversation records.

A curated fact may combine information from several source turns or sessions. S3 can therefore place facts that were distant in the raw history into one thematic section. It may also express an updated state more directly than the original dialogue. These properties can reduce local dispersion and provide stronger entity and topic cues.

Curation also introduces a new failure surface. The Management Agent may omit a fact, remove a qualifier, merge different entities, flatten a temporal sequence, or retain a stale state. A later retriever cannot recover information that was never written into S3. Furthermore, curation does not necessarily reduce storage size. In this implementation, detailed consolidation and repeated provenance can make curated facts longer than their source fragments. S3 should therefore be understood as a complete agent-curated memory representation, not as a guaranteed compression method.

Figure 3.3 makes the structural differences among S1, S2, and S3 explicit while retaining their common canonical source basis.

Table 3.3. Comparison of the three memory representations.

Property	S1: Flat Raw Sessions	S2: Foldered Raw Sessions	S3: Agent-Curated Memory
Primary stored unit	Raw session file	Raw session file	Curated factual item
Source wording rewritten	No	No	Yes
Main structure	One flat directory	Four topic folders	Entity files and H2 topic groups
Chronology	Preserved directly	Preserved directly	Represented through facts, dates, and locators
Provenance	Direct source locators	Direct source locators	Inline source locators
Main intended benefit	Maximum source fidelity	Additional navigation cues	Thematic grouping and cross-session consolidation
Principal risk	Relevant facts remain dispersed	Folder taxonomy may be imperfect	Omission, distortion, or stale consolidation

3.6 Three Retrieval Systems

R1, R2, and R3 are evidence-retrieval systems. They do not generate the final benchmark answer. Each retriever observes a question, searches one declared memory representation, and selects evidence. The selected material is then validated and converted into the common EvidenceBundle described in Section 3.7.

3.6.1 R1: Agentic Filesystem Retrieval

R1 is an LLM-controlled filesystem Retrieval Agent based on the filesystem-search paradigm in [4]. It receives the question, store-specific search instructions, and read-only access to one mounted memory store. Its role is to locate supporting source material rather than answer the question.

The Agent can use four filesystem tools:

- `view`, which inspects a file or directory;
- `grep`, which finds text matches across files;
- `toc`, which exposes headings and document structure;
- `section_read`, which reads a selected document section.

These tools provide access but do not determine the search strategy. In particular, `grep` performs deterministic text matching. The Agent must decide which words or expressions to search, which files require inspection, and whether the returned context is sufficient.

A typical R1 episode follows an observation–action loop. The Agent first surveys the available store. It then formulates a search direction, calls one or more read-only tools, inspects the returned observations, and refines its search when evidence remains incomplete. This process continues until the Agent saves sufficient evidence or reaches a stopping condition.

One accepted action-bearing provider completion is defined as one R1 retrieval round. A completion may request several independent filesystem reads, so one round is not equivalent to one tool call. However, every accepted round must use exactly one action mode: filesystem operations, `take_note`, or `finish_search`. These modes cannot be mixed in one completion. Evidence selection and search termination therefore consume rounds of their own. Protocol-correction attempts and context-compaction calls are counted separately and do not create retrieval rounds. Store-specific round limits are reported in Chapter 4.

R1 adds two host-side orchestration actions to the four filesystem tools. These actions are project-specific and are not tools proposed by *Filesystem-Based Memory* [4].

The first action, `take_note`, records an evidence selection. The Agent may identify only line ranges that have already appeared in a successful observation. It cannot supply a rewritten quotation or model-generated summary as evidence. The host reopens the frozen file, verifies that the selected range was observed, extracts the exact text, resolves its source locators, and creates an EvidenceItem.

The second action, `finish_search`, terminates the retrieval episode. It carries a structured stopping reason, but it cannot contain an answer or introduce new evidence. Its purpose is to close retrieval and hand the validated EvidenceBundle to the downstream Answerer. Thus:

$$\texttt{finish_search} \neq \text{answer generation}.$$

R1 uses different routing guidance for the three store shapes. On S1, the Agent searches a flat collection of raw sessions. On S2, it can use topic folders before inspecting the original session text. On S3, it navigates entity files and topic sections. The underlying filesystem capability remains read-only in all three cases.

3.6.2 R2: ReFind-Style Raw-Chat Retrieval

R2 is an iterative lexical retrieval system for raw dialogue. It adapts the ReFind search principle [6] to the canonical exchange units defined in Section 3.4. The local exchange is the functional counterpart of ReFind's paired dialogue turn, not the single-speaker LoCoMo source-turn unit. Its retrieval hierarchy is:

$$\text{exchange} \longrightarrow \text{session}.$$

An exchange is the first-stage scoring unit, while a session is the higher-level grouping unit. R2 operates over 292 exchanges derived from the 568 source turns in the 30-session conversation.

At each search step, the Retrieval Agent generates one lexical query and may also specify a date range. The query is tokenised and scored jointly against every eligible exchange. R2 does not automatically execute a separate search for each query word.

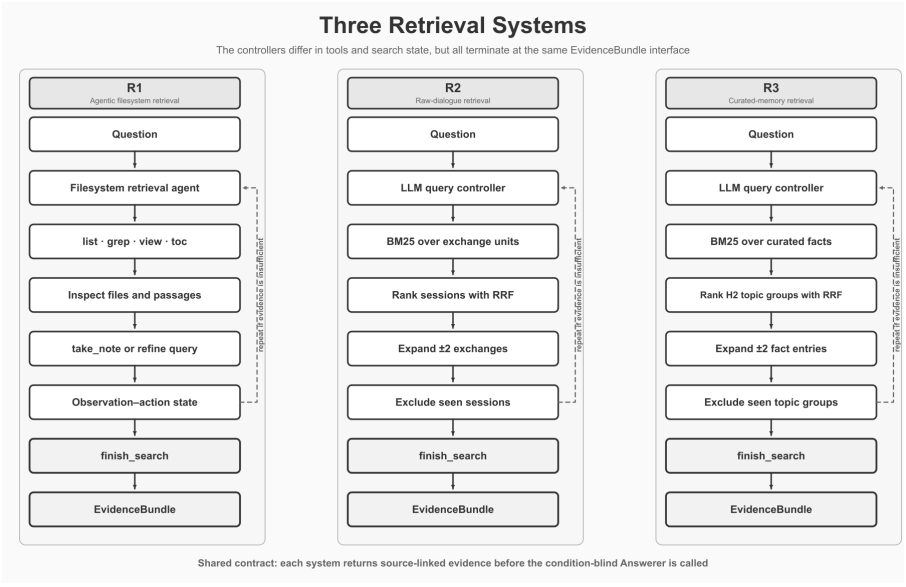


Figure 3.4. Control flow of R1, R2, and R3. R1 navigates a read-only filesystem through iterative observations and actions, R2 ranks raw exchanges and expands within sessions, and R3 ranks curated facts and expands within H2 topic groups. Each path terminates retrieval through `finish_search` and returns a validated EvidenceBundle rather than a benchmark answer.

The indexed exchange text includes speaker names, original dialogue, and available image captions. It excludes gold information and experimental labels. Source locators and folder paths are retained for provenance but do not contribute lexical ranking terms.

R2 first assigns a BM25 score to each eligible exchange [12]. Positive exchange scores are then aggregated within each session to produce a session-level score. Each exchange consequently receives both an exchange rank and the rank of its parent session. The two views are combined by Reciprocal Rank Fusion [13]:

$$\mathrm{RRF}(x) = \frac{1}{60 + r_x} + \frac{1}{60 + r_{\sigma(x)}},$$

where r_x is the exchange rank and $r_{\sigma(x)}$ is the rank of its parent session. The system returns the five highest-ranked centre exchanges.

A centre exchange is not presented in isolation. It is expanded with up to two preceding and two following exchanges from the same session. Expansion never crosses a session boundary. This local window provides the Retrieval Agent with nearby speaker and event context that may not share the query terms.

R2 also maintains a seen-session state. Once a session appears in a search result, that session is excluded from later searches in the same question episode. This rule encourages the Agent to inspect new parts of the conversation rather than repeatedly retrieve the same session.

The controller supports three actions:

1. `search_chatrecord`;
2. `take_note`;
3. `finish_search`.

Each of the four permitted provider completions may produce one valid action. Search, evidence selection, and stopping all consume this same cap. A valid four-completion episode therefore does not necessarily contain four searches. A malformed or empty completion also consumes one completion even though it produces no accepted action.

When the Agent selects a returned context block, the host reopens the raw store and resolves the block to canonical source turns. Overlapping selections are merged deterministically, duplicated source turns are removed, and provenance is validated before an EvidenceItem enters the EvidenceBundle. Model-written text cannot replace the source evidence.

R2 uses the same dialogue text for S1 and S2. Their initial backend ranking is therefore identical for the same query. Later behaviour may differ because S2 paths are visible in observations and can influence subsequent query reformulation. This distinction allows E2 and E4 to test whether folder cues affect an iterative retrieval process even when they are excluded from first-stage lexical scoring.

3.6.3 R3: Retrieval Adapted to Curated Memory

R3 adapts the main R2 retrieval pattern [6] to the structure of S3. It is not a third algorithm proposed by ReFind. The appropriate description is **ReFind-inspired retrieval over agent-curated memory**.

R2 cannot be applied unchanged to S3 because S3 does not consist of raw exchanges grouped by sessions. Its retrieval hierarchy is instead:

$$\text{curated fact} \longrightarrow \text{H2 topic group}.$$

The frozen S3 index contains 390 locator-bearing fact units under 35 H2 topic groups. YAML front matter is excluded because it provides file-level description rather than factual evidence.

Each fact is indexed using three components: the entity name, the H2 topic heading, and the fact text with source-locator syntax removed. Entity and topic terms allow S3's organisation to contribute to lexical retrieval. The separately stored relative-path and source-locator metadata do not become ranking terms. However, a literal cross-reference path written inside the curated fact remains part of the fact text and can affect lexical scoring.

R3 assigns a BM25 score to each eligible fact [12]. Positive fact scores are aggregated within their H2 groups. Fact rank and topic-group rank are then combined using the same Reciprocal Rank Fusion formula as R2 [13]. The five highest-ranked centre facts are returned.

Each centre fact is expanded with up to two neighbouring facts in the same H2 section. Expansion cannot cross an H2 or file boundary. This restriction preserves the thematic context created during memory construction.

R3 tracks seen H2 groups rather than seen sessions. Once a topic group appears in a result, that group is excluded from later searches in the same episode. Other topics for the same entity remain eligible. This is important because one entity file may contain many distinct topic groups.

Temporal filtering also requires an adaptation. A raw exchange belongs to one session date, whereas one curated fact may cite several source turns from different dates. R3 resolves all source dates through the fact's locators. A centre fact is eligible for ranking when at least one of these dates overlaps the requested interval. The subsequent ± 2 neighbourhood expansion preserves local topic context and does not reapply the date predicate, so a neighbouring fact may lie outside the requested interval.

The controller retains the same three-action interface and four-completion cap as R2: `search_chatrecord`, `take_note`, and `finish_search`. Invalid completions also consume the cap. When the Agent selects a result, the host reopens the frozen S3 file, verifies the fact range and H2 boundary, parses its locators, resolves the corresponding canonical records, and constructs the EvidenceItem.

R3 can improve access to information that exists in S3, but it cannot repair a write omission. It is also sensitive to the structure created during curation. Very large topic groups may receive stronger aggregate scores, and a long curated fact may introduce more material than the question requires. These properties form part of the retrieval mechanism rather than evidence of superior or inferior performance.

Figure 3.4 compares the control flow of the three retrieval systems and their common handoff boundary.

Table 3.4. Comparison of the three retrieval systems.

Property	R1	R2	R3
Formal name	Agentic Filesystem Retrieval	ReFind-Style Raw-Chat Retrieval	Retrieval Adapted to Curated Memory
Compatible stores	S1, S2, and S3	S1 and S2	S3
Main retrieval unit	File region or section	Raw exchange	Curated fact
Higher-level group	Filesystem path or section	Session	H2 topic group
Search mechanism	Agent-controlled filesystem tools	BM25, session aggregation, and RRF	BM25, topic aggregation, and RRF
Local context	Agent-selected reads	Up to ± 2 exchanges within one session	Up to ± 2 facts within one H2 group
Seen-state	No explicit backend exclusion state; prior observations remain in the trace	Previously returned sessions	Previously returned H2 groups
Evidence selection	Observed line ranges	Returned exchange windows	Returned fact windows
Stopping mechanism	`finish_search` or round limit	`finish_search` or action limit	`finish_search` or action limit
Principal limitation	Tool-use cost and behavioural variation	Query mismatch or missed implicit relations	Dependence on S3 content and topic structure

3.7 Shared EvidenceBundle and Answerer

The retrieval systems differ internally, but their outputs must reach the Answerer through a common interface. Otherwise, an apparent retrieval effect could instead be caused by a different evidence format or answering prompt. All selective conditions therefore convert their selected material into the same EvidenceBundle schema.

3.7.1 Retrieval-Neutral Evidence Interface

An EvidenceBundle is an ordered collection of validated EvidenceItems associated with one question and one condition. It hides retriever-specific control details from the Answerer while preserving the text and provenance required for answering.

R1 contributes observed filesystem ranges, R2 contributes raw exchange windows, and R3 contributes curated fact windows. These internal units remain visible in the retrieval record, but the Answerer receives them through one common item format. Retrieval scores, search queries, tool observations, and stopping reasons are not part of the answering input.

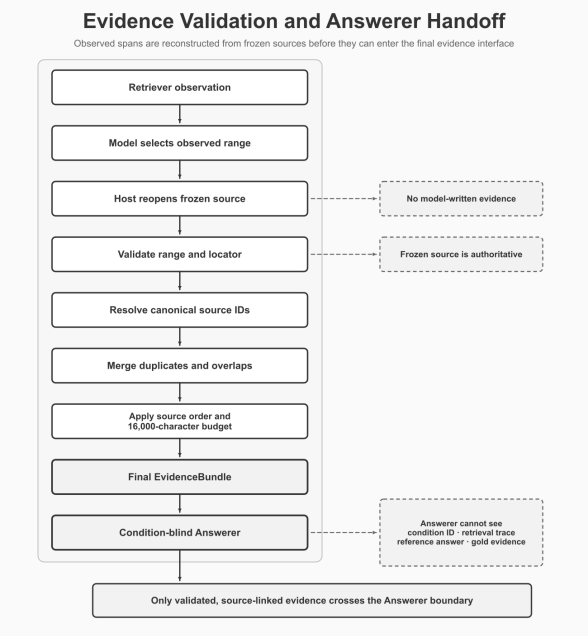


Figure 3.5. Evidence validation and Answerer handoff. Model-selected evidence references are resolved against frozen source bytes, validated, ordered, merged, and budgeted before the condition-blind Answerer generates a structured candidate response.

3.7.2 EvidenceItem Structure

Each EvidenceItem contains:

- a stable evidence identifier;
- the exact selected text;
- the source kind;
- the source path or structural group;
- source locators;
- canonical source-record identifiers;
- source dates and speakers where applicable;
- selection order; and
- a deterministic content identity.

This schema preserves two levels of provenance. The representation-level fields identify where the evidence was read, such as a raw session path or a curated topic group. The canonical fields identify the underlying source turns. Both are required because a curated fact may differ in wording from the source records on which it is based.

3.7.3 Evidence Validation

The Retrieval Agent chooses evidence references rather than authoring evidence text. For R1, it selects previously observed file ranges. For R2 and R3, it selects returned result blocks. The host then reopens the declared frozen store and reconstructs the evidence from source bytes.

This validation boundary prevents a model-generated paraphrase from being treated as retrieved evidence. It also ensures that every source locator resolves through the common provenance layer. Invalid, unobserved, or unresolvable selections are rejected rather than silently repaired.

3.7.4 Evidence Ordering, Merging, and Budget

EvidenceItems retain a deterministic order derived from their accepted selections. Same-path selections with identical, contained, overlapping, or adjacent source ranges may be rejected or merged when the provenance permits it. Repeated anchors are also rejected in R2 and R3. Identical text at two distinct provenance locations is not globally deduplicated.

Each single-source condition from E1 to E6 uses a maximum evidence budget of:

$$\beta = 16{,}000 \text{ characters}.$$

The budget is applied after deterministic evidence merging. Complete EvidenceItems are retained or skipped; their text is not silently cut in the middle. This common limit prevents one selective retriever from obtaining a larger Answerer input simply because it emits more evidence. Chapter 4 treats the budget as a fixed experimental control.

3.7.5 Shared Answerer Input

The Answerer receives the question, the ordered EvidenceBundle, and the shared answering instructions. Its prompt presents evidence through stable identifiers so that the generated answer can cite the supplied material. The Answerer has no retrieval tools and cannot request more memory after the bundle has been finalised.

Full Context uses the same Answerer renderer, answer-generation objective, and structured response contract. To reuse this interface, the 30 complete S1 session bodies are wrapped as 30 content-addressed EvidenceItems in a non-selective full-context bundle. The Answerer therefore cites ordinary evidence identifiers. What Full Context bypasses is question-dependent retrieval and the selective 16,000-character budget, not evidence packaging or citation validation.

3.7.6 Condition-Blind Answering

The Answerer does not receive the condition identifier, store label, retriever name, retrieval trace, retrieval scores, gold answer, gold evidence, or evaluation labels. It is therefore blind to the explicit condition label. It is not blind to evidence provenance: the rendered items expose source kind, text, timestamps, and locators. In E7, the presence of both raw and curated items is consequently visible even though the label "E7" is not.

This condition-label-blind boundary keeps answer generation comparable. E1–E7 may change the memory path and the evidence delivered, but they do not change the role of the Answerer. Any remaining difference enters through the evidence content and provenance or through stochastic model behaviour, not through an explicit condition instruction.

3.7.7 Output Contract

The Answerer returns a structured object with three fields:

- `answer`;
- `citations`;
- `insufficient_evidence`.

A citation may refer only to an evidence identifier present in the input bundle. The `insufficient_evidence` field records whether the Answerer judges the supplied evidence to be inadequate. It does not trigger another retrieval attempt.

If the initial response violates the output schema, the system may perform a constrained format repair. The repair cannot retrieve new material or change the EvidenceBundle. Its prompt instructs the model to preserve the answer content, use only allowed evidence identifiers, and add no new facts or citations. The purpose is to recover a valid representation of the original response; the repair is not treated as another retrieval or answering stage.

Figure 3.5 summarises the validation, packaging, and answering boundary shared by the selective conditions.

3.8 Mapping of E1–E7 Conditions

3.8.1 Complete Condition Map

The seven experimental conditions describe complete paths from a memory representation to the shared Answerer. They are not seven independent retrieval algorithms.

Table 3.5. Storage, retrieval, and evidence composition of E1–E7.

Condition	Memory representation or source	Retrieval or processing	Evidence supplied to the Answerer	Primary methodological role
E1	S1: Flat Raw Sessions	R1	Raw filesystem spans	Flat raw-filesystem reference path
E2	S1: Flat Raw Sessions	R2	Ranked raw exchange windows	Enhanced retrieval over flat raw dialogue
E3	S2: Foldered Raw Sessions	R1	Raw filesystem spans	Effect of directory organisation on R1
E4	S2: Foldered Raw Sessions	R2	Ranked raw exchange windows	Interaction between folder cues and R2
E5	S3: Agent-Curated Memory	R1	Curated filesystem spans	General filesystem retrieval over curated memory
E6	S3: Agent-Curated Memory	R3	Ranked curated fact windows	Retrieval adapted to curated structure
E7	E2 and E6 EvidenceBundles	Deterministic fusion	Interleaved raw and curated evidence	Dual-source complementarity and interference

3.8.2 Controlled Raw-Memory Core

E1–E4 form the controlled raw-memory core. E1 and E2 share S1 but use different retrieval systems. E3 and E4 share S2 but use different retrieval systems. E1 and E3 use R1 over the same raw session contents under different directory structures. E2 and E4 apply R2 to the same raw exchange corpus while exposing different provenance paths in the observations.

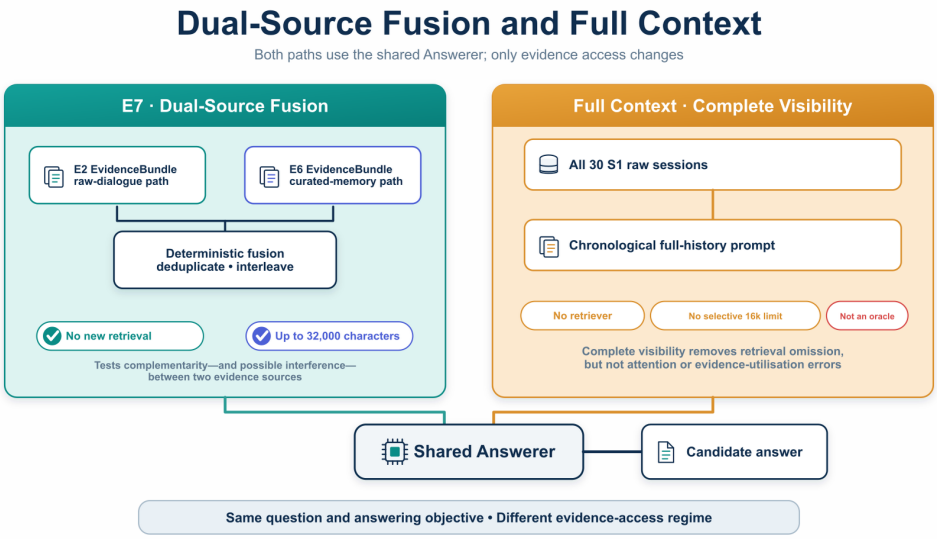


Figure 3.6. Comparison of E7 dual-source fusion and the Full Context diagnostic path. E7 deterministically combines the completed E2 and E6 EvidenceBundles without new retrieval, whereas Full Context exposes the complete ordered raw history without the selective 16,000-character limit. Both paths use the shared Answerer.

These relations support the closest component-level comparisons in the study. They distinguish the effect of directory organisation from the effect of the retrieval family without changing the underlying dialogue.

3.8.3 Curated-Memory Extension

E5 and E6 use the same frozen S3 snapshot. E5 asks a general filesystem Agent to navigate curated files and sections. E6 applies retrieval that is explicitly adapted to S3's fact and topic structure.

These conditions compare access methods over the same curated store. Comparisons between E5 or E6 and raw-store conditions are broader system comparisons. S3 changes wording, granularity, organisation, and the presentation of provenance, so such comparisons cannot be interpreted as a single-feature storage ablation.

3.8.4 Dual-Source Condition

E7 is constructed from the completed E2 and E6 EvidenceBundles. It does not add another memory representation or retrieval system. Its purpose is to test whether already selected raw and curated evidence provide complementary information or create interference when presented together.

The parent pair is methodologically meaningful. E2 supplies evidence from raw dialogue through the retrieval system designed for raw chat. E6 supplies evidence from curated memory through the adapter designed for curated facts. Their fusion therefore contrasts two complete, representation-appropriate evidence paths.

3.8.5 Interpretation Boundaries

The condition set is theory-driven rather than an exhaustive Cartesian product. R3 is a representation-specific adaptation of the R2 retrieval principle, not a universal retriever that must be paired with every store. E7 is a fusion condition, not a new retrieval algorithm. Full Context is a diagnostic path outside E1–E7.

The strength of an inference also depends on the comparison. E1 versus E3 and E2 versus E4 isolate directory organisation most directly because the raw file contents are held constant. E1 versus E2, E3 versus E4, and E5 versus E6 compare retrieval systems on one fixed store. Comparisons between raw and curated conditions concern complete system paths because curation changes several properties at once.

Replicating the same condition graph with another backbone model does not create new E-condition definitions. It tests whether the observed effects remain stable when the model changes.

3.9 Alternative Evidence-Access Paths

E7 and Full Context depart from the standard single-store selective path in different ways. E7 retains selective retrieval but combines two completed evidence sources. Full Context removes external evidence selection and sends the complete raw conversation to the Answerer.

3.9.1 E7 Dual-Source Fusion

E7 requires a completed E2 EvidenceBundle and a completed E6 EvidenceBundle for the same question. The two parent bundles represent different evidence sources:

- E2 provides raw dialogue evidence selected through R2;
- E6 provides curated facts selected through R3.

Before fusion, the system verifies that both bundles correspond to the same question and the expected parent conditions. Their provenance and single-source budgets must also remain valid. If either parent is missing or mismatched, E7 is not produced.

Fusion is deterministic. The order inside each parent bundle is preserved, while the two streams are interleaved:

$$\text{raw}_1,\ \text{curated}_1,\ \text{raw}_2,\ \text{curated}_2,\ \ldots$$

This rule prevents one source from occupying the entire beginning of the Answerer input.

Exact duplicates are identified using the combination of source kind, canonical source identifiers, and evidence-text identity:

$$\big(\text{source kind},\ \text{source identifiers},\ \text{text hash}\big).$$

Source kind remains part of the key because raw and curated items may cite the same source turns while expressing them differently. Such items are retained when their text differs. This allows E7 to preserve the distinction between original dialogue and its curated representation.

Each parent bundle retains the single-source budget:

$$|\mathcal{B}_{E2}| \le \beta, \qquad |\mathcal{B}_{E6}| \le \beta.$$

The fused evidence is therefore bounded by:

$$|\mathcal{B}_{E7}| \le 2\beta = 32{,}000 \text{ characters}.$$

The fused bundle is then passed to the shared Answerer once. Fusion does not formulate a query, rerun R1, R2, or R3, invoke an LLM, or create an intermediate summary. E7 therefore measures post-retrieval complementarity and interference rather than the performance of a new joint retriever.

3.9.2 Full Context Diagnostic Path

Full Context uses a separate evidence-access path. For each question, it constructs one deterministic input containing all 30 raw sessions and all 568 source turns. Sessions and turns remain in fixed chronological order. Original dialogue and available image captions are retained. No question-dependent filtering, summarisation, or reordering is performed.

The Full Context answer can be represented as:

$$\hat{a}_{FC} = A_\theta(q, \mathcal{C}),$$

where $\mathcal{C}$ is the complete canonical raw conversation. In contrast, a selective condition uses:

$$\hat{a} = A_\theta(q, \mathcal{B}),$$

where $\mathcal{B}$ is a question-specific EvidenceBundle.

Full Context does not use S2, S3, R1, R2, R3, or E7 fusion. It packages the 30 complete S1 session bodies as a non-selective full-context bundle so that it can use the shared Answerer interface. It does not apply the 16,000-character selective-evidence limit. Applying that limit would convert it into a truncated-context condition rather than Full Context.

The Answerer retains the same answer-generation objective and structured output contract where applicable. The principal change is the information-access path: the model must locate and combine relevant information inside the complete history instead of receiving externally selected evidence.

Full Context is a diagnostic path, not a fourth memory store, a fourth retriever, or an eighth E-condition. It is also not a theoretical oracle. Complete visibility does not ensure correct use, and long-context performance can depend on evidence position [5]. The Answerer may still be affected by repeated facts, changing states, entity similarity, temporal confusion, or failed multi-session composition.

The comparison is informative in both directions. If Full Context repairs a selective-retrieval error, missing evidence is a plausible explanation. If selective retrieval succeeds while Full Context fails, evidence reduction may have improved focus. If both paths receive the required information but fail, the remaining difficulty is more likely to involve evidence utilisation or reasoning [7]. Chapter 4 defines how these paths are evaluated, and Chapter 5 examines their observed behaviour.

Figure 3.6 contrasts the two alternative evidence-access paths while holding the answering objective constant.

This chapter has defined the complete system graph. E1–E4 form the controlled raw-memory core, E5–E6 extend the design to agent-curated memory, E7 combines two completed evidence sources, and Full Context removes explicit retrieval as a diagnostic comparison. All selective conditions converge on the same EvidenceBundle and Answerer boundary.

Chapter 4 instantiates this architecture as a controlled experiment. It consolidates the method constants defined here and specifies the frozen question set, backbone models, remaining run controls, execution protocol, evaluation metrics, and paired statistical analysis.

Chapter 4. Experimental Design and Evaluation Methodology

Chapter 3 defined the memory representations, retrieval systems, EvidenceBundle interface, Answerer, dual-source fusion mechanism, and Full Context path. This chapter explains how those components are instantiated as a controlled experiment. It defines the frozen test set, experimental matrix, model roles, controlled conditions, evaluation metrics, paired analysis, and error-review method.

The chapter separates design from outcome interpretation. It specifies which comparisons are valid and what each comparison can identify. It does not report final scores or select a winning condition. Those results are presented in Chapter 5.

4.1 Research Questions, Variables, and Identification Strategy

4.1.1 Mapping Research Questions to Experimental Comparisons

The six research questions introduced in Chapter 1 require different experimental evidence. RQ1 concerns storage, retrieval, and their interaction. RQ2 asks whether the same architectural effects remain stable across backbone models. RQ3 examines dual-source fusion. RQ4 compares selective retrieval with Full Context. RQ5 locates likely failures along the information path. RQ6 combines quality, stability, cost, and maintenance considerations.

No single condition answers all six questions. The study therefore assigns each research question a defined comparison family. This mapping prevents an overall condition ranking from replacing the analysis required by each question.

Table 4.1. Mapping from research questions to experimental comparisons and evidence.

Research question	Main experimental material	Principal comparisons	Main evidence
RQ1: storage and retrieval effects	E1–E6	E3−E1, E4−E2, E2−E1, E4−E3, E6−E5, and the storage–retrieval interaction	Paired answer-quality differences, evidence coverage, and retrieval traces
RQ2: cross-backbone stability	Qwen and DeepSeek E1–E7	The same condition contrast under both backbone models	Effect direction, magnitude, question-level agreement, and contrast-of-contrasts
RQ3: dual-source fusion	E2, E6, and E7	E7−E2 and E7−E6	Repair, preserve, harm, unresolved cases, evidence overlap, and cost
RQ4: Full Context versus selective retrieval	DeepSeek selective conditions and DeepSeek-family Full Context	Paired Full Context comparisons, with the raw-memory conditions as the principal diagnostic family	Answer quality, context size, latency, and paired success or failure states
RQ5: failure location	Stores, retrieval traces, EvidenceBundles, candidate answers, and evaluation outputs	Stage-based review within and across conditions	Storage fidelity, evidence coverage, budget records, answer utilisation, and manual cases
RQ6: practical choice	All formal conditions	Quality–cost and stability comparisons	Answer quality, calls, tokens, latency, construction cost, recovery cost, and maintenance implications

The comparisons in Table 4.1 are paired by question whenever both conditions produce an answer for the same Main-60 item. The pair, rather than the aggregate condition mean alone, is the basic comparative unit.

Figure 4.1 summarises the complete experimental and evaluation workflow. It also makes the evaluation boundary explicit: gold answers and gold evidence enter only after candidate-answer generation.

4.1.2 Independent, Dependent, and Controlled Variables

The experiment varies five main factors. The first is the **memory representation**: S1, S2, or S3. The second is the **retrieval system**: R1, R2, or R3, subject to the compatible pairings defined in Chapter 3. The third is the **backbone model** used by the Retrieval Agent and Answerer. The fourth is the **evidence-access regime**, which distinguishes selective retrieval from Full Context. The fifth is **evidence composition**, which distinguishes a single EvidenceBundle from E7 dual-source fusion.

The dependent variables cover four levels. Answer-level lexical measures are token Precision, Recall, and F1. Answer-level semantic measures are Strict Accuracy, Partial Rate, and Soft Semantic Score. Evidence-level measures are Evidence Recall, Any-hit, and All-hit. Process measures include calls, tokens, latency, retrieval rounds or actions, failed attempts, recovery work, and construction cost.

The principal controls are the unified Main-60 test set, its fixed order, the canonical conversation, frozen S1–S3 snapshots, condition definitions, the EvidenceBundle contract, the Answerer contract, and the offline evaluation rules. These controls are applied wherever they are meaningful. Full Context is not subject to a selective-evidence budget because doing so would change it into a truncated-context condition.

4.1.3 Identifiable and Non-Identifiable Effects

The raw-memory conditions provide the cleanest component comparisons. S1 and S2 contain the same raw session bodies. E3−E1 and E4−E2 therefore examine directory organisation under R1 and R2, respectively. E2−E1 and E4−E3 compare the two retrieval systems while holding the raw store fixed. Together, E1–E4 support a storage–retrieval interaction analysis.

These contrasts still have explicit limits. R1 and R2 are complete retrieval systems with different observation and action interfaces. Their difference cannot be reduced to one ranking algorithm. In E2 and E4, directory names do not enter the initial lexical score, but returned provenance paths may affect a later Retrieval Agent action. E4−E2 can therefore capture adaptive use of folder cues, not a direct effect on first-stage BM25 scoring.

E5 and E6 use the same frozen S3 store. E6−E5 compares a general filesystem retriever with retrieval adapted to curated facts and topic groups. Comparisons between S3 and a raw store are broader. S3 changes wording, information granularity, organisation, and provenance presentation together. A raw-versus-curated difference is therefore a complete-system difference rather than a single-component effect.

E7 evaluates fusion. It reuses the completed E2 and E6 EvidenceBundles and performs no new retrieval. Its comparison with either parent tests whether a second evidence source repairs, preserves, or harms the answer. It does not estimate the value of a third retrieval algorithm.

The backbone comparison also changes more than one role. Within an E1–E7 pipeline, the same backbone model serves as both Retrieval Agent and Answerer. A Qwen–DeepSeek difference is therefore an end-to-end query-time pipeline difference. It does not isolate retrieval capability from answer-generation capability.

Full Context is diagnostic. It removes explicit retrieval and expands the information visible to the Answerer. It also changes input scale. It is not a strict one-factor ablation, an eighth E-condition, or an oracle. In addition, Full Context uses a later served identifier within the same DeepSeek model family. Comparisons with the DeepSeek selective conditions must therefore be interpreted as strong descriptive diagnostics rather than strict single-factor causal estimates.

4.1.4 Primary and Exploratory Analyses

The primary analyses use all 60 questions and retain same-question pairing. They cover the raw-memory storage and retrieval contrasts, the S3 access contrast, the E7 parent comparisons, matched architectural effects across backbone models, and the Full Context diagnostic comparisons. The analysis emphasises effect direction, effect magnitude, and the distribution of question-level changes.

Category-level analyses use the four non-adversarial LoCoMo categories. They are exploratory mechanism analyses. They indicate where an effect may be concentrated, but they do not replace the full-set result. Correlation analyses, anomaly inspection, and individual cases are also exploratory.

The term **primary comparison** is used to describe analytical priority. It does not imply that the full report was preregistered before every run. Claims in Chapter 5 must therefore distinguish frozen design choices from later exploratory interpretation.

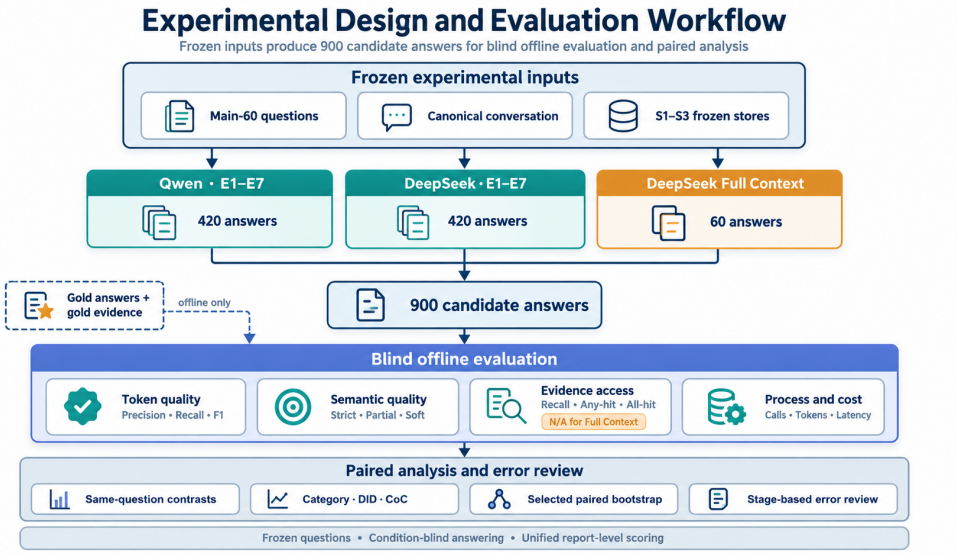


Figure 4.1. Experimental design and evaluation workflow. Frozen inputs are processed through the Qwen E1–E7 matrix, the DeepSeek E1–E7 matrix, and the DeepSeek Full Context path. Their 900 candidate answers then enter condition-blind offline evaluation, paired analysis, and stage-based error review. Gold answers and gold evidence are restricted to offline evaluation.

4.2 Dataset and the Frozen Main-60 Test Set

4.2.1 LoCoMo Source and Conversation Scope

The experiment uses the question-answering material from LoCoMo [1]. The frozen source file was obtained from the official LoCoMo repository at commit `cbfbc1dba6bc53d00625212a0f22d55ffee7c1fc` and canonicalised under the local data-contract version `locomo-canonical-v1`. The study focuses on the released conversation identified locally as `conv-50`. This conversation contains 30 ordered sessions and 568 source turns. The source preserves speaker identity, session order, dates, dialogue text, and the released image captions where present.

The study does not rerun a vision model. Existing captions are treated as part of the released textual record. All storage representations and retrieval indices originate from the same canonicalised conversation described in Chapter 3.

Using one conversation supports detailed provenance checks and exact same-question comparisons. It also limits external validity. The results describe controlled variation within this conversation and question set. They do not estimate variation across LoCoMo conversations, users, or datasets.

The source-file fingerprint, licence record, ordered question identifiers, source manifest, and evidence-eligibility fields are recorded in Appendix B.

4.2.2 Composition of Main-60

Main-60 is one frozen test set of 60 questions. It contains only the four non-adversarial categories used in this study. The fixed distribution is shown in Table 4.2.

Table 4.2. Composition and evidence eligibility of Main-60.

Category	Label used in this report	Questions	Evidence-eligible questions	Role in analysis
C1	Multi-hop	13	13	Requires information from more than one part of the history
C2	Temporal	13	13	Requires time, order, update, or state tracking
C3	Open-domain	7	5	Combines dialogue information with wider knowledge or inference
C4	Single-hop	27	27	Uses a more localised fact or relation
Total	**Unified Main-60**	**60**	**58**	**Primary full-set analysis unit**

The question order is fixed across conditions and backbone models. Selection did not use candidate answers, evaluation scores, or model outputs. The 60 questions are analysed as one set. The report does not divide its main results by when a question was added to the experimental suite.

The category counts are unequal. C3 contains only seven questions. Category comparisons are therefore descriptive and should be interpreted with particular care for small groups.

4.2.3 Gold Answers and Gold Evidence

Each Main-60 question has a reference answer used for offline answer evaluation. LoCoMo also supplies source-turn annotations for many questions [1]. The two forms of gold information have different roles. The reference answer supports token and semantic evaluation. The source identifiers support evidence-access evaluation.

Fifty-eight Main-60 questions have usable annotated evidence for the evidence metrics. Two questions do not. The denominator for Evidence Recall, Any-hit, and All-hit is therefore 58, not 60. This eligibility rule is fixed before condition-level evidence aggregation.

Annotated source turns are evidence locations, not complete reasoning chains. A valid answer may use an alternative supporting passage, especially for open-domain questions. Evidence measures are therefore diagnostic indicators of annotated-source coverage. They are not a complete definition of retrieval correctness.

Gold answers, gold evidence, question categories, and evaluation labels are unavailable to the online memory pipeline. They are attached only after candidate-answer generation. The Retrieval Agent, deterministic retrieval backend, fusion procedure, and Answerer cannot use them.

4.3 Experimental Matrix and Core Comparisons

4.3.1 Complete Experimental Matrix

The formal study contains three answer sets. Qwen produces one answer for every Main-60 question under E1–E7. DeepSeek produces the same E1–E7 matrix. A DeepSeek-family Full Context path produces one further answer per question. The complete scale is shown in Table 4.3.

Table 4.3. Formal experimental matrix.

Answer set	Conditions or path	Questions per condition	Candidate answers	Query-time model scope
Model 1	Qwen E1–E7	60	420	Qwen Retrieval Agent and Qwen Answerer
Model 2	DeepSeek E1–E7	60	420	DeepSeek Retrieval Agent and DeepSeek Answerer
Diagnostic baseline	DeepSeek-family Full Context	60	60	DeepSeek-family Answerer; no Retrieval Agent
Total	**Two E1–E7 matrices plus Full Context**	—	**900**	—

The 900 answers are not 900 independent test questions. They are repeated observations over the same 60 questions. Statistical comparisons must preserve this structure.

Every E1–E7 answer is associated with its storage path, retrieval trace, EvidenceBundle, Answerer record, and process measurements. Full Context has no retrieval trace. It records the complete source input, Answerer output, and applicable process measurements.

4.3.2 E1–E4: Controlled Storage–Retrieval Comparisons

E1–E4 form the controlled raw-memory core. As defined in Table 3.5, E1 and E2 use S1 with R1 and R2, respectively, while E3 and E4 use S2 with the same two retrieval systems.

The storage-organisation contrasts are E3−E1 and E4−E2. The retrieval-system contrasts are E2−E1 and E4−E3. The same four cells also support a difference-in-differences description of whether directory organisation affects R1 and R2 differently.

For a per-question metric m_i, the storage–retrieval interaction is defined as:

$$DID_i^{(m)} = \big(m_i(E4) - m_i(E2)\big) - \big(m_i(E3) - m_i(E1)\big)\,.$$

The raw interaction is the mean of these paired question-level values. For cross-metric directional interpretation, it is oriented using the metric-direction term o_m defined in Section 4.7.2:

$$DID_{i,+}^{(m)} = o_m\, DID_i^{(m)}.$$

A positive oriented value means that the S1-to-S2 change is more favourable under R2 than under R1. The interaction does not identify which internal R2 action caused the difference.

4.3.3 E5–E6: Curated-Memory Systems

E5 and E6 share the same frozen S3 Agent-Curated Memory. E5 uses R1, while E6 uses R3. The contrast E6−E5 therefore compares two access methods on one curated representation.

This contrast is narrower than a raw-versus-curated comparison, but it is still a complete retrieval-system comparison. R1 navigates files and sections through filesystem tools. R3 ranks curated facts and topic groups, expands local context, and uses topic-level seen-state. Their output is standardised only at the EvidenceBundle boundary.

Comparisons between E5 or E6 and E1–E4 are retained as complete-system comparisons. They may show whether a curated-memory path is useful as deployed. They cannot isolate the effects of summarisation, fact consolidation, directory structure, or provenance formatting.

4.3.4 E7: Dual-Source Fusion

E7 reuses the completed E2 and E6 EvidenceBundles for the same question. E2 supplies selected raw-dialogue evidence. E6 supplies selected curated-memory evidence. Fusion validates the parent identities, removes exact provenance-aware duplicates, preserves within-source order, and interleaves the two evidence streams.

The two principal comparisons are E7−E2 and E7−E6. They test whether adding the second evidence source repairs an error, preserves a correct answer, introduces harm, or leaves an error unresolved. E7 does not make a new search call and does not generate an intermediate summary.

Its cost has two interpretations. The recorded E7 operation measures the post-retrieval fusion and fused Answerer call because the parent bundles already exist in the experiment. A deployable online E7 system would need to execute both parent retrieval paths before the fused Answerer call. Chapter 5 reports these two cost views separately.

4.3.5 Full Context Diagnostic

For each question, Full Context supplies all 30 raw S1 sessions in fixed chronological order. It performs no explicit retrieval, filtering, summarisation, or evidence reordering. It also does not apply the selective EvidenceBundle limit.

Full Context is compared with the DeepSeek selective conditions on the same questions. The raw-memory E1–E4 family provides the principal diagnostic anchor because it avoids the additional content transformation introduced by S3. Results against E5–E7 are also reported, but they answer broader system-level questions.

Full Context has complete source visibility by construction, but this property is not Evidence Recall because no selection process exists. Section 4.6.3 defines the resulting metric-applicability rule.

The diagnostic comparison is bidirectional. A Full Context success paired with a selective-retrieval failure is consistent with retrieval omission or evidence-budget loss. A selective-retrieval success paired with a Full Context failure is consistent with a benefit from evidence selection or reduced noise. Neither pattern establishes a cause without trace and case review.

4.4 Models and Evaluation Roles

4.4.1 Backbone Models and Query-Time Roles

The experiment uses two pipelines with different backbone models for E1–E7. Model 1 uses a Qwen served model. Model 2 uses a DeepSeek served model. Within each pipeline, the same backbone model performs both Retrieval Agent and Answerer roles. The prompts, stores, condition graph, EvidenceBundle interface, and evaluation rules remain fixed.

Full Context uses the DeepSeek V4.1 Flash family but a later served identifier. It contains only the Answerer role because explicit retrieval is absent. Table 4.4 records the model identities returned by the service.

Table 4.4. Backbone models and evaluation roles.

Report role	Requested service alias	Recorded served identity	Applicable conditions	Role in the pipeline
Model 1: Qwen	`coding`	`qwen3.8:27b`	E1–E7	Retrieval Agent and Answerer
Model 2: DeepSeek	`x-test-2`	`deepseek-v4.1-flash`	E1–E7	Retrieval Agent and Answerer
DeepSeek-family Full Context	`x-test-2`	`deepseek-v4.1-flash-1017`	Full Context	Answerer only
Independent LLM judge	`x-test-1`	`glm-5.3-flash`	All 900 candidate answers	Offline semantic evaluator

The served identities are deployment records, not independently verified descriptions of model weights or training data. The Full Context comparison is described at the model-family level because its served identity is not identical to that of Model 2 E1–E7.

Detailed model metadata and fixed configuration values belong in Appendix F.

4.4.2 Independent LLM Judge

Semantic evaluation uses one independent LLM judge from a different model family. The judge receives only the question, reference answer, and candidate answer. It does not receive the condition identifier, backbone model, storage representation, retrieval system, trace, EvidenceBundle, cost, token score, or prior judgement.

The judge applies one fixed three-label rubric. A response is `correct` when it contains all required information and has no material contradiction. It is `partial` when it contains a relevant but incomplete answer, or when correct material is mixed with a material unsupported claim. It is `incorrect` when the required answer is missing, contradicted, or replaced by a substantively wrong response.

One judgement produces all three semantic measures defined in Section 4.6.2. The study does not use one model for strict scoring and another for partial scoring. The LLM judge supplements deterministic token evaluation because semantic equivalence may not preserve lexical overlap. It is not treated as error-free ground truth, since LLM-based evaluation can exhibit position, verbosity, and reasoning biases [8].

The complete judge prompt, rubric, output schema, and prompt identity belong in Appendix A. Judge identity, blindness controls, parsing rules, and evaluation configuration belong in Appendices F and G.

4.5 Controlled Experimental Conditions

Fair comparison requires both shared controls and explicit exceptions. Table 4.5 separates factors that are held fixed, deliberately varied, or not applicable.

Table 4.5. Held-fixed, varied, and non-applicable experimental factors.

Factor	Held-fixed component	Intended variation	Non-applicable or bounded exception
Questions	Unified Main-60 membership and order	None	None
Source conversation	Canonical `conv-50` records	None	None
Memory stores	Frozen S1–S3 snapshots across query-time runs	S1, S2, or S3 according to condition	Full Context reads complete S1 session bodies rather than a selective store path
Retrieval	Frozen R1–R3 definitions and condition pairing	Retrieval system according to E1–E6	E7 reuses parent bundles; Full Context has no retriever
Evidence input	EvidenceItem schema, provenance validation, deterministic ordering, and shared renderer	Retrieved content and E7 source composition	Full Context is not subject to the selective-evidence limit
Answer generation	Shared Answerer objective and structured output contract	Backbone model and supplied evidence	Full Context uses the same objective with complete raw history
Backbone model	One model is used consistently within each E1–E7 pipeline	Qwen versus DeepSeek	Full Context uses only the DeepSeek family
Evaluation	Frozen reference data, metric definitions, eligibility rules, and LLM judge rubric	None	Retrieval metrics are `N/A` for Full Context

Selective conditions use the fixed evidence budget and retriever-specific search limits introduced in Chapter 3. These values were frozen before formal evaluation and were not tuned from Main-60 outcomes. The same condition definitions are used under Qwen and DeepSeek.

The controls do not make every comparison a single-factor experiment. S3 changes several representation properties together. R1, R2, and R3 expose different retrieval mechanisms. Changing the backbone model changes both retrieval and answering. Full Context changes the access mechanism and context scale. These limits are part of the interpretation, not implementation noise.

Exact budgets, caps, decoding settings, model identities, and condition contracts belong in Appendix F. Complete prompts belong in Appendix A. Local process-management details are not part of the scientific comparison and are omitted from the report.

4.6 Evaluation Metrics and Cost Accounting

The evaluation uses complementary answer, evidence, and process measures. Each metric has a defined unit, denominator, aggregation rule, and applicability range. No single score is used as a complete measure of system quality.

4.6.1 Token Precision, Recall, and F1

Token evaluation follows the category-aware LoCoMo scoring basis [1, 14]. Category-specific alignment is performed where required. Each aligned candidate–reference pair is then normalised before token comparison. The implementation converts text to lowercase, removes punctuation and commas, removes the articles and conjunctions `a`, `an`, `the`, and `and`, collapses whitespace, and applies Porter stemming.

All 900 candidate answers are rescored offline with this single evaluator before report-level aggregation. Scores emitted by individual generation runners are retained only as execution diagnostics and are not mixed into the reported aggregates. This common rescoring step is required for direct Precision and Recall comparisons between E1–E7 and Full Context.

Let P_i be the normalised candidate-token multiset for question i, and let G_i be the aligned reference-token multiset. Let c_i be their multiset overlap count. For the ordinary token comparison, per-question Precision and Recall are:

$$\text{Precision}_i = \frac{c_i}{|P_i|}, \qquad \text{Recall}_i = \frac{c_i}{|G_i|}.$$

If either aligned multiset is empty, Precision and Recall are both defined as zero for that comparison. This rule prevents division by zero and is applied before F1 is calculated.

The per-question F1 score is:

$$\text{F1}_i = \begin{cases} \dfrac{2\,\text{Precision}_i\,\text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}, & \text{Precision}_i + \text{Recall}_i > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The implementation retains LoCoMo's category-aware alignment. For C1, let g_{ik} be reference item k and let p_{ij} be candidate item j, after splitting both strings on commas. Each reference item is matched to the candidate item that gives the highest item-level F1:

$$j^*(k) = \arg\max_j \text{F1}(p_{ij}, g_{ik}).$$

The question-level C1 measures average the matched item-level measures:

$$m_i = \frac{1}{K_i} \sum_{k=1}^{K_i} m\Big(p_{i,j^*(k)}, g_{ik}\Big), \qquad m \in \{\text{Precision}, \text{Recall}, \text{F1}\}.$$

If no valid reference item remains after parsing, all three C1 measures are defined as zero.

For C3, only the part of the reference before the first semicolon is used. C2, C3, and C4 otherwise use the ordinary normalised token comparison. Exact parsing and edge-case rules are recorded in Appendix G.

For any per-question metric m_i, the condition-level macro-average is:

$$\overline{m} = \frac{1}{N} \sum_{i=1}^{N} m_i.$$

Thus, reported aggregate F1 is the mean of the 60 per-question F1 values. It is not recomputed from aggregate Precision and aggregate Recall.

Precision and Recall are project-defined diagnostics on the same matching basis. Precision helps identify additional or non-matching answer text. Recall helps identify omitted reference content. F1 provides a balanced lexical score and supports alignment with LoCoMo-style evaluation. All three remain sensitive to wording and answer length.

4.6.2 LLM Judge Metrics

Let the LLM judge label for question i be:

$$y_i \in \{\texttt{correct}, \texttt{partial}, \texttt{incorrect}\}.$$

Strict Accuracy gives credit only to `correct` responses:

$$\text{StrictAccuracy} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{I}(y_i = \texttt{correct}).$$

Partial Rate reports the proportion of answers labelled `partial`:

$$\text{PartialRate} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{I}(y_i = \texttt{partial}).$$

Soft Semantic Score assigns full credit to `correct`, half credit to `partial`, and zero credit to `incorrect`:

$$s_i^{\text{soft}} = \begin{cases} 1, & y_i = \texttt{correct}, \\ 0.5, & y_i = \texttt{partial}, \\ 0, & y_i = \texttt{incorrect}, \end{cases}$$

and:

$$\text{SoftSemanticScore} = \frac{1}{N}\sum_{i=1}^{N} s_i^{\text{soft}}.$$

Partial Rate is descriptive. A high Partial Rate is not necessarily desirable because it may reflect movement from `incorrect` to `partial` or from `correct` to `partial`. It must be interpreted with Strict Accuracy, Soft Semantic Score, and label-transition counts.

The semantic measures and token measures answer different questions. A semantically correct paraphrase may receive a low token score. A response may also repeat reference terms while expressing a wrong relation. Chapter 5 therefore reports their agreement and disagreement rather than treating one family as the ground truth for the other.

4.6.3 Evidence-Access Metrics

Evidence evaluation compares annotated gold source turns with the canonical source identifiers linked to the retrieved EvidenceItems. Let G_i^E be the set of eligible annotated gold source identifiers for question i. Let R_i^E be the union of canonical source identifiers represented in its EvidenceBundle.

Per-question Evidence Recall is:

$$\text{EvidenceRecall}_i = \frac{|G_i^E \cap R_i^E|}{|G_i^E|}.$$

Any-hit records whether at least one annotated source is present:

$$\text{AnyHit}_i = \mathbf{I}\Big(|G_i^E \cap R_i^E| > 0\Big).$$

All-hit records whether every annotated source is present:

$$\text{AllHit}_i = \mathbf{I}\Big(G_i^E \subseteq R_i^E\Big).$$

Condition-level evidence metrics are macro-averaged over the 58 eligible questions. The underlying set operation removes repeated canonical identifiers. It does not treat duplicate evidence items as additional recall.

Evidence Recall measures source coverage rather than evidence precision or downstream use. A bundle can cover all annotated sources and still contain distracting material. The Answerer may also fail despite All-hit, or answer correctly from valid support not listed in the annotations. Evidence metrics are therefore used to diagnose access, not to replace answer evaluation.

Full Context receives the whole source history and does not perform retrieval. Its Evidence Recall, Any-hit, and All-hit values are `N/A`. Complete visibility by construction must not be reported as perfect retrieval.

4.6.4 Process and Efficiency Metrics

Process records distinguish retrieval, answering, fusion, recovery, and evaluation. For each selective answer, the study records Retrieval Agent model calls, filesystem or orchestration calls where applicable, input tokens, output tokens, total tokens, elapsed time, retrieval rounds or actions, failed attempts, and format-repair or retry activity. The Answerer has its own call and token record.

The accounting separates four cost classes:

1. **Memory-construction cost:** one-time work required to create S2 or S3 from the canonical source.
2. **Query-time cost:** retrieval, evidence packaging, and Answerer work required for one question.
3. **Failure and recovery cost:** failed requests, retries, protocol correction, and format repair.
4. **Evaluation-only cost:** token and call cost of the independent LLM judge.

Calls, tokens, monetary charges, and latency are not added into one synthetic scalar. Let d denote one additive cost dimension, such as model calls, input tokens, output tokens, total tokens, or monetary charge. For condition c and question i, each dimension is accounted for separately:

$$C_{i,c,d}^{\text{query}} = C_{i,c,d}^{\text{retrieval}} + C_{i,c,d}^{\text{packaging}} + C_{i,c,d}^{\text{answering}} + C_{i,c,d}^{\text{recovery}}.$$

The packaging term covers EvidenceBundle validation, ordering, merging, and budget enforcement. These operations are deterministic. Their value is therefore zero for model-call, model-token, and API-charge dimensions unless a deployment records a separate non-zero charge. Their wall-clock contribution remains part of the observed latency discussed below.

Judge cost is excluded from deployment query cost:

$$C_{i,c,d}^{\text{recorded}} = C_{i,c,d}^{\text{query}} + C_{i,c,d}^{\text{judge}},$$

with the query and judge components reported separately for every dimension.

For a store that requires one-time construction and is queried Q times, an amortised view is:

$$C_{c,d}(Q) = C_{c,d}^{\text{construction}} + Q\,\overline{C_{c,d}^{\text{query}}}.$$

This expression is used for scenario analysis rather than as a claim about a universal workload.

E7 requires an additional distinction. Its experiment-time record reuses two completed parent bundles. A realistic online E7 cost includes both parent retrieval paths and one fused Answerer call:

$$\begin{aligned} C_{i,E7,d}^{\text{online}} = {} & C_{i,E2,d}^{\text{retrieval}} + C_{i,E2,d}^{\text{packaging}} \\ & + C_{i,E6,d}^{\text{retrieval}} + C_{i,E6,d}^{\text{packaging}} \\ & + C_{i,E7,d}^{\text{fusion}} + C_{i,E7,d}^{\text{answering}} \\ & + C_{i,E2,d}^{\text{recovery}} + C_{i,E6,d}^{\text{recovery}} + C_{i,E7,d}^{\text{recovery}}. \end{aligned}$$

The fusion term covers deterministic deduplication and interleaving of the two parent bundles. As with packaging, it is zero for model-only cost dimensions unless separately charged. The parent Answerer calls are not required to construct a deployed E7 answer. They remain part of the experiment because E2 and E6 are independently evaluated conditions.

Latency is reported as an observed operational measure rather than inserted into the additive equations above. For a hypothetical online E7 deployment, latency depends on whether the two parent retrieval paths run sequentially or in parallel. Any scenario estimate states that scheduling assumption explicitly. Small observed latency differences may also reflect network and remote-service variation, so they are not automatically attributed to the architecture.

4.6.5 Metric Aggregation and Applicability

Table 4.6 summarises the calculation unit and interpretation of each metric family.

Table 4.6. Evaluation metrics, aggregation, and applicability.

Metric family	Per-question unit	Aggregate	Eligible denominator	Applicable paths	Analysis status	Main role
Token Precision	Normalised candidate-reference token overlap	Macro mean	60 E1–E7 and Full Context		Diagnostic answer metric	Lexical concision and extra text
Token Recall	Normalised candidate-reference token overlap	Macro mean	60 E1–E7 and Full Context		Diagnostic answer metric	Lexical answer coverage
Token F1	Category-aware per-question F1 from matched token overlap	Macro mean	60 E1–E7 and Full Context		Primary lexical metric	LoCoMo-aligned lexical answer quality
Strict Accuracy	`correct` indicator	Mean	60 E1–E7 and Full Context		Primary semantic metric	Complete semantic correctness
Partial Rate	`partial` indicator	Mean	60 E1–E7 and Full Context		Diagnostic semantic metric	Semantic label distribution
Soft Semantic Score	1, 0.5, or 0	Mean	60 E1–E7 and Full Context		Primary graded semantic metric	Graded semantic quality
Evidence Recall	Fraction of annotated source IDs retrieved	Macro mean	58 E1–E7 EvidenceBundles		Diagnostic evidence metric	Annotated-source coverage
Any-hit	At least one annotated source retrieved	Mean	58 E1–E7 EvidenceBundles		Diagnostic evidence metric	Minimum evidence access
All-hit	Every annotated source retrieved	Mean	58 E1–E7 EvidenceBundles		Diagnostic evidence metric	Complete annotated-source access
Calls, tokens, latency, rounds, and actions	Recorded process fields	Total, mean, median, or distribution as appropriate	All completed records with field availability	Path-dependent	Process and cost measure	Efficiency and process diagnosis
Memory construction	Store-level records	One-time total and amortised scenarios	Store or workload level	Relevant storage systems	Process and cost measure	Deployment trade-off

The main answer metrics are reported for every answer set. Evidence measures are reported for eligible questions with question-selected EvidenceBundles, including E7's fused parent bundles. They are `N/A` for Full Context. Process fields are not forced into one common unit when their mechanisms differ. An R1 round and an R2 action, for example, are described separately rather than treated as equivalent operations. Memory-construction effort is measured where records are available. Long-term maintenance is outside the executed experiment, so maintenance consequences are discussed qualitatively rather than reported as measured cost.

4.7 Statistical Analysis Plan

The analysis is primarily descriptive and paired. Main-60 contains the same questions under each compared condition. Preserving this pairing provides more useful evidence than treating condition means as unrelated samples. Confidence intervals and tests support selected comparisons; they do not organise the entire results chapter.

4.7.1 Unit of Analysis and Paired Design

The primary analysis unit is `question_id`. Let $m_i(A)$ and $m_i(B)$ be the value of metric m for question i under conditions A and B. The paired difference is:

$$d_i^{(m)}(A,B) = m_i(A) - m_i(B).$$

The average paired effect is:

$$\overline{d}^{(m)}(A,B) = \frac{1}{N}\sum_{i=1}^{N} d_i^{(m)}(A,B).$$

The median paired difference is reported beside the mean. The mean describes the average signed change. The median indicates whether the typical question moves in the same direction. Both are needed because a small number of large changes can shift the mean.

Architecture comparisons pair answers with the same question and backbone model. Backbone comparisons pair the same question and E-condition across Qwen and DeepSeek. E7 comparisons pair E7 with E2 or E6 for the same question and backbone. Full Context comparisons pair its answer with the selected DeepSeek condition for the same question.

4.7.2 Overall Paired Differences and Win/Tie/Loss

For each main comparison, metric direction is fixed before assigning wins and losses. Let $o_m = 1$ for higher-is-better metrics, such as F1 or Strict Accuracy, and let $o_m = -1$ for lower-is-better measures, such as calls, tokens, latency, or cost. The oriented difference is:

$$d_{i,+}^{(m)}(A,B) = o_m\, d_i^{(m)}(A,B).$$

A question is then a win, tie, or loss according to the sign of the oriented difference:

$$\text{state}_i(A,B) = \begin{cases} \text{win}, & d_{i,+}^{(m)}(A,B) > 0, \\ \text{tie}, & d_{i,+}^{(m)}(A,B) = 0, \\ \text{loss}, & d_{i,+}^{(m)}(A,B) < 0. \end{cases}$$

The report presents the mean difference, median difference, win/tie/loss counts, and the distribution of d_i. This combination distinguishes broad improvement from an average driven by a few large gains. It also exposes systems that repair some questions while harming others.

Win/tie/loss is metric-specific. A condition may win on Recall and lose on Precision for the same question. Semantic label transitions are analysed separately in Section 4.7.6.

4.7.3 Category-Level Analysis

The paired analysis is repeated descriptively for C1 multi-hop, C2 temporal, C3 open-domain, and C4 single-hop questions. The same estimators are used within each category, but category results are not treated as independent replications.

Category analysis supports mechanism interpretation. A retrieval change may primarily affect multi-hop evidence coverage. A curated representation may help temporal consolidation but lose local wording. Full Context may repair dispersed evidence while adding noise for simple questions.

The category groups are small and unequal. C3 contains seven questions. The report therefore emphasises effect direction, paired cases, and consistency with trace evidence rather than standalone category-level significance claims.

4.7.4 Storage–Retrieval Interaction and Backbone Stability

The per-question storage–retrieval interaction for metric m was defined in Section 4.3.2. Its aggregate form is:

$$\overline{DID}^{(m)} = \frac{1}{N}\sum_{i=1}^{N}\left[\big(m_i(E4) - m_i(E2)\big) - \big(m_i(E3) - m_i(E1)\big)\right].$$

For a direction-of-benefit summary, the report also uses:

$$\overline{DID}_{+}^{(m)} = o_m\, \overline{DID}^{(m)}.$$

This estimator asks whether the effect of changing from S1 to S2 differs between R1 and R2. It is reported with the two underlying storage contrasts. A DID value without those component effects can be misleading.

Backbone stability concerns the same architectural contrast under Qwen and DeepSeek. For conditions A and B, let:

$$\Delta_Q^{(m)}(A,B) = \overline{m_Q(A) - m_Q(B)},$$

and:

$$\Delta_D^{(m)}(A,B) = \overline{m_D(A) - m_D(B)}.$$

The contrast-of-contrasts is:

$$CoC^{(m)}(A,B) = \Delta_D^{(m)}(A,B) - \Delta_Q^{(m)}(A,B).$$

Stability is assessed from effect direction, magnitude, paired win/tie/loss patterns, and the contrast-of-contrasts. A result that crosses an arbitrary significance threshold under one backbone but not the other is not, by itself, evidence of model dependence. The difference between the effects must be examined directly.

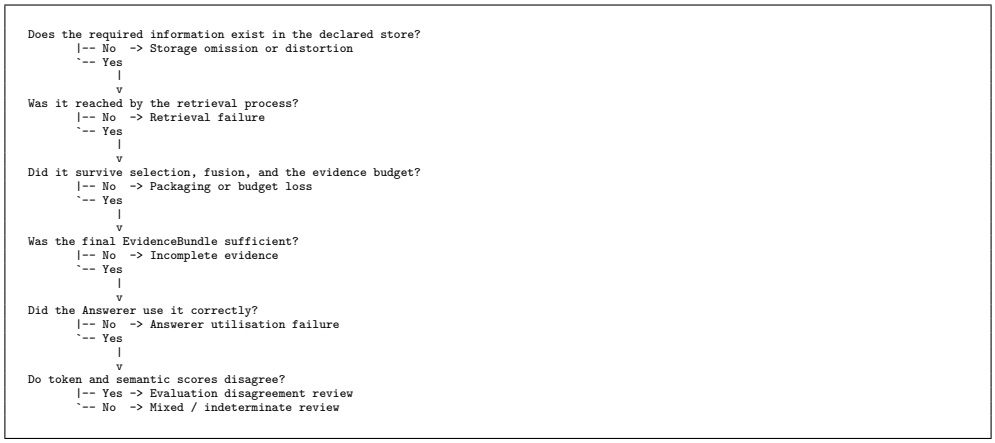


Figure 4.3. Stage-based error-attribution decision flow.

4.7.5 Auxiliary Paired Bootstrap Confidence Intervals

Paired bootstrap confidence intervals are used only for a limited set of central contrasts. In the implemented analysis, they are reported for the main raw-memory storage and retrieval comparisons. E7 and the principal Full Context comparisons remain paired central analyses, but their main evidence consists of descriptive paired effects and Strict-label transitions rather than additional bootstrap intervals.

For one bootstrap replicate, the analysis samples N question identifiers with replacement from the comparison set. All condition values associated with a sampled question move together. The statistic is then recomputed on the paired sample. If $\widehat{\delta}^{*}(1), \ldots, \widehat{\delta}^{*}(B)$ are the bootstrap estimates, the percentile interval is:

$$CI_{95\%} = \left[Q_{0.025}\left(\widehat{\delta}^{*}\right), Q_{0.975}\left(\widehat{\delta}^{*}\right) \right].$$

The interval describes resampling stability across the 60 observed Main-60 question units. It does not estimate variation across conversations because the study contains one conversation. Bootstrap results are supporting evidence. They do not replace the paired distributions, category patterns, or manual review.

The number of replicates, random seed, missing-record policy, and implementation details belong in Appendix H.

4.7.6 Strict-Label Transitions

Selected semantic comparisons use question-level transitions between Strict outcomes. Let $s_i(A)$ equal one when condition A is labelled `correct` and zero otherwise. Four transition counts are reported:

$$n_{ab} = \sum_{i=1}^{N} \mathbf{I}\big(s_i(B) = a \wedge s_i(A) = b\big), \qquad a, b \in \{0, 1\}.$$

Here, B is the baseline and A is the target condition. Therefore, n_{01} is an incorrect-to-correct gain and n_{10} is a correct-to-incorrect harm. The counts n_{00} and n_{11} represent unchanged incorrect and unchanged correct outcomes.

These transitions are central to E7 and Full Context diagnosis. They show whether a higher mean Strict Accuracy reflects a broad excess of repairs, a small net excess of repairs, or larger offsetting numbers of gains and harms. For a small number of central binary comparisons, an exact McNemar test may be reported using the discordant counts n_{01} and n_{10}. It remains supplementary.

4.7.7 Multiple Comparisons and Interpretation Boundaries

The study does not treat every combination of condition, metric, category, and backbone as an independent confirmatory test. Primary comparison families are defined by the research questions. Category, correlation, anomaly, and case analyses are exploratory.

Interpretation emphasises direction, magnitude, distribution, and cross-backbone consistency. The report does not rank systems by a large set of unadjusted p-values. If a multiple-comparison correction is applied to a formal test family, that family and procedure are stated explicitly in Appendix H.

Table 4.7. Principal comparison families and analysis methods.

Analytical purpose	Main contrasts	Primary unit	Main summaries	Supporting evidence
Directory organisation	E3−E1 and E4−E2	Question	Mean and median paired differences; win/tie/loss	Evidence coverage, traces, and selected bootstrap intervals
Retrieval system	E2−E1 and E4−E3	Question	Mean and median paired differences; win/tie/loss	Evidence coverage, calls, tokens, and traces
Storage–retrieval interaction	$(E4 - E2) - (E3 - E1)$	Question	DID and component contrasts	Category patterns and selected bootstrap interval
Curated-memory access	E6−E5	Question	Paired differences and label transitions	Fact/topic retrieval traces and cost
Backbone stability	Matched contrast under Qwen and DeepSeek	Question	Effect direction, magnitude, and contrast-of-contrasts	Question-level agreement and disagreement cases
Dual-source fusion	E7−E2 and E7−E6	Question	Paired differences and Strict transitions	Repair/harm review, evidence overlap, and online cost
Full Context diagnosis	Full Context versus DeepSeek selective conditions	Question	Paired differences and Strict transitions	Context scale, evidence state, latency, and manual review

4.8 Error Analysis Method

Answer scores show whether a candidate differs from the reference. They do not identify where information was lost or misused. The error analysis therefore follows the complete information path defined in Chapter 3.

4.8.1 Error Categories

The coding scheme is stage-based. A case is assigned to the earliest well-supported failure stage when the available records permit that decision. Cases with interacting causes may receive a mixed label. Cases without enough evidence remain indeterminate.

Table 4.8. High-level error taxonomy.

Error category	Operational question	Main evidence inspected
Storage omission or distortion	Was the required information absent, altered, stale, or incorrectly combined in the declared store?	Canonical source, S1/S2/S3 contents, and provenance locators
Retrieval failure	Was the required information present in the store but absent from the retrieved observations or selected evidence?	Store content, retrieval trace, ranked results, and EvidenceBundle
Incomplete multi-hop evidence	Was only part of the required annotated or inferential evidence retrieved?	Gold sources, retrieved source set, local context, and question requirements
Evidence-budget or packaging loss	Did relevant material appear upstream but disappear during selection, merging, ordering, or budget application?	Retrieval observations, accepted notes, merge records, and budget records
Answerer utilisation failure	Did the EvidenceBundle contain sufficient support while the candidate answer remained wrong or incomplete?	EvidenceBundle, candidate answer, citations, and reference answer
Fusion interference	Did E7 become worse after adding the second parent evidence source?	E2/E6 parent bundles, fused order, duplicate handling, and E7 answer
Long-context utilisation failure	Did Full Context fail despite complete visibility, especially where selective evidence succeeded?	Full Context input, paired selective evidence, answer, and source locations
Lexical–semantic evaluation disagreement	Did token and semantic measures disagree because of paraphrase, verbosity, omitted qualifiers, or keyword overlap?	Candidate, reference, token breakdown, and LLM judge label
Mixed or indeterminate	Are several stages implicated, or is the record insufficient for one supported attribution?	Complete case record and reviewer notes

Evidence coverage alone does not assign an error label. Low Evidence Recall with a wrong answer supports a retrieval hypothesis, but an alternative valid passage may exist. All-hit with a wrong answer supports an utilisation hypothesis, but the annotated source may still be insufficient for the full reasoning chain. Each attribution therefore requires inspection of the relevant artefacts.

Figure 4.3 presents the core decision sequence for selective-retrieval conditions. It is deliberately concise. E7 additionally requires a fusion-interference check, while Full Context bypasses the retrieval, EvidenceBundle, and evidence-budget stages and instead receives a long-context utilisation check. These condition-specific branches follow the definitions in Table 4.8.

The abbreviated labels in Figure 4.3 map to the taxonomy in Table 4.8: *Incomplete evidence* includes incomplete multi-hop evidence; *Packaging or budget loss* denotes evidence-budget or packaging loss; and *Evaluation disagreement review* denotes lexical–semantic evaluation disagreement. If the Answerer uses sufficient evidence correctly and the evaluation families agree, the case exits the error flow with no attributable error. *Mixed / indeterminate* is reserved for unresolved erroneous cases, interacting causes, or insufficient records.

4.8.2 Review Procedure

Manual review proceeds in a fixed order:

1. Read the question and reference answer.
2. Locate the annotated source turns, when available.
3. Check whether the required information exists in the declared memory representation.

4. Inspect the retrieval trace and returned observations, where applicable.
5. Inspect selected EvidenceItems, merging, order, and budget outcomes, where applicable. For Full Context, verify complete and chronological context assembly instead.
6. Compare the evidence input—an EvidenceBundle or the complete Full Context input—with the candidate answer and citations.
7. Inspect token scores, the LLM judge label, and any disagreement between them.
8. Assign a primary error category, a secondary category when justified, or `indeterminate`.

The review sample covers both backbone models, all condition families, and all question categories. It includes evidence–answer mismatches, E7 repairs and harms, Full Context repairs and failures, and token–semantic scoring disagreements. Cases are selected to explain aggregate patterns and important counterexamples. They are not presented as a random estimate of population-wide error frequencies unless a separate random-sampling procedure is applied.

The complete codebook records inclusion rules, exclusion rules, required artefacts, and decision examples. Reviewers must preserve uncertain cases rather than force one cause. Any agreement or adjudication statistic is reported only if more than one reviewer independently applies the codebook.

Chapter 5 reports the observed error distribution and representative cases. The complete statistical settings and error-analysis protocol belong in Appendix H. Extended per-question results and the full casebook belong in Appendix I.

This protocol links each research question to a controlled comparison, an applicable metric set, and an explicit interpretation boundary. It produces the answer, evidence, process, and review records analysed in Chapter 5.

Chapter 5. Results and Analysis

5.1 Analysis Scope and Research Questions

Chapter 3 defined the storage representations, retrieval systems, EvidenceBundle interface, and shared Answerer contract. Chapter 4 then specified the unified Main-60 test set and evaluation protocol. This chapter reports what the experiments show. It analyses the same 60 questions under seven selective evidence-access conditions for each of two backbone models, Qwen and DeepSeek, together with one DeepSeek-family Full Context diagnostic path.

Table 5.1 restates the condition mapping so that the experimental labels can be read without returning to Chapter 3. S1 is the flat raw-session store, S2 is the foldered raw-session store, and S3 is Agent-Curated Memory. R1 is Agentic Filesystem Retrieval, R2 is ReFind-style raw-chat retrieval, and R3 is retrieval adapted to the structure of S3.

Table 5.1. Experimental conditions used in Chapter 5. Full Context is a separate diagnostic path rather than E8.

Condition	Evidence source	Retrieval or processing path	Analytical role
E1	S1	R1	Flat raw-session filesystem baseline
E2	S1	R2	ReFind-style raw-chat retrieval on S1
E3	S2	R1	Effect of folder organisation under R1
E4	S2	R2	Foldered raw sessions combined with R2
E5	S3	R1	Agent-Curated Memory accessed as a filesystem
E6	S3	R3	Retrieval adapted to Agent-Curated Memory
E7	E2 and E6 EvidenceBundles	Fixed deduplication and interleaving, followed by the shared Answerer	Dual-source complementarity and interference; no new retrieval
Full Context	All 30 S1 raw sessions	Complete chronological context processed under the shared Answerer contract	DeepSeek-family diagnostic comparison without selective retrieval or a 16,000-character EvidenceBundle; not an exact served-model ablation

The six research questions form a four-stage argument. RQ1 and RQ2 identify the architecture effects and test whether their directions persist across backbone pipelines. RQ3 and RQ4 examine the benefits and limits of expanding evidence access through dual-source fusion and Full Context. RQ5 traces non-Correct paths through the information chain to identify the main confirmed bottlenecks. RQ6 then combines quality, online cost, construction cost, recovery burden, and existing memory assets to support conditional system selection. Table 5.2 provides the chapter-level roadmap. Each section follows the same sequence: research question, direct answer, main evidence, interpretation, and boundary.

Table 5.2. Research questions and their locations in Chapter 5.

Research question	Main question	Section
RQ1	How should memory representation and retrieval be paired to support effective selective access?	5.2
RQ2	Which architectural effects remain stable across Qwen and DeepSeek, and which depend on the backbone model?	5.3
RQ3	Does E7 convert the complementary E2 and E6 access paths into better answers while preserving parent-correct outcomes?	5.4
RQ4	Does complete context visibility outperform selective evidence access, or do the two access regimes provide different advantages?	5.5
RQ5	Across the non-Correct answer paths, where is the earliest evidence-supported break, and which stage is the largest confirmed system-internal bottleneck?	5.6
RQ6	Given the backbone and available memory assets, which condition offers the most useful balance between answer quality, query cost, construction cost, and recovery burden?	5.7

Strict Accuracy and Soft Semantic Score are the primary answer-quality measures. Partial Rate supports interpretation of the semantic labels, while Token Precision, Token Recall, Token F1, and normalised answer length diagnose lexical overlap and answer style. These answer measures use all 60 questions. Evidence Recall and gold-source coverage evaluate selective evidence access over the 58 questions with valid source-turn annotations; diagnostics that do not require gold annotations, such as source-ID overlap, use all 60 questions. Full Context has no retrieval-level Evidence Recall because it bypasses selective evidence construction. RQ6 reports successful-query tokens and model calls separately from construction and recovery costs. All condition comparisons reuse the same Main-60 questions and are interpreted as paired comparisons rather than independent samples.

At a high level, architecture quality depends on the match among storage representation, retrieval path, and backbone model, with R1 providing the most consistent quality-oriented access in the tested systems. Expanding evidence access through E7 or Full Context creates additional opportunities but does not guarantee correct evidence utilisation. No condition is globally optimal: the practical choice depends on the quality target, online budget, and whether S2 or S3 already exists.

5.2 RQ1: Matching Memory Representation with Retrieval

Research question. Section 5.2 addresses RQ1:

> **How should memory representation and retrieval be paired to support effective selective access?**

This question treats storage and retrieval as a coupled design problem. A representation may organise information into useful topics, paths, and evidence units, but those affordances affect the final answer only if the retrieval system can exploit them. The analysis therefore proceeds through three linked questions. First, does reorganising the same raw conversation content improve performance? Second, which retrieval path most effectively converts stored information into answer-relevant evidence? Third, when paired with the stronger observed retrieval path, does Agent-Curated Memory improve end-to-end quality?

Direct answer. Folder organisation does not produce an automatic or general gain: its clearest positive effect appears for Qwen with R1 and is concentrated on temporal questions. Across the tested raw and curated stores, R1 is the more consistent quality-oriented access path, whereas R2 and R3 use less recorded query-time effort but return weaker annotated-evidence coverage and semantic outcomes. When S3 is accessed through R1, it improves semantic quality and Evidence Recall relative to foldered raw sessions, but lowers Token F1. The central result is therefore that the effective unit of design is the **storage–retrieval pairing**, rather than either component in isolation.

All differences below are target minus baseline. Token F1, Strict Accuracy, and Soft Semantic Score use all 60 Main-60 questions. Evidence Recall and mean EvidenceBundle length use the 58 questions with valid source-turn annotations. The other two questions are open-domain questions without annotated source-turn IDs; they remain in all answer-quality calculations. The annotations identify valid evidence locations but are not guaranteed to constitute a minimal or exhaustive proof set. E7 is excluded because its fusion effect is analysed under RQ3. Formal cross-backbone stability is addressed in RQ2, detailed failure attribution in RQ5, and the final quality–cost decision in RQ6.

5.2.1 Does Storage Organisation Help When the Raw Information Is Unchanged?

Research question. The first analysis asks whether changing S1 flat raw sessions into S2 foldered raw sessions helps when the underlying conversation content remains raw. The primary comparison is E3−E1, which holds R1 fixed. E4−E2 is retained only as an auxiliary comparison because R2 does not receive the same form of directory access as R1.

Direct answer. The results do not support a general directory advantage. Folder organisation produces a positive answer-quality profile only for Qwen with R1. This pattern is not reproduced by DeepSeek, and the effects observed under R2 are small and inconsistent. Directory structure is therefore a navigation resource that a particular retrieval–backbone pairing may exploit, not an automatic source of improvement.

Table 5.3 reports the primary E3−E1 comparison under R1 and the auxiliary E4−E2 path-cue comparison under R2.

Table 5.3. Effects of storage organisation on answer quality and annotated-evidence coverage. Values are target minus baseline. Answer metrics use N = 60; Evidence Recall uses N = 58. Bracketed values are 95% paired-bootstrap intervals.

Backbone	Contrast	Analytical role	ΔToken F1 [95% interval]	ΔStrict [95% interval]	ΔSoft	ΔEvidence Recall
Qwen	E3−E1	Primary R1 comparison	+0.0462 [−0.0160, +0.1109]	+0.0333 [−0.0500, +0.1171]	+0.0250	+0.0072
DeepSeek	E3−E1	Primary R1 comparison	−0.0041 [−0.0499, +0.0431]	−0.0500 [−0.1333, +0.0167]	−0.0500	−0.0302
Qwen	E4−E2	Auxiliary R2 path-cue comparison	−0.0071 [−0.0678, +0.0533]	+0.0333 [−0.0500, +0.1333]	+0.0083	−0.0014
DeepSeek	E4−E2	Auxiliary R2 path-cue comparison	−0.0141 [−0.0742, +0.0471]	−0.0333 [−0.1000, +0.0333]	0.0000	−0.0287

For Qwen, E3−E1 increases Token F1 by 0.0462, Strict Accuracy by 0.0333, and Soft Semantic Score by 0.0250. Token F1 improves on 19 questions, is unchanged on 30, and declines on 11. Strict correctness records five repairs and three harms, giving a net increase of two correct answers. The corresponding DeepSeek effects are −0.0041, −0.0500, and −0.0500. The Qwen result is therefore a positive observed signal for one retrieval–backbone pairing, rather than a storage effect that generalises across both pipelines. The paired intervals also show that none of the storage effects is statistically resolved in Main-60.

Figure 5.1 visualises the corresponding E1–E4 condition means. Within each backbone, the R1 line connects E1 to E3 and the R2 line connects E2 to E4. The slopes therefore show descriptive changes from S1 to S2 under each retrieval path; they are not inferential interaction estimates.

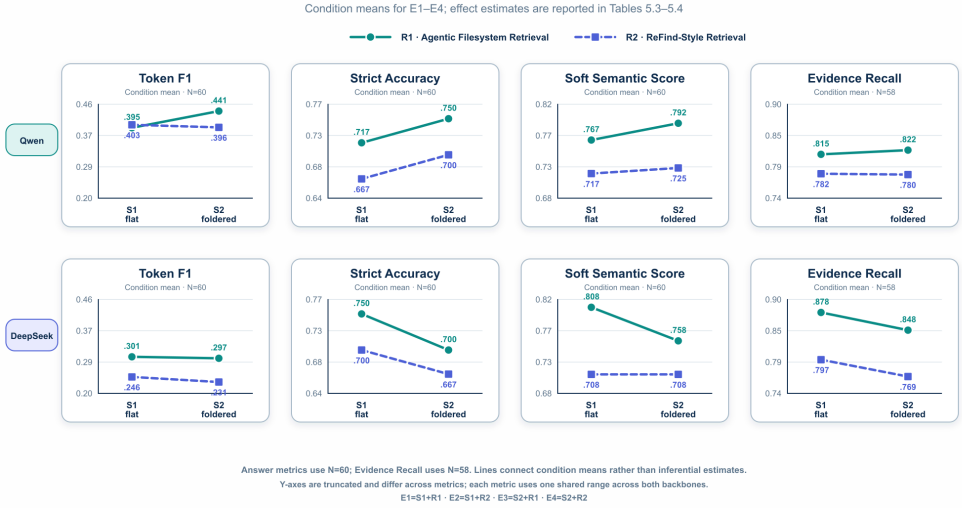


Figure 5.1. Storage–retrieval interaction across backbones. The panels report Token F1, Strict Accuracy, Soft Semantic Score, and Evidence Recall for E1–E4. Answer metrics use N = 60 and Evidence Recall uses N = 58. The y-axes are truncated and differ across metrics, while each metric retains one common range across the two backbones. The related storage and retrieval contrasts are reported in Tables 5.3–5.4.

The category results further localise the Qwen signal. On the 13 temporal questions, E3−E1 increases Token F1 by 0.1994, Strict Accuracy by 0.3077, and Soft Semantic Score by 0.1923. Evidence Recall nevertheless decreases by 0.0385 within the same category. The aggregate benefit is therefore concentrated on temporal questions and is not explained by broader coverage of the annotated source turns.

This pattern is consistent with directory structure improving how R1 navigates or interprets already sufficient evidence. Folder and path cues can provide a more informative search entry point, reduce the apparent search space, or help preserve relationships among time-related statements. They need not increase the number of annotated turns retrieved to improve the final answer. These are architecture-level explanations consistent with the observed metrics; trace-level attribution is deferred to RQ5.

E4−E2 requires a narrower interpretation. E2 and E4 use the same raw-exchange index, and folder names do not enter R2's first-stage lexical ranking. The S2 relative path becomes visible mainly through provenance observations and may influence later query actions. E4−E2 therefore tests the auxiliary effect of exposing folder/path cues within the current R2 pipeline, not the full directory-navigation effect tested by E3−E1. Its mixed answer effects and near-zero Evidence Recall changes show no stable benefit from this limited exposure.

The difference-in-differences analysis does not strengthen the claim. The interaction, DID = (E4−E2)−(E3−E1), has Token F1 estimates of −0.0533 for Qwen and −0.0100 for DeepSeek. All paired-bootstrap intervals for the Token F1, Strict Accuracy, and Soft Semantic Score interactions span zero. Moreover, R1 and R2 do not receive equivalent access to S2's directory information, so the DID is an interface-level descriptive comparison rather than a symmetric test of folder navigation. The bounded conclusion is that folder organisation can help a particular retrieval–backbone combination, especially Qwen+R1 on temporal questions, but Main-60 does not establish a universal benefit from directory structure.

5.2.2 Which Retrieval Path Converts Stored Information into Useful Evidence?

Research question. The second analysis holds the representation fixed and asks which retrieval path more effectively identifies and retains the evidence required by the Answerer. It compares R2 with R1 on S1 and S2, and R3 with R1 on the frozen S3 store.

Direct answer. Across these comparisons, R1 is the more consistent quality-oriented path. R2 and R3 use fewer recorded query-time model calls and tokens, but this efficiency is accompanied by lower Evidence Recall and weaker semantic outcomes. The key distinction is therefore not how much text a retriever returns, but whether the resulting EvidenceBundle contains the decisive evidence in a usable form.

Table 5.4 reports the retrieval-path effects while holding each storage representation fixed.

Table 5.4. Retrieval-path effects on fixed storage representations. Values are target minus baseline. E2−E1 and E4−E3 compare R2 with R1; E6−E5 compares R3 with R1. Answer metrics use N = 60; Evidence Recall and bundle length use N = 58.

Storage	Backbone	Contrast	ΔToken F1	ΔStrict	ΔSoft	ΔEvidence Recall	ΔBundle characters
S1	Qwen	E2−E1	+0.0082	−0.0500	−0.0500	−0.0330	+4,040.8
S1	DeepSeek	E2−E1	−0.0557	−0.0500	−0.1000	−0.0805	+7,142.3
S2	Qwen	E4−E3	−0.0451	−0.0500	−0.0667	−0.0417	+3,885.6
S2	DeepSeek	E4−E3	−0.0657	−0.0333	−0.0500	−0.0790	+6,678.9
S3	Qwen	E6−E5	−0.0135	−0.0500	−0.0583	−0.0891	+5,109.1
S3	DeepSeek	E6−E5	−0.0364	−0.0833	−0.0917	−0.2371	+5,457.3

On S1, Qwen E2−E1 produces a small Token F1 increase of 0.0082, but Strict Accuracy and Soft Semantic Score both fall by 0.0500 and Evidence Recall falls by 0.0330. This is a lexical–semantic disagreement, not evidence that R2 improves overall answer quality. DeepSeek shows a clearer negative profile on the same store.

On S2, both backbones show lower Token F1, Strict Accuracy, Soft Semantic Score, and Evidence Recall under R2. This is the clearest raw-memory result: replacing R1 with R2 does not improve any of the four principal quality measures. On S3, R3 likewise remains below R1 across all four measures. The Evidence Recall decrease is especially large for DeepSeek at −0.2371. No paired-bootstrap interval was pre-declared for E6−E5, so this S3 result is descriptive of the observed Main-60 pipelines.

Bundle size moves in the opposite direction from evidence quality. R2 adds approximately 3,886–7,142 characters to the mean eligible EvidenceBundle, while R3 adds approximately 5,109–5,457 characters. Nevertheless, Evidence Recall declines in all six comparisons. More retrieved text therefore does not imply greater coverage of the annotated evidence, and neither property guarantees a better answer.

The process logs reveal a quality–efficiency trade-off. Across the four raw-memory comparisons, R2 reduces recorded model calls by 6.68–7.98 and total query-time model tokens by 121,809.1–183,717.1 per question. On S3, R3 reduces the corresponding totals by 3.22 calls and 244,976.5 tokens under Qwen, and by 3.90 calls and 339,242.3 tokens under DeepSeek. These totals include retrieval and answering, apply to successful-query records, and exclude separately reported failed-attempt and recovery costs.

The observed pattern is consistent with a difference in evidence selection rather than evidence volume. R1's observation–action loop can inspect files, revise a search after observing partial results, follow a new path, and decide whether further evidence is needed. By contrast, the current R2 and R3 pipelines are more tightly constrained by initial query formulation, ranking, local expansion, action limits, and stopping decisions. A large bundle may consequently contain relevant but non-decisive material, reach the correct session while omitting an adjacent decisive turn, or increase the Answerer's evidence-selection burden. RQ5 determines how often these mechanisms occur; RQ1 only establishes the aggregate architecture pattern.

These contrasts compare complete access systems. They do not isolate BM25, grep, query generation, local expansion, or any other individual component. R1 rounds and R2/R3 actions are also not commensurate units. The supported conclusion is therefore limited to the implemented Main-60 pipelines: R1 more consistently converts the tested representations into annotated evidence and answers with higher semantic quality, whereas the lower recorded query-time effort of R2 and R3 is not a quality-neutral substitution. Whether that reduction justifies the quality loss is considered in RQ6.

5.2.3 Does Curated Memory Add Value When Paired with Effective Retrieval?

Research question. The third analysis asks whether Agent-Curated Memory adds value when both representations are accessed through the stronger observed retrieval path. The principal comparison is E5−E3: E3 uses S2+R1, while E5 uses S3+R1.

Direct answer. The two backbones show the same directional profile. S3+R1 improves Strict Accuracy, Soft Semantic Score, and Evidence Recall, but lowers Token F1. Its benefit is therefore semantic and evidence-oriented rather than uniformly reflected by lexical overlap.

Table 5.5 reports the E5−E3 representation contrast under the shared R1 access path.

Table 5.5. Effect of replacing foldered raw sessions with Agent-Curated Memory under R1. E5−E3 is a descriptive complete-system representation contrast. Answer metrics use N = 60; Evidence Recall uses N = 58.

Backbone	ΔToken Precision	ΔToken Recall	ΔToken F1	ΔStrict	ΔSoft	ΔEvidence Recall	Mean normalised answer length, E3→E5
Qwen	−0.0436	+0.0380	−0.0463	+0.0333	+0.0250	+0.1034	11.97→16.13
DeepSeek	−0.0385	+0.0566	−0.0367	+0.0500	+0.0417	+0.1149	21.72→29.17

Under Qwen, E5−E3 increases Strict Accuracy by 0.0333, Soft Semantic Score by 0.0250, and Evidence Recall by 0.1034, while Token F1 decreases by 0.0463. Under DeepSeek, the corresponding changes are +0.0500, +0.0417, +0.1149, and −0.0367. The agreement in direction across the two observed pipelines distinguishes this result from the conditional storage-organisation effect in Section 5.2.1.

Figure 5.2 places the representation contrast E5−E3 beside the same-store retrieval contrast E6−E5. It therefore separates the value observed for the complete S3+R1 path from the effect of replacing R1 with R3 on the frozen S3 store.

The lexical result can be decomposed into lower Token Precision and higher Token Recall. Mean normalised answer length increases from 11.97 to 16.13 tokens under Qwen, approximately 35%, and from 21.72 to 29.17 under DeepSeek, approximately 34%. This pattern is consistent with S3 co-locating related facts across sessions and allowing the Answerer to produce a more complete or differently worded response. Such answers can receive higher semantic scores while losing Token F1 because of additional explanation or reduced lexical precision. Answer length was not independently controlled, however, and should not be treated as the sole cause of the F1 decline.

Table 5.6 locates E5 within the complete E1–E6 RQ1 condition set.

Panel A. Qwen.

Table 5.6. Absolute condition means within the E1–E6 RQ1 scope. Answer metrics use N = 60; Evidence Recall uses N = 58. Bold values are the highest value within each backbone panel, including ties.

Condition	Storage + retrieval	Token F1	Strict Accuracy	Soft Semantic Score	Evidence Recall
E1	S1 + R1	0.3949	0.7167	0.7667	0.8147
E2	S1 + R2	0.4031	0.6667	0.7167	0.7816
E3	S2 + R1	**0.4411**	0.7500	0.7917	0.8218
E4	S2 + R2	0.3960	0.7000	0.7250	0.7802

Continued from the previous page

Condition	Storage + retrieval	Token F1	Strict Accuracy	Soft Semantic Score	Evidence Recall
E5	S3 + R1	0.3948	**0.7833**	**0.8167**	**0.9253**
E6	S3 + R3	0.3813	0.7333	0.7583	0.8362

Panel B. DeepSeek.

Condition	Storage + retrieval	Token F1	Strict Accuracy	Soft Semantic Score	Evidence Recall
E1	S1 + R1	**0.3013**	**0.7500**	**0.8083**	0.8779
E2	S1 + R2	0.2455	0.7000	0.7083	0.7974
E3	S2 + R1	0.2971	0.7000	0.7583	0.8477
E4	S2 + R2	0.2314	0.6667	0.7083	0.7687
E5	S3 + R1	0.2604	**0.7500**	0.8000	**0.9626**
E6	S3 + R3	0.2240	0.6667	0.7083	0.7256

Under Qwen, E5 has the highest Strict Accuracy, Soft Semantic Score, and Evidence Recall, whereas E3 has the highest Token F1. Under DeepSeek, E5 has the highest Evidence Recall and ties E1 for the highest Strict Accuracy, but E1 retains the highest Token F1 and Soft Semantic Score. E5 is therefore the strongest of the tested curated-memory paths and an important semantic/evidence leader, but it is not a universal winner across metrics and backbones. Its advantage over E6 also shows that S3's value depends on how the representation is accessed.

E5−E3 should not be interpreted as a pure curation effect. Although R1 is held fixed, the move from S2 to S3 changes wording, granularity, organisation, deduplication, and provenance presentation together. No new paired-bootstrap interval was pre-declared for this descriptive S3 comparison. The defensible conclusion is that the complete S3+R1 representation–access path improves semantic quality and annotated-evidence coverage relative to S2+R1, while introducing a lexical-overlap trade-off. S3 construction cost and its effect on deployment choice are addressed in RQ6.

Taken together, RQ1 shows that memory structure does not improve performance automatically. Folder organisation provides a conditional benefit only when the retrieval path and backbone can exploit its navigation cues. Across the tested raw and curated representations, R1 more consistently converts stored information into annotated evidence and answers with higher semantic quality, whereas R2 and R3 mainly reduce recorded query-time effort and incur observable quality losses. When paired with R1, Agent-Curated Memory improves semantic quality and evidence coverage relative to foldered raw sessions, but lowers Token F1. The appropriate design target is therefore the storage–retrieval pairing rather than either component considered in isolation. RQ2 next tests which of these architecture patterns persist across the two backbone pipelines.

5.3 RQ2: Robustness across Backbone Models

Research question. Which findings from RQ1 remain directionally consistent across the Qwen and DeepSeek pipelines, and which depend on the backbone model?

The robustness analysis uses five main E1–E6 contrasts from RQ1. E4−E2 is excluded from the primary matrix because R2 does not use the S2 folder hierarchy as a symmetric navigation intervention. E7 appears only in the aggregate answer profile; its fusion effect is analysed in RQ3. Full Context is excluded because no matched Qwen run is available.

Direct answer. Three findings are directionally stable across the two tested pipelines. On S2, R1 produces higher answer quality and Evidence Recall than R2. On S3, R1 also outperforms the current R3 implementation. In addition, S3+R1 consistently trades lower Token F1 for higher semantic quality and Evidence Recall relative to S2+R1.

Other findings depend more strongly on the backbone. On S1, R2 lowers semantic quality and Evidence Recall under both backbones, but its Token F1 effect changes direction. The benefit of replacing S1 with S2 under R1 appears only in the Qwen pipeline.

The backbone comparison also reveals an important measurement distinction. Qwen has much higher Token F1, but the two pipelines have similar mean Strict Accuracy and Soft Semantic Score. The lexical gap therefore cannot be interpreted as an equally large semantic-quality gap. Strict Accuracy and Soft Semantic Score provide the primary quality interpretation below. Evidence Recall supports the evidence-access interpretation, while Token F1 is treated as a diagnostic lexical measure.

5.3.1 Which RQ1 Findings Remain Directionally Stable?

Table 5.7 reports the five principal architecture contrasts. Each value is the target-condition mean minus the baseline-condition mean. The classifications describe point-estimate directions across the two tested pipelines; they do not imply statistical significance or equal effect size.

Table 5.7. Directional stability of the principal RQ1 architecture effects. Answer metrics use N = 60. Evidence Recall uses the 58 evidence-eligible questions. Q/D denotes Qwen and DeepSeek.

Contrast	Qwen ΔStrict / ΔSoft	DeepSeek ΔStrict / ΔSoft	ΔEvidence Recall, Q/D	ΔToken F1, Q/D Interpretation
E4−E3	−0.0500 / −0.0667	−0.0333 / −0.0500	−0.0417 / −0.0790	−0.0451 / −0.0657 Stable negative effect
E6−E5	−0.0500 / −0.0583	−0.0833 / −0.0917	−0.0891 / −0.2371	−0.0135 / −0.0364 Stable negative effect
E5−E3	+0.0333 / +0.0250	+0.0500 / +0.0417	+0.1034 / +0.1149	−0.0463 / −0.0367 Stable lexical–semantic trade-off
E2−E1	−0.0500 / −0.0500	−0.0500 / −0.1000	−0.0330 / −0.0805	+0.0082 / −0.0557 Semantic and evidence effects stable; lexical effect backbone-dependent
E3−E1	+0.0333 / +0.0250	−0.0500 / −0.0500	+0.0072 / −0.0302	+0.0462 / −0.0041 Backbone-dependent

Three contrasts support stable conclusions. E4−E3 is negative across all reported metrics, so R1 provides higher answer quality and annotated-evidence coverage than R2 on the fixed S2 store. E6−E5 is also uniformly negative: on the frozen S3 store, the current R3 implementation remains below R1. E5−E3 reproduces the same trade-off in both pipelines: S3+R1 raises semantic quality and Evidence Recall but lowers Token F1. This last comparison is a complete-system representation contrast and does not isolate curation as its sole cause.

E2−E1 is only partly stable. On S1, R2 lowers Strict Accuracy, Soft Semantic Score, and Evidence Recall under both backbones. Its Token F1 rises slightly under Qwen and falls under DeepSeek. The semantic and evidence conclusions are therefore stable, whereas the lexical conclusion depends on the backbone.

E3−E1 is fully backbone-dependent. Replacing S1 with S2 under R1 improves the Qwen results but does not improve the DeepSeek results. Folder organisation should therefore be treated as a useful match for the tested Qwen+R1 pipeline, not as a general storage advantage.

Figure 5.3 visualises the five principal contrasts together with E4−E2 as a supplementary path-cue comparison.

5.3.2 Why Does Token F1 Diverge More Than Semantic Quality?

Table 5.8 separates lexical overlap, semantic quality, and answer style. The values are equal-weight descriptive means across E1–E7. The seven conditions reuse the same 60 questions and must not be interpreted as 420 independent observations per backbone.

Table 5.8. Cross-backbone measurement profile across E1–E7.

Metric	Qwen	DeepSeek
Token Precision	0.3411	0.1957
Token Recall	0.6774	0.6884
Token F1	0.3972	0.2636
Strict Accuracy	0.7262	0.7119
Soft Semantic Score	0.7619	0.7560
Mean normalised answer length	14.03	25.35

The clearest measured difference is answer style. DeepSeek answers are approximately 80.7% longer. Its mean Token Recall is slightly higher, but its Token Precision is lower by 0.1454. Extra explanation, qualification, and paraphrastic wording increase the number of candidate tokens without necessarily increasing overlap with a short reference answer. Because Token F1 combines Token Precision and Token Recall, the small Recall advantage cannot offset the much larger Precision deficit.

This pattern explains only part of the difference. In E1, E2, and E7, DeepSeek combines lower Token F1 with higher Token Recall and equal or higher Strict Accuracy, which is consistent with lexical under-credit for longer answers. In E3–E6, however, DeepSeek also has lower Token Recall and lower semantic scores. These conditions suggest additional differences in retrieval behaviour, evidence selection, or EvidenceBundle utilisation. RQ2 does not assign those differences to a specific component because the backbone controls both the Retrieval Agent and the Answerer. Their root causes are examined in RQ5.

5.3.3 Answer to RQ2 and Boundary

Retrieval-path ordering is more robust across the two tested backbone pipelines than the benefit of changing storage organisation. R1 remains the stronger quality-oriented access path on S2 and S3, and S3+R1 preserves the same lexical–semantic trade-off under both backbones. In contrast, the benefit of folder organisation under R1 is backbone-dependent.

This conclusion applies only to the two tested end-to-end pipelines. Because the backbone is changed for both retrieval and answering, the experiment cannot isolate which component produces a cross-backbone difference. Directional agreement also does not establish statistical significance, equal effect size, or a model-independent law.

RQ3 next examines whether the two complementary access paths can be combined without losing correct parent outcomes.

5.4 RQ3: Dual-Source Fusion

Research question. Does deterministic fusion of the E2 and E6 EvidenceBundles convert their complementary evidence into better answers without losing answers already solved by either parent?

E2 retrieves from raw dialogue through R2, whereas E6 retrieves from Agent-Curated Memory through R3. E7 reuses the two completed parent EvidenceBundles, performs no new retrieval, applies a fixed deterministic merge, and invokes the shared Answerer once. RQ3 therefore evaluates a fusion condition, not a third retriever.

Direct answer. E2 and E6 provide measurable answer-level and annotated-source-access complementarity. E7 expands Evidence Recall under both backbones, but the answer benefit differs. Under Qwen, E7 improves Strict Accuracy and Soft Semantic Score relative to E2, ties E6 on both semantic measures, and lowers Token F1 relative to both parents. Under DeepSeek, E7 improves Token Precision, Token Recall, Token F1, Strict Accuracy, Soft Semantic Score, and Evidence Recall relative to both parents.

The opportunity is not fully converted. E7's net Strict-correct count is three questions below the post-hoc parent oracle under Qwen and one question below under DeepSeek. It produces both Repair and Harm, so dual-source fusion is a conditional improvement rather than a universal replacement for either parent.

Figure 5.4 shows how the fixed E7 condition is constructed.

The analysis contains 120 paired E2–E6–E7 triples: the same 60 questions under each backbone. Answer metrics and mean source-ID Jaccard use all 60 questions. Evidence Recall and gold-source coverage use the 58 evidence-eligible questions. All reported changes are descriptive paired effects; RQ3 introduces no additional significance test.

5.4.1 Main Evidence: Benefit, Opportunity, and Conversion

5.4.1.1 Observed Fusion Benefit

Table 5.9, Panel A reports E7 minus each parent. Positive values favour E7.

Panel A. Paired E7-minus-parent effects.

Table 5.9. Dual-source fusion benefit, complementarity opportunity, and conversion outcomes.

Backbone	Parent	ΔToken Precision	ΔToken Recall	ΔToken F1	ΔStrict	ΔSoft	ΔEvidence Recall
Qwen	E2	−0.0425	+0.0194	−0.0340	+0.0667	+0.0417	+0.1178
Qwen	E6	−0.0028	−0.0131	−0.0122	0.0000	0.0000	+0.0632
DeepSeek	E2	+0.0408	+0.0341	+0.0398	+0.0500	+0.0917	+0.0977
DeepSeek	E6	+0.0498	+0.1280	+0.0614	+0.0833	+0.0917	+0.1695

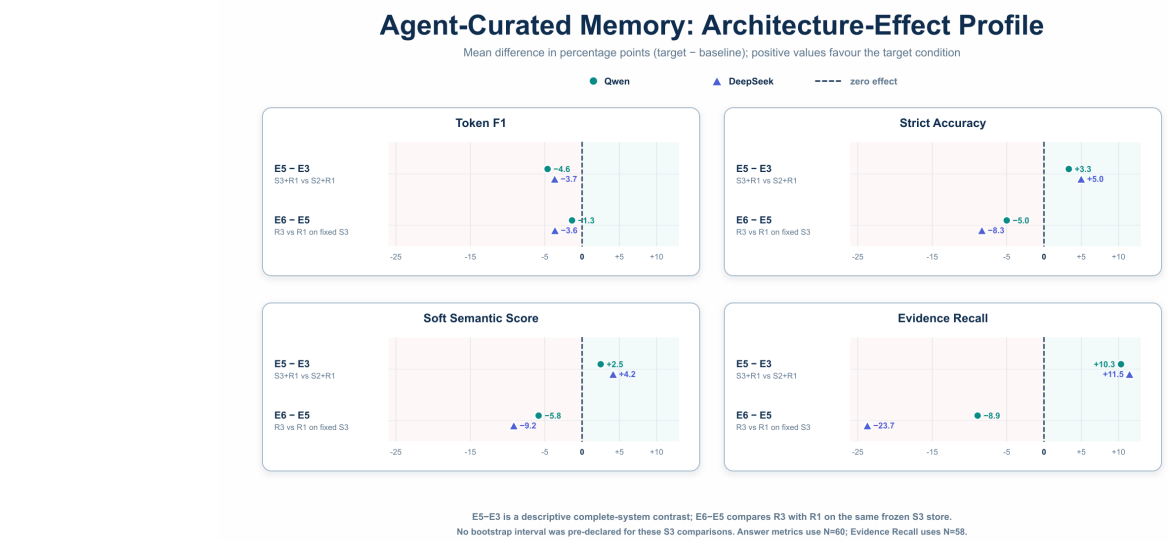


Figure 5.2. Agent-Curated Memory storage–retrieval pairing profile. Values are mean target-minus-baseline differences in percentage points. E5−E3 is a descriptive complete-system representation contrast, whereas E6−E5 compares R3 with R1 on the same frozen S3 store. Answer metrics use N = 60 and Evidence Recall uses N = 58. No paired-bootstrap interval was pre-declared for these S3 comparisons.

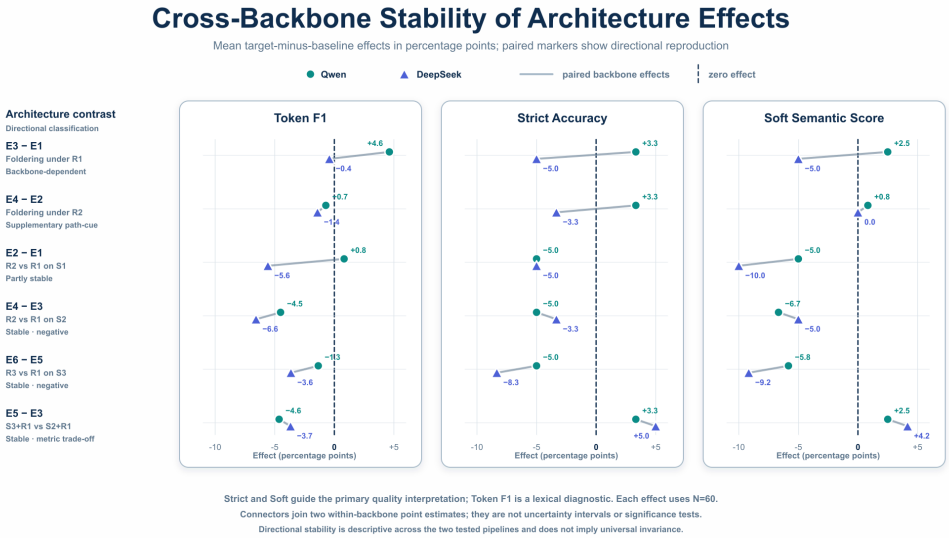


Figure 5.3. Cross-backbone stability of five primary architecture effects and one supplementary path-cue comparison. Each value is a target-minus-baseline mean difference over the same 60 questions, shown in percentage points. Circles denote Qwen and triangles denote DeepSeek. Each connector joins the two backbone estimates for one contrast; it is not a confidence interval or significance test. Table 5.7 reports the five primary RQ2 contrasts; E4−E2 is shown only as supplementary context.

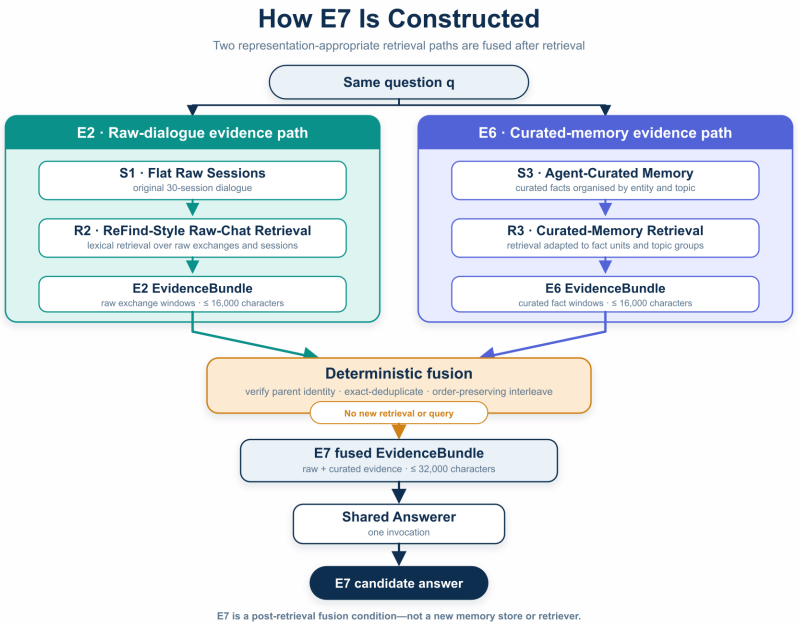


Figure 5.4. Composition of the E7 dual-source fusion condition. E2 combines S1 Flat Raw Sessions with R2 ReFind-Style Raw-Chat Retrieval, whereas E6 combines S3 Agent-Curated Memory with R3 Curated-Memory Retrieval. E7 validates and deterministically fuses the two completed EvidenceBundles, performs no new query or retrieval, and invokes the shared Answerer once.

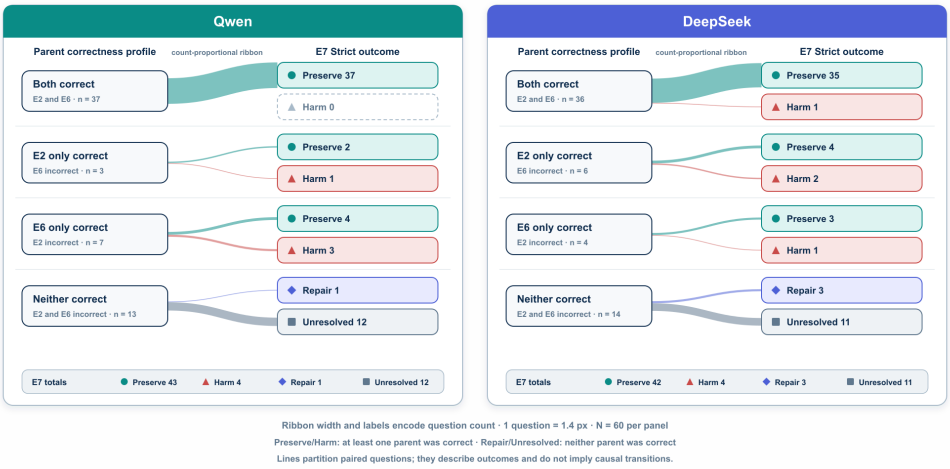


Figure 5.5. Parent Strict-correctness profiles and E7 fusion outcomes. Each panel maps the 60 questions from the four E2/E6 correctness profiles to Repair, Preserve, Harm, or Unresolved. Flow widths and labels represent question counts. Outcomes use Strict Accuracy only. Harm identifies a possible-interference case and does not establish a causal failure mechanism.

The Qwen benefit is parent-dependent. Relative to E2, E7 gains four Strict-correct answers and raises Soft Semantic Score and Evidence Recall, but loses Token F1 because Token Precision falls more than Token Recall rises. Relative to E6, E7 expands annotated-source coverage without improving semantic quality, and its Token F1 is lower. The DeepSeek result is broader: every reported answer metric and Evidence Recall increases relative to both parents.

5.4.1.2 Complementarity Opportunity

Answer-level complementarity asks whether E2 and E6 solve different questions. Panel B shows that exactly one parent is Strict-correct on ten questions under each backbone.

Panel B. Parent Strict-correctness profiles and descriptive oracle benchmark.

Backbone	Both correct	E2 only	E6 only	Neither	Exactly one parent	Parent oracle	E7 correct
Qwen	37	3	7	13	10	47	44
DeepSeek	36	6	4	14	10	46	45

Evidence-access complementarity asks whether the two paths reach different annotated locations. Panel C reports low source-ID overlap and unique gold-source coverage from both parents.

Panel C. Evidence-source overlap and union coverage.

Backbone	E2 Evidence Recall	E6 Evidence Recall	Union/E7 Evidence Recall	Mean source-ID Jaccard	Questions with unique gold coverage from either parent	Union complete
Qwen	0.7816	0.8362	0.8994	0.1956	14/58	49/58
DeepSeek	0.7974	0.7256	0.8951	0.1587	19/58	49/58

The answer sets and retrieved source-ID sets are therefore not redundant. However, source-ID Jaccard measures overlap in recorded locations, not textual or semantic similarity. Gold source IDs also identify annotated locations rather than a minimal or exhaustive reasoning chain.

5.4.1.3 Conversion of the Opportunity

The conversion analysis uses four mutually exclusive Strict-based states. **Repair** means that neither parent is Strict-correct but E7 is correct. **Preserve** means that at least one parent is correct and E7 remains correct. **Harm** means that at least one parent is correct but E7 is not. **Unresolved** means that neither parent nor E7 is correct. Partial and Incorrect judge labels are both non-correct under this definition.

Panel D. Strict-based fusion outcomes.

Backbone	Repair	Preserve	Harm	Unresolved	Preserve/oracle-correct	Repair/neither-correct	Harm/oracle-correct	E7−oracle
Qwen	1	43	4	12	43/47 (91.49%)	1/13 (7.69%)	4/47 (8.51%)	−3
DeepSeek	3	42	4	11	42/46 (91.30%)	3/14 (21.43%)	4/46 (8.70%)	−1

These counts satisfy three identities: E7 correct equals Preserve plus Repair; parent-oracle correct equals Preserve plus Harm; and the E7–oracle gap equals Repair minus Harm. Thus, Qwen yields 44 E7-correct and 47 oracle-correct questions, while DeepSeek yields 45 and 46. Among the ten exactly-one-parent-correct questions, E7 preserves six under Qwen and seven under DeepSeek.

Figure 5.5 visualises the conversion from parent-correctness profiles to the four E7 outcomes.

E7 preserves most oracle-correct questions and also repairs some questions missed by both parents. It does not preserve every available correct parent outcome. The oracle gap is therefore a conversion gap, not evidence that the parents lack complementarity.

5.4.2 Interpretation of the Observed Pattern

5.4.2.1 Why Fusion Has an Opportunity to Help

Low source-ID overlap, unique gold-source coverage, and the ten exactly-one-parent-correct questions under each backbone establish distinct access and success profiles. The raw-dialogue and curated-memory paths may expose different factual detail, temporal structure, or cross-session links. RQ3 does not verify these semantic roles; it shows only that the paths reach different recorded locations and solve partly different question sets.

5.4.2.2 Why Complementarity Does Not Guarantee a Correct Fused Answer

Higher coverage does not automatically become a correct answer. E7 fails to preserve some oracle-correct cases and repairs only a subset of the cases missed by both parents. Repetition, competing states, additional answer length, or Answerer evidence selection may contribute to this gap. These are candidate explanations rather than causal findings; trace-level attribution is deferred to RQ5.

5.4.2.3 Question-Type Heterogeneity

Three subgroup patterns are most informative:

- **DeepSeek temporal:** the 13 questions produce Repair/Preserve/Harm/Unresolved counts of 2/9/1/1. Token F1, Strict Accuracy, Soft Semantic Score, and Evidence Recall all increase relative to both parents.
- **Qwen multi-hop:** the 13 questions produce 1/9/0/3. Strict Accuracy and Soft Semantic Score increase relative to both parents, but Token F1 falls.
- **Qwen single-hop:** the 27 questions produce 0/21/3/3. This category contains three of the four Qwen Harm cases and no Repair, making it the strongest observed possible-interference signal.

These are exploratory subgroup patterns, not established question-type interactions. The complete category matrix is retained in the audit material. Open-domain contains only seven answer questions and five evidence-eligible questions, so it does not support an independent general claim.

5.4.3 Answer to RQ3 and Boundary

E2 and E6 provide measurable answer-level and annotated-source-access complementarity, and E7 converts part of that opportunity into broader evidence coverage and better answers. The Qwen benefit is limited and parent-dependent; the DeepSeek benefit is broader. E7 nevertheless remains below the post-hoc parent oracle and produces four Harm cases under each backbone. Dual-source fusion is therefore useful under specific backbone and question conditions, not uniformly superior.

This conclusion has seven boundaries:

- E7 performs no new retrieval and represents one fixed deterministic fusion design, not fusion methods in general.
- Answer metrics and source-ID overlap use 60 questions per backbone; Evidence Recall and gold-source coverage use 58 evidence-eligible questions.
- Repair and Harm use Strict Accuracy. Partial and Incorrect labels are both treated as non-correct.
- Source-ID Jaccard and gold source IDs measure annotated-location coverage, not semantic complementarity or conflict.
- The parent oracle is post-hoc and non-deployable. It is not a strict mathematical upper bound because E7 can repair questions missed by both parents.
- Each parent EvidenceBundle is limited to 16,000 characters, while the combined E7 input can reach 32,000 characters. The observed E7 effect therefore combines source diversity with greater total evidence capacity.
- RQ3 reports descriptive effects. Harm is possible interference rather than a proven causal mechanism; trace attribution is addressed in RQ5 and full dual-path cost in RQ6.

Section 5.5 next compares selective retrieval with Full Context as a separate diagnostic access condition.

5.5 RQ4: Full Context versus Selective Retrieval

Research question. Does complete context visibility outperform selective evidence access, or do the two access regimes provide different advantages?

Full Context supplies all 30 raw sessions to the Answerer in chronological order. It performs no question-dependent retrieval and does not construct an EvidenceBundle. E7 is the main selective comparison because it combines the E2 and E6 evidence paths; E1 and E5 provide additional quality-oriented anchors.

Direct answer. Full Context achieves the highest mean Strict Accuracy and Soft Semantic Score, but its advantage is small: it answers 46 of 60 questions correctly, compared with 45 for E1, E5, and E7. It has lower Token F1 than all three anchors, and its same-question comparison with E7 contains both Full Context-only and E7-only successes. Full Context therefore does not

dominate selective retrieval. The observed pattern is a coverage–focus trade-off: complete visibility can help when relevant evidence is distributed across sessions, whereas selective retrieval can help when the answer depends on isolating a specific fact or temporal cue. Complete visibility recovers some questions missed by selective retrieval, but it does not remove long-context reasoning or evidence-utilisation errors.

All answer metrics use the same 60 questions. Full Context has no retrieval-level Evidence Recall because it has no retrieved source set; complete visibility must not be reported as Evidence Recall equal to one. The selective conditions used deepseek-v4.1-flash, whereas Full Context used deepseek-v4.1-flash-1017. The comparison is therefore a same-family diagnostic rather than a strict single-variable ablation.

5.5.1 Does Full Context Dominate Selective Retrieval?

Table 5.10 compares Full Context with the three main selective anchors. Strict Accuracy and Soft Semantic Score are the primary quality measures; Token F1 is retained as a lexical diagnostic.

Table 5.10. Overall quality of Full Context and the principal selective anchors. All metrics use the same 60 questions. Correct is the number of Strict-correct answers.

Condition	Storage and access path	Correct	Strict	Soft	Token F1
Full Context	Complete S1 raw history; no retrieval	**46/60**	**0.7667**	**0.8250**	0.2445
E1	S1 + R1	45/60	0.7500	0.8083	**0.3013**
E5	S3 + R1	45/60	0.7500	0.8000	0.2604
E7	Fused E2 + E6 evidence	45/60	0.7500	0.8000	0.2854

Full Context leads each anchor by only one Strict-correct answer. Its Soft advantage is also limited: +0.0167 relative to E1 and +0.0250 relative to E5 and E7. By contrast, its Token F1 is lower than all three selective anchors. The highest semantic mean therefore does not imply superiority on every quality measure.

The paired Full Context–E7 comparison gives the clearest test of dominance. Table 5.11 partitions the same 60 questions by Strict outcome.

Table 5.11. Same-question Strict outcomes for Full Context and E7. The four cells are mutually exclusive and sum to 60.

Strict outcome	Questions
Both correct	42
Full Context only	4
E7 only	3
Neither correct	11

The net Full Context advantage is $4 - 3 = 1$ question, which matches the aggregate Strict difference. More importantly, each regime solves questions missed by the other. Full Context therefore has the highest semantic mean without forming a one-way dominance relation. The lexical result points in the same direction: relative to E7, Full Context lowers Token F1 by 0.0408 even while increasing Strict Accuracy by 0.0167 and Soft Semantic Score by 0.0250. Token overlap and semantic quality capture different properties of the answer.

5.5.2 Why Does Neither Access Regime Dominate?

The results are consistent with one central explanation: complete coverage and selective focus address different information-access problems. Table 5.12 summarises the question-type asymmetry and the scale of the evidence or context payload supplied to the Answerer.

Panel A. Category-specific semantic quality and Strict outcomes.

Table 5.12. Question-type asymmetry and evidence/context payload. Panel A reports descriptive Full Context–E7 results; the subgroup analysis is exploratory. Panel B reports evidence or raw-context characters supplied to the Answerer, not the complete prompt size or total online cost.

Category	N	Full Context Strict	E7 Strict	Full Context Soft	E7 Soft	Both / FC only / E7 only / Neither
Multi-hop	13	**0.8462**	0.6923	**0.8846**	0.8077	8 / 3 / 1 / 1
Temporal	13	0.7692	**0.8462**	0.8077	**0.8462**	10 / 0 / 1 / 2
Open-domain	7	**0.4286**	0.2857	**0.5000**	0.2857	2 / 1 / 0 / 4
Single-hop	27	0.8148	**0.8519**	0.8889	**0.9074**	22 / 0 / 1 / 4

Panel B. Evidence/context payload supplied to the Answerer.

Access regime	Payload (characters)	Share of Full Context
Full Context	108,960.0	100.00%
E1	1,720.9	1.58%
E7	19,758.6	18.13%

The coverage side is clearest for Multi-hop questions. Full Context is Strict-correct on 11 of 13 questions, compared with 9 for E7, and it produces three Full Context-only successes against one E7-only success. Its Soft Semantic Score also rises from 0.8077 to 0.8846. This pattern is consistent with complete visibility helping when an answer must combine facts distributed across sessions. It does not establish that retrieval omission caused every E7 error; case-level attribution is deferred to RQ5.

The focus side appears in the Temporal and Single-hop groups. Neither group contains a Full Context-only success, while each contains one E7-only success. E7 also has higher Strict and Soft scores in both groups. Selective evidence may reduce competition from unrelated dialogue, repeated events, similar entities, or different temporal states, thereby making a decisive attribute or time anchor easier to locate. The payload comparison shows the scale of this concentration: E7 exposes 18.13% of the Full Context characters, while E1 exposes 1.58%, yet both remain within one Strict-correct answer of Full Context overall. These values describe evidence concentration, not total prompt tokens or a deployment-cost ranking; cost is analysed in RQ6.

The Open-domain subgroup favours Full Context semantically, but it contains only seven questions and is retained as an exploratory observation. Across all categories, the results support a coverage–focus trade-off rather than a universal rule that either more context or less context is always better.

5.5.3 Answer to RQ4 and Boundary

Full Context provides a small average semantic advantage, but it is not uniformly superior to selective retrieval. Its clearest observed advantage appears when relevant evidence is dispersed across the conversation. Selective retrieval remains useful when the task requires precise localisation, temporal focus, or protection from competing information. Full Context is therefore a complementary evidence-access regime rather than an oracle or a replacement for retrieval.

Complete visibility also does not guarantee a correct answer. Full Context produces 46 Correct, seven Partial, and seven Incorrect judgements, leaving 14 non-Correct answers. RQ4 establishes only that these failures remain despite complete source visibility. Their detailed attribution to long-context processing, reasoning, or evidence utilisation is addressed in RQ5.

This conclusion has four boundaries:

- Full Context is not E8, a new storage representation, or an oracle; it is a diagnostic path that bypasses selective retrieval.
- Retrieval-level Evidence Recall is not applicable to Full Context.
- Full Context and E1–E7 use different DeepSeek served-model identifiers, and there is no Qwen Full Context run. The comparison is not a backbone-matched single-variable ablation.
- The experiment covers one fixed 30-session conversation. The observed patterns are consistent with coverage and focus mechanisms; they do not prove the cause of individual errors. Root-cause attribution is deferred to RQ5, and quality–cost selection to RQ6.

Section 5.6 next traces the remaining failures through storage, retrieval, packaging, Answerer utilisation, and long-context processing.

5.6 RQ5: Error Chain and Bottleneck Diagnosis

Research question. RQ5 asks: across the non-Correct answer paths, where is the earliest evidence-supported break in the information chain, and which stage is the largest confirmed system-internal bottleneck?

The analysis covers the 250 answers that were not labelled Correct among the 900 frozen answers. These comprise 205 E1–E6 selective-retrieval paths, 31 E7 paths, and 14 Full Context paths. The unit is one `backbone model x condition x question` path, so repeated questions across conditions are not independent observations. Each path receives one primary attribution at the earliest supported break. Later mechanisms may co-occur, while unsupported cases remain Mixed/Indeterminate.

Direct answer. Among the 123 confirmed system-internal failures, Answerer utilisation is the largest observed bottleneck, with 68 paths (55.3%). Retrieval is second, with 42 paths (34.1%). Packaging accounts for seven paths (5.7%), and long-context processing accounts for six (4.9%). No decisive store-specific omission is confirmed. This internal denominator excludes 53 paths whose primary attribution is Evaluation-sensitive and 74 Mixed/Indeterminate paths; the result is therefore a diagnosis of confirmed failures, not a claim that every error has one known cause.

Table 5.13. Primary attribution under the two RQ5 diagnostic denominators. The confirmed-internal denominator excludes Evaluation-sensitive and Mixed/Indeterminate paths. All six Fusion/long-context cases are Full Context long-context failures; no E7 path has fusion as its primary cause.

Primary attribution	All non-Correct paths	Share of 250	Confirmed internal failures	Share of 123
Storage	0	0.0%	0	0.0%
Retrieval	42	16.8%	42	34.1%
Packaging/budget	7	2.8%	7	5.7%
Answerer utilisation	68	27.2%	68	55.3%
Fusion/long-context	6	2.4%	6	4.9%
Evaluation-sensitive	53	21.2%	–	–
Mixed/Indeterminate	74	29.6%	–	–
Total	**250**	**100.0%**	**123**	**100.0%**

The 68 Answerer, seven Packaging, and six long-context attributions are supported directly by the final trace. The 42 Retrieval attributions are convergent diagnoses based on the observed break and matched-path evidence.

Figure 5.6 separates the full error population from the confirmed system-internal subset.

5.6.1 Where Does the Information Chain First Break?

The store audit finds differences in organisation rather than a decisive store-specific omission. S1 preserves the rendered session content verbatim but leaves cross-session composition to later stages. S2 preserves the same content and adds directory cues, although those cues do not always match the local answer-bearing topic. S3 improves thematic co-location and provenance, but a curated fact is not necessarily answer-sufficient. One shared source-scope limitation affects a single question: the common renderer omits image-search query metadata that contains a required detail. It recurs in 12 selective error paths and 14 paths overall. Because every store inherits this limitation and one E7 path succeeds despite it, the issue is retained as a co-occurring upstream limitation rather than a store-specific primary attribution. Thus, `Storage = 0` in Table 5.13 does not mean that every representation is perfectly lossless.

The next diagnostic follows official source locators through selective retrieval. Among the 191 evidence-eligible E1–E6 errors, 111 contain every annotated locator in the final Evidence-Bundle. Seventy-three fail to carry a complete annotated set through final selection, and seven lose selected evidence at the bundle boundary.

Table 5.14. Annotated-evidence state among the 191 evidence-eligible E1–E6 Strict-noncorrect paths. Retrieval-stage attrition covers failure to reach, observe, completely observe, or select every official locator. Selected-to-bundle loss occurs after complete selection.

Retriever	Eligible errors	Complete annotated bundle	Retrieval-stage attrition	Selected-to-bundle loss
R1	85	68 (80.0%)	17 (20.0%)	0 (0.0%)
R2	72	28 (38.9%)	44 (61.1%)	0 (0.0%)
R3	34	15 (44.1%)	12 (35.3%)	7 (20.6%)
Overall	**191**	**111 (58.1%)**	**73 (38.2%)**	**7 (3.7%)**

Figure 5.7 makes the retriever-specific pattern explicit.

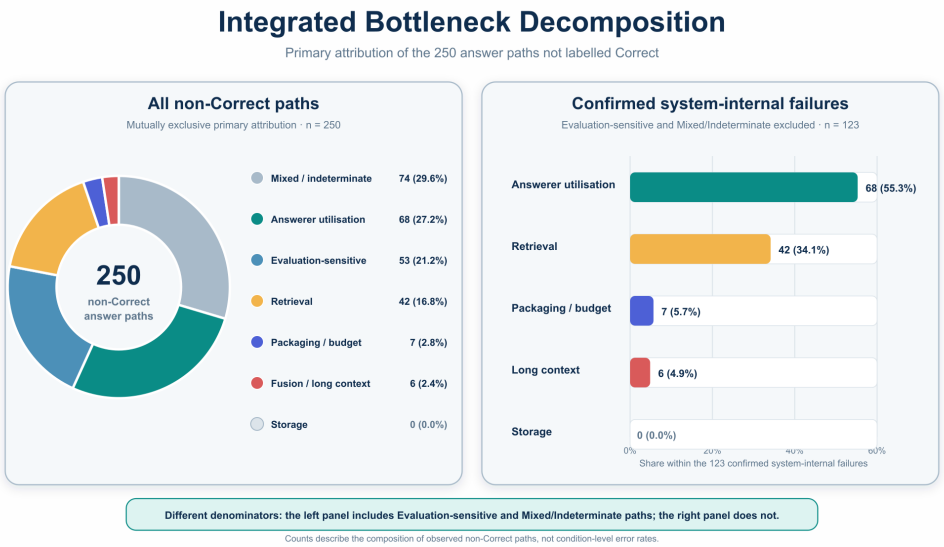


Figure 5.6. Primary attribution of all 250 non-Correct paths (left) and the 123 confirmed system-internal failures after excluding 53 Evaluation-sensitive and 74 Mixed/Indeterminate paths (right). Counts describe error-path composition, not condition-level failure rates; all six Fusion/long-context cases are Full Context long-context failures.

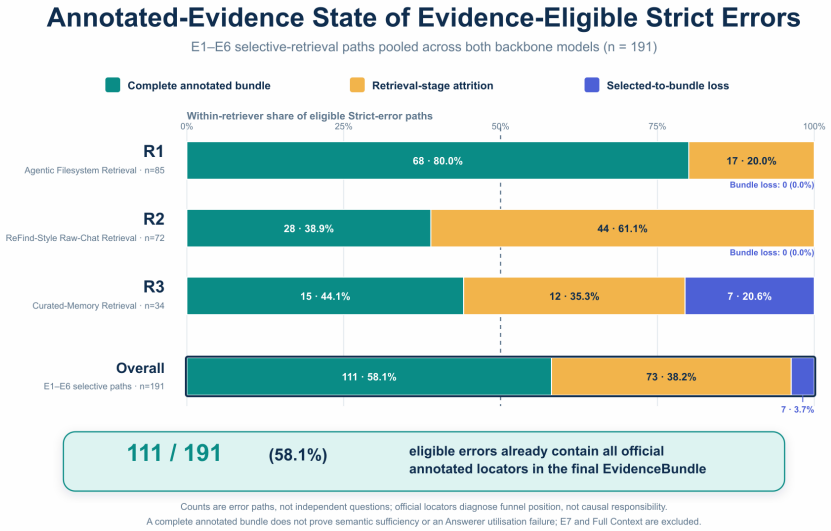


Figure 5.7. Annotated-evidence states for 191 evidence-eligible E1–E6 Strict-noncorrect paths, normalised within retriever. These are diagnostic states, not primary causes; complete locator coverage does not imply semantically sufficient input. E7 and Full Context are excluded.

These states locate evidence loss but do not assign its primary cause. Of the 73 selective retrieval-attrition paths, 41 receive Retrieval as the primary attribution; one inherited-parent break in E7 raises the integrated Retrieval count to 42. Likewise, the seven bundle-boundary cases overlap only three of the seven Packaging primaries. Four other Packaging primaries occur during incomplete selection; of the remaining four boundary cases, two retain sufficient alternative input and are attributed to the Answerer, while two remain Mixed/Indeterminate. Across primary and secondary labels, nine paths show demonstrated packaging loss.

Locator completeness also differs from semantic sufficiency. Of the 205 E1–E6 selective errors, 88 provide sufficient final input, 111 provide insufficient input, and six remain indeterminate. Among the 111 complete annotated bundles, only 63 are semantically sufficient; conversely, 19 of the 80 non-All-hit paths contain sufficient alternative support. Six further sufficient paths have no official locator and are excluded from Table 5.14. Official locators therefore diagnose evidence position, not a minimal or exhaustive proof set. Table 5.15 reports the corresponding final-input states.

Table 5.15. Final-input sufficiency among the 205 E1–E6 Strict-noncorrect paths. Shares use the error-conditioned selective-path denominator.

Final-input state	Qwen	DeepSeek	Total	Share
Sufficient	43	45	88	42.9%
Insufficient	52	59	111	54.1%
Indeterminate	4	2	6	2.9%
Total	**99**	**106**	**205**	**100.0%**

Within the 88 sufficient-input errors, 56 contain an explicit Answerer failure: 36 misuse available evidence and 20 abstain despite adequate support. Adding nine E7 Answerer failures and three Full Context Answerer failures gives the 68 primary Answerer attributions in Table 5.13. This link between input sufficiency and final use supports the main RQ5 result: evidence acquisition remains important, but the largest confirmed residual bottleneck is the Answerer's use of available evidence.

Evaluation disagreement does not overturn this ordering. A directional diagnostic excludes nine evaluation-sensitive questions from 17 registered paired contrasts, reducing each comparison from 60 to 51 questions. Token F1 direction remains unchanged in all 17 contrasts, and Strict Accuracy direction remains unchanged in 15. Qwen E7 minus E6 changes from a tie to `+1/51`, while Full Context minus E5 changes from `+1/60` to `−1/51`. These small margins require qualification. The nine excluded questions and the 53 answer paths whose primary attribution is Evaluation-sensitive use different units and definitions and must not be interchanged.

5.6.2 Architecture-Specific Failure Signatures

The aggregate result masks different failure locations across access architectures. R1 errors are usually observed after complete annotated evidence has reached the EvidenceBundle. R2 errors concentrate before complete observation under the tested query, ranking, expansion, and four-action-cap design; reaching the cap is not itself an error label. R3 can reach the correct curated region yet fail to expose or retain every locator under the shared evidence budget. E7 performs no new retrieval and has no primary fusion failure, but fusion changes the evidence presentation received by the Answerer. Full Context bypasses selective retrieval and packaging, yet its remaining errors still include localisation and cross-event composition failures. Table 5.16 summarises these architecture-specific signatures.

Table 5.16. Architecture-specific failure signatures. Counts describe the composition of observed error paths, not component failure rates over all requests.

Architecture	Main observed signature	Primary-cause evidence	Interpretation limit
R1	68/85 eligible errors already contain a complete annotated bundle	34 Answerer and three Packaging primaries; no Retrieval primary	This does not imply that R1 never misses useful evidence
R2	44/72 eligible errors show retrieval-stage attrition	33 Retrieval and one Packaging primary	The result applies to the tested query, ranking, expansion, and action-cap design
R3	12/34 eligible errors show retrieval attrition and 7/34 show bundle-boundary loss	Eight Retrieval and three Packaging primaries	Reaching a curated topic does not guarantee locator exposure or retention
E7	Parent EvidenceBundles are reused without new retrieval	Nine Answerer and one inherited Retrieval primary; no Fusion primary	Harm is consistent with evidence competition but does not prove a fusion defect
Full Context	Every session is visible, but evidence must still be located and composed	Six Long-context and three Answerer primaries among 14 errors	The result does not establish a universal lost-in-the-middle mechanism

Table 5.17 uses five representative traces to illustrate these signatures without reproducing question or answer text.

Table 5.17. Representative trace-reviewed cases. Each case reports the strongest supported attribution; later mechanisms may still co-occur.

Path	Case	Trace evidence	Earliest supported break
R1 / Qwen E1	q037	A focused bundle contains the complete annotated evidence, but the answer returns only part of the required date	Answerer utilisation
R2 / Qwen E2	q022	Query and ranking fail to reach either annotated session before the action limit	Retrieval
R3 / Qwen E6	q052	The key fact is found, but the selected key item is removed by the evidence budget	Packaging/budget



Figure 5.8. Soft Semantic Score versus mean successful-query tokens on a logarithmic cost axis. Teal points and the orange Full Context diamond lie on the frontier of their respective panels; pale points are dominated. Panel A compares Qwen E1–E7. Panel B adds Full Context as a mixed-served-model diagnostic point. Query cost excludes construction, formal failed attempts, and LLM judge evaluation; E7 cost comprises E2 retrieval, E6 retrieval, and one fused Answerer.

Continued from the previous page

Path	Case	Trace evidence	Earliest supported break
E7 / Qwen	q051	E2 is Correct and all parent items are preserved, but the fused input remains insufficient under the audit and E7 becomes non-Correct	Mixed/Indeterminate; possible evidence competition is not established as primary
Full Context / DeepSeek	q019	All sessions are visible, but the answer combines only one of two required events	Long-context localisation and composition

The traces support one mechanism-level conclusion. Improving search alone is most directly relevant to R2 and part of R3. For many R1 errors, and for confirmed E7 and Full Context errors with sufficient input, the remaining need is more reliable evidence selection, temporal or entity composition, and answer generation.

5.6.3 Answer to RQ5 and Boundary

RQ5 therefore has a conditional answer. Among the 123 confirmed system-internal failures, Answerer utilisation is the largest observed bottleneck, followed by retrieval. Packaging and long-context processing are smaller but directly supported. Within these confirmed attributions, supported breaks concentrate after store construction; no decisive store-specific omission is confirmed.

This conclusion has six boundaries. First, the counts are error-conditioned path compositions, not component failure rates over all requests. Second, the same question can recur across conditions and backbone models, so paths are not independent samples. Third, earliest-supported attribution prevents double-counting but does not exclude later co-occurring mechanisms. Fourth, official locators are not an exhaustive semantic rubric. Fifth, 53 Evaluation-sensitive and 74 Mixed/Indeterminate paths remain explicit rather than being forced into a system component. Finally, RQ5 diagnoses failure location; quality–cost selection remains the role of RQ6.

RQ6 therefore converts the quality evidence and failure burden into conditional system choices.

5.7 RQ6: Quality, Cost, and Practical System Choice

Research question. Which tested condition offers the most appropriate quality–cost trade-off after the backbone model, available memory assets, and quality target are specified? Strict Accuracy and Soft Semantic Score are the primary quality measures; Token F1 is a lexical diagnostic. Mean successful-query tokens are the primary online cost proxy, while model calls, construction, and recovery are reported separately. No composite utility score is formed because monetary price, latency, maintenance cost, and metric weights were not defined.

E7 online cost is reconstructed as E2 retrieval plus E6 retrieval plus one fused Answerer call. The two parent Answerer calls are excluded. LLM judge usage is evaluation-only and is also excluded.

Direct answer. There is no global winner. E2 defines the mean successful-query token floor. Under the stated semantic-quality and token priorities, E3 is the recommended Qwen balance, while E5 provides the highest Qwen semantic quality at much greater cost. For DeepSeek, E7 is the preferred selective balance when S3 already exists; otherwise, E1 is the simpler baseline without additional S2/S3 construction. Full Context is a competitive diagnostic or candidate escalation path under the tested setting, not a universal replacement for retrieval.

5.7.1 Online Quality–Cost Frontier

Question. Before construction and recovery are considered, which conditions occupy the useful online quality–cost frontier?

Table 5.18 reports answer quality and successful-query cost separately for the two backbone pipelines.

Panel A. Qwen E1–E7.

Table 5.18. Answer quality and successful-query cost by backbone. Each quality metric uses all 60 questions. Query cost includes retrieval and the Answerer but excludes construction, recovery, and LLM judge usage. In each E1–E7 panel, bold marks the best selective value, including ties. In Panel B, bold Full Context values separately identify diagnostic values that are better than every selective condition in the favourable direction.

Condition	Token F1	Strict Accuracy	Soft Semantic Score	Mean query tokens	Mean model calls
E1	0.395	0.717	0.767	148,920	12.50
E2	0.403	0.667	0.717	**18,419**	4.93
E3	**0.441**	0.750	0.792	140,288	11.67
E4	0.396	0.700	0.725	18,479	4.98
E5	0.395	**0.783**	**0.817**	281,354	8.05
E6	0.381	0.733	0.758	36,378	**4.83**
E7	0.369	0.733	0.758	53,637	8.77

Panel B. DeepSeek E1–E7 and the Full Context diagnostic point. Full Context used `deepseek-v4.1-flash-1017`, whereas E1–E7 used `deepseek-v4.1-flash`; it is not an exact served-model ablation.

Condition	Token F1	Strict Accuracy	Soft Semantic Score	Mean query tokens	Mean model calls
E1	**0.301**	**0.750**	**0.808**	201,030	12.98
E2	0.246	0.700	0.708	**17,313**	**5.00**
E3	0.297	0.700	0.758	189,646	12.70
E4	0.231	0.667	0.708	17,458	**5.00**
E5	0.260	**0.750**	0.800	374,700	8.90
E6	0.224	0.667	0.708	35,457	**5.00**
E7	0.285	**0.750**	0.800	51,581	9.00
Full Context	0.245	**0.767**	**0.825**	36,646	**1.00**

E2 uses 87.6% fewer query tokens than E1 under Qwen and 91.4% fewer under DeepSeek, but Strict Accuracy falls by 0.050 under both backbones. Soft Semantic Score falls by 0.050 under Qwen and 0.100 under DeepSeek. E2 is therefore a cost-first option, not a cost-free replacement.

For Qwen, E3 improves on E1 by 0.046 Token F1, 0.033 Strict, and 0.025 Soft while using 5.8% fewer query tokens. E5 reaches the highest Strict (0.783) and Soft (0.817), but uses about twice the query tokens of E3. E3 is the stronger balance; E5 is the semantic-quality maximum. E4 and E6 remain intermediate Soft-frontier points but do not change this scenario recommendation.

For DeepSeek, E7 matches E1's Strict Accuracy of 0.750, remains 0.008 lower on Soft and 0.016 lower on Token F1, and uses 74.3% fewer query tokens. It is therefore the stronger selective balance when S3 exists. Without S3, E1 remains the simpler quality baseline. Full Context records the highest DeepSeek-family Strict (0.767) and Soft (0.825), but its different served-model identifier prevents a strict dominance claim.

Figure 5.8 summarises the Soft Semantic Score frontier. Panel A is the exact Qwen E1–E7 comparison. Panel B deliberately adds Full Context to the DeepSeek conditions; it is therefore diagnostic and does not replace the selective-only result that E7 is Pareto-efficient within DeepSeek E1–E7.

5.7.2 Construction and Recovery Modifiers

Question. Do one-time construction and observed recovery burden change the online recommendation?

Finding. Yes. S2 can be amortised after modest reuse, whereas S3 requires substantially greater reuse or an external reason to value the asset.

S2 construction uses 395,749 tokens. Under Qwen, E3 crosses E1's cumulative token cost after 46 queries and is higher on all three quality measures. S3 construction uses 30,864,958 tokens. Under DeepSeek, E7 crosses E1 after 207 queries, but only Strict is equal; Soft and Token F1 remain slightly lower. E7 never becomes cheaper than E2 because both its construction and per-query costs are higher. S1 has no model-based construction cost, but deterministic normalisation cost was not measured.

The runs contain 41 formal failed attempts. Usage is known for 39 attempts, totalling 10,126,358 recovery tokens; two Qwen E5 DNS attempts have unknown usage. All observed attempts occur in R1 conditions E1, E3, and E5. This is an observed-run result, not a general R1 failure-rate estimate.

Construction and recovery therefore modify the online frontier. S2 is a modest structural investment. S3 is a reusable infrastructure asset and should not be treated as a free input to E5–E7.

5.7.3 Scenario-Based Recommendation and Answer to RQ6

Table 5.19 converts the quality and cost evidence into scenario-based recommendations.

Table 5.19. Scenario-based system selection. Recommendations use observed Main-60 quality and cost; they do not imply unmeasured latency or monetary advantages.

Deployment objective	Recommended condition	Evidence	Main qualification
Minimum online token cost	E2 under either backbone	Lowest mean query tokens: 18,419 for Qwen; 17,313 for DeepSeek	Lower semantic quality than stronger R1 or fused paths
Recommended Qwen balanced frontier point	E3	F1 0.441, Strict 0.750, Soft 0.792; 5.8% fewer query tokens than E1	Requires S2; build amortises after 46 queries against E1
Maximum Qwen semantic quality	E5	Highest Qwen Strict (0.783) and Soft (0.817)	Very high S3 construction, query, and recovery cost
Recommended DeepSeek selective balance with existing S3	E7	Matches E1 Strict at 0.750 with 74.3% fewer query tokens	Requires S3 and two online retrieval paths
DeepSeek baseline without S2/S3 construction	E1	Strict 0.750, Soft 0.808, F1 0.301; direct raw-source traceability	Deterministic normalisation cost was not measured; query and observed recovery cost are high
Diagnostic or candidate escalation assessment	Full Context	Complete visibility and competitive tested semantic quality	Different served model; no universal recovery or scaling claim

The recommendations reflect different sources of work. R2 reduces calls and tokens through a constrained retrieval process, with a quality cost. S2 is a relatively inexpensive organisational asset. S3 is expensive, and E7 additionally executes both parent retrieval paths before one fused answer. Full Context removes retriever calls but transfers localisation and composition to a long Answerer input. RQ5 further shows that Answerer utilisation accounts for 68 of 123 confirmed internal failures, so additional retrieval complexity is not automatically the best next investment.

RQ6 therefore supports conditional routing rather than one global champion. Use E2 when online token cost dominates. Use E3 as the Qwen default once S2 can be reused. Use E5 only when maximum Qwen semantic quality justifies S3. For DeepSeek, use E7 when S3 already exists; otherwise use E1. Treat Full Context as a diagnostic or candidate escalation path under the tested setting.

These recommendations have four boundaries. Tokens are a cost proxy, not monetary price or latency, and LLM judge cost is excluded. S3 maintenance and update costs were not measured; known recovery tokens are a lower bound. Full Context uses a different served-model identifier and one fixed 30-session conversation. Pareto membership depends on the chosen quality metric, and token break-even does not imply quality equivalence.

5.8 Chapter Summary and Consolidated Answers to RQ1–RQ6

This chapter examined how memory representation, retrieval design, backbone model, evidence fusion, and complete-context access affect long-term memory question answering. Table 5.20 consolidates the final answer to each research question together with its strongest evidence and principal boundary.

5.8.1 Consolidated Answers to RQ1–RQ6

Table 5.20. Consolidated answers to RQ1–RQ6.

RQ	Final answer	Anchor evidence and principal boundary
RQ1	The effective design unit is the storage–retrieval pairing. Folder organisation is not automatically beneficial; R1 is the more consistent quality-oriented path, while S3+R1 improves semantic quality and Evidence Recall over S2+R1 at the cost of lower Token F1.	E3−E1 is positive only for Qwen, E4−E3 and E6−E5 are negative on the main semantic and evidence measures under both backbones, and E5−E3 shows the same lexical–semantic trade-off in both pipelines. Main-60 does not establish a universal directory benefit, and the complete-system contrasts do not isolate a single causal component.
RQ2	Retrieval-path ordering is more stable across the two tested pipelines than the benefit of storage organisation. R1 remains stronger on S2 and S3, whereas the benefit of S2 under R1 is backbone-dependent. Qwen's much higher Token F1 does not correspond to an equally large semantic-quality advantage.	Of the five principal contrasts, three are directionally stable, E2−E1 is partly stable, and E3−E1 is backbone-dependent. Mean Token F1 is 0.3972 for Qwen and 0.2636 for DeepSeek, while Strict Accuracy is 0.7262 versus 0.7119 and Soft Semantic Score is 0.7619 versus 0.7560. Only two end-to-end pipelines were tested, and both the Retrieval Agent and Answerer changed with the backbone.
RQ3	E2 and E6 provide measurable complementarity in answer outcomes and annotated-source access, and E7 converts part of that opportunity into better answers. The benefit is broader under DeepSeek but conditional under Qwen, and fusion does not preserve every correct parent outcome.	E7/union Evidence Recall reaches 0.8994 under Qwen and 0.8951 under DeepSeek. The parent oracle yields 47 and 46 correct answers, compared with 44 and 45 for E7, and each backbone has four Harm cases. E7 is a fixed deterministic fusion with no new retrieval; because it can use up to 32,000 evidence characters rather than 16,000 for either parent, its effect cannot be attributed to source diversity alone.
RQ4	Full Context records the highest observed mean Strict Accuracy and Soft Semantic Score in the DeepSeek-family diagnostic comparison, but it does not dominate selective retrieval. The two access regimes express a coverage–focus trade-off.	Full Context reaches 46/60 Strict-correct answers, compared with 45/60 for E1, E5, and E7. Against E7, 42 questions are correct under both, four only under Full Context, three only under E7, and 11 under neither. Full Context uses a different served-model identifier, has no retrieval-level Evidence Recall, and is evaluated on one fixed 30-session conversation.
RQ5	The largest confirmed internal bottleneck is Answerer utilisation, followed by retrieval; the dominant failure location nevertheless varies by architecture.	Among 123 confirmed system-internal failures, 68 (55.3%) are attributed to Answerer utilisation, 42 (34.1%) to retrieval, seven (5.7%) to packaging, and six (4.9%) to long-context processing. A further 53 paths are evaluation-sensitive and 74 are mixed or indeterminate. These are error-conditioned path counts, not general component failure rates.
RQ6	No condition is best under every objective. System choice should depend on the backbone, existing memory assets, semantic-quality target, and online token budget.	E2 has the lowest successful-query token cost but lower semantic quality. E3 is the recommended Qwen balance. With an existing S3 store, DeepSeek E7 uses 74.3% fewer successful-query tokens than E1 while matching its Strict Accuracy, although its Soft score is 0.008 lower; without S3, E1 is the simpler quality baseline. Construction and recovery costs remain material, and Full Context is a diagnostic rather than an exact served-model ablation.

5.8.2 Overall Synthesis and Transition

Three conclusions connect the six research questions.

First, architecture quality depends on the match among memory representation, retrieval path, and backbone model. R1 is the most consistent quality-oriented retrieval path in the tested systems, but neither folder organisation nor Agent-Curated Memory produces an automatic gain. Their value depends on whether the access path can exploit the available structure. Token F1 must also remain separate from semantic quality because the two backbone pipelines differ much more in lexical overlap than in Strict Accuracy or Soft Semantic Score.

Second, evidence access is not equivalent to evidence utilisation. E7 and Full Context expand access in different ways, yet neither converts every additional opportunity into a correct answer. Among confirmed internal failures, Answerer utilisation accounts for 55.3%, making it a larger observed bottleneck than retrieval. Improving retrieval alone is therefore insufficient; the system must also select, compose, and use available evidence reliably.

Third, deployment requires conditional selection rather than one global winner. E2 provides the online token floor but sacrifices semantic quality. E3 is the recommended Qwen balance once S2 can be reused. E5 is appropriate only when its higher Qwen semantic quality justifies S3, while DeepSeek E7 is attractive when S3 already exists. Otherwise, E1 remains the simpler DeepSeek quality baseline. Full Context is better treated as a diagnostic or candidate escalation path than as a universal replacement for selective retrieval.

The next chapter derives broader implications for evidence-controlled long-term memory system design. Chapter 7 then examines the limitations and validity threats that bound those implications.

Chapter 6. General Discussion: Towards Evidence-Controlled Memory Systems

The results in Chapter 5 converge on one general problem. Information may exist in memory, become visible to the system, and still fail to support a reliable answer. Improving storage or retrieval alone cannot close every failure in this chain.

> **Central argument.** Long-term memory systems should be organised around explicit evidence control, rather than around storage or retrieval alone.

The chapter follows one argument:

> **evidence-conversion gap → Evidence Controller → staged evidence-access policy**

The staged policy begins with selective retrieval, adds another source only when needed, uses Full Context as recovery, and abstains when the evidence remains unsafe. The controller is a design proposal derived from the experiments, not a component implemented or evaluated in this study.

6.1 The Evidence-Conversion Gap

The principal gap lies between **information access** and **evidence readiness**. A conventional memory pipeline is often written as:

> memory store → retrieval → answer

This pipeline treats retrieved text as if it were already usable evidence. A more accurate chain is:

> stored information → accessible information → answer-ready evidence → correctly used evidence

Storage must preserve the required fact and its qualifiers. Retrieval must reach the appropriate source. Evidence packaging must retain the relevant entities, relations, time anchors, and context. The Answerer must then resolve and combine them correctly. Failure at any transition can produce the same incorrect answer.

Chapter 5 shows that these states cannot be collapsed. Decisive storage omission was not the dominant observed failure. Many error paths reached all annotated source locations, but the final EvidenceBundle was not always semantically sufficient. Some EvidenceBundles were sufficient but still produced an incorrect or unsupported answer. Full Context removed explicit retrieval selection without removing localisation, temporal ordering, cross-event composition, or Answerer utilisation failures. Dual-source fusion exposed complementary evidence, but extra evidence could also introduce repetition, competing states, or weaker focus.

The **evidence-conversion gap** is therefore the gap between information that a system stores or exposes and evidence that is sufficiently focused, coherent, and usable for the current question. It is not one more error label. It is the central transformation problem connecting storage, retrieval, packaging, and answering.

This changes the design objective. A memory system should preserve answer-bearing relationships, not only maximise stored content or retrieved matches. A Temporal question may require an ordered update rather than two independently relevant passages. A Multi-hop question may require an explicit connection between evidence items. A factual question may require one unambiguous entity–attribute relation. The best access path therefore depends on the question and the current evidence state, not only on the underlying store.

The general lesson is direct: memory quality must be judged by the complete conversion from historical information to usable evidence. This extends the emphasis on store shape and tool harnesses in Filesystem-Based Memory [4], the iterative access process in ReFind [6], and the distinction between retrieval and downstream utilisation in recent diagnostic work [7].

6.2 The Missing Architectural Layer: An Evidence Controller

The missing architectural component is an **Evidence Controller** between evidence access and the Answerer. It would inspect the question, the current EvidenceBundle, its provenance, and the remaining resource budget before deciding whether the system is ready to answer. It would not replace the memory store, Retrieval Agent, or Answerer. Its purpose would be to decide whether the present evidence justifies answering or requires another access action.

The controller would perform four functions.

Sufficiency checking. The controller would test whether the evidence covers the entities, relations, time constraints, and intermediate facts required by the question. A relevant passage is not necessarily a complete answer path.

Conflict checking. It would detect incompatible claims and competing temporal states. Rather than remove them through blind deduplication, it would compare their source and recency or seek disambiguating evidence.

Provenance checking. It would verify that answer-critical claims remain traceable to source records. This allows the system to distinguish a source event from a curated interpretation and to investigate disagreement between them.

Action and stopping control. It would choose whether to answer, continue searching, switch or add a source, fuse evidence, escalate to Full Context, or abstain. Stopping would depend on evidence readiness rather than only on an action cap or the presence of several matches.

The controller could not use gold answers, gold source turns, Evidence Recall, or LLM judge labels at deployment time. It would require observable proxies such as question decomposition, entity and time-slot coverage, cross-source agreement, provenance completeness, evidence novelty, and calibrated uncertainty. Chapter 5 uses gold evidence to diagnose completed runs; an online controller would have to estimate readiness without those annotations.

The resulting system would replace a one-way pipeline with a closed control loop:

> access evidence → inspect evidence state → choose the next action → answer or access again

The distinction from a Retrieval Agent is important. A Retrieval Agent decides how to search one store. An Evidence Controller decides whether the system should remain on that path at all. It coordinates access modes and stops further search when the evidence is already adequate. This avoids adding retrieval complexity when the main remaining risk is downstream utilisation.

6.3 Adaptive Evidence Access as a Control Policy

The controller should implement one clear policy: **select the least expansive access mode that can produce an adequate evidence state**. This turns selective retrieval, fusion, Full Context, and abstention into stages of one control process.

Level 1: selective retrieval by default. The system should begin with focused, traceable evidence. The exact storage–retrieval pairing may vary by backbone model and question type. The controller should stop at this level when the EvidenceBundle is sufficient and coherent.

Level 2: multi-source fusion when complementarity is plausible. A second source should be added only when the first path is incomplete and another representation may supply the missing relation, event, or temporal cue. Fusion must preserve source identity, remove redundancy without erasing distinct states, check conflicts, and control the combined budget. E7 demonstrates the opportunity and the risk: complementary evidence can repair an answer, but more evidence can also weaken focus.

Level 3: Full Context as recovery. The system should escalate when selective omission remains plausible, the question requires broad cross-session inspection, or retrieved fragments cannot be resolved locally. Full Context restores source visibility but is not an oracle. Localisation, temporal reasoning, composition, and Answerer utilisation can still fail.

Abstention when evidence remains unsafe. If evidence remains incomplete or irreconcilable, the system should abstain, expose the uncertainty, or request clarification rather than produce a falsely confident answer.

The four principal evidence states and their corresponding actions are summarised below.

Table 6.1. Conceptual adaptive evidence-access policy. The policy is a design implication, not an online routing procedure evaluated in the present experiments.

Current evidence state	Controller action	Intended benefit	Required safeguard
Focused, coherent, and sufficient	Send the EvidenceBundle to the Answerer	Avoid unnecessary search and context growth	Retain answer-critical provenance
Partial, with a plausible complementary source	Query the alternative source or apply controlled multi-source fusion	Recover a missing relation, event, or temporal cue	Deduplicate, preserve source identity, and check conflicts
Selective omission remains plausible or broad cross-session inspection is required	Escalate to Full Context recovery	Restore complete source visibility	Recheck localisation, temporal order, and cross-event composition
Conflicting or insufficient after escalation	Abstain, flag uncertainty, or request clarification	Avoid an unsupported or falsely confident answer	Report the unresolved claims and their provenance

The policy should be cost-aware but not cost-driven. Budgets can limit escalation, but they do not establish evidence sufficiency. Because the backbone model affects both retrieval behaviour and evidence utilisation, controller thresholds should be calibrated for the complete backbone pipeline rather than transferred unchanged between models.

The resulting principle is concise: **retrieval should be a controlled evidence-acquisition process, not a one-time preprocessing step**. The Evidence Controller would assess whether evidence is sufficient, coherent, and traceable, then choose the least expansive safe action. This proposal does not establish that one controller implementation will improve performance. It identifies the architectural problem exposed by the experiments and the most direct design response. Chapter 7 examines the limitations and validity threats that bound this conclusion.

Bibliography

[1] A. Maharana, D.-H. Lee, S. Tulyakov, M. Bansal, F. Barbieri, and Y. Fang, "Evaluating very long-term conversational memory of LLM agents," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Bangkok, Thailand, Aug. 2024, pp. 13 851–13 870.

[2] Y. Hu, Y. Wang, and J. McAuley, "Evaluating memory in LLM agents via incremental multi-turn interactions," in *The Fourteenth International Conference on Learning Representations*, 2026. [Online]. Available: https://openreview.net/forum?id=DT7JyQC3MR

[3] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Kuttler, M. Lewis, W.-T. Yih, T. Rocktaschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474. [Online]. Available: https://proceedings.neurips.cc/paper/2020/hash/6b493230-Abstract.html

[4] S. Zhou, S. Yu, H. Wei, J. Wu, S. Ouyang, Y. Jiao, S. Pan, J. McAuley, Y. Zhang, T. Yu, and J. Han, "Filesystem-based memory for LLM agents: Organization, evolution, and sustainability," *arXiv preprint arXiv:2607.26637*, 2026.

[5] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024.

[6] R. Li, L. Zhang, B. Xu, M. Du, Z. Fu, and W. Chen, "When your agent opens the chat app: Agent-controlled search over raw chat logs rivals structured memory," *arXiv preprint arXiv:2608.12888*, 2026.

[7] B. Yuan, Y. Su, and K. Yao, "Diagnosing retrieval vs. utilization bottlenecks in LLM agent memory," in *ICLR 2026 Workshop on Memory for LLM-Based Agentic Systems (MemAgents)*, 2026.

[8] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica, "Judging LLM-as-a-judge with MT-Bench and chatbot arena," in *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 46 595–46 623.

[9] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez, "MemGPT: Towards LLMs as operating systems," *arXiv preprint arXiv:2310.08560*, 2023.

[10] P. Chhikara, D. Khant, S. Aryan, T. Singh, and D. Yadav, "Mem0: Building production-ready AI agents with scalable long-term memory," in *ECAI 2025—28th European Conference on Artificial Intelligence*, ser. Frontiers in Artificial Intelligence and Applications, vol. 413, 2025, pp. 2993–3000.

[11] W. Xu, Z. Liang, K. Mei, H. Gao, J. Tan, and Y. Zhang, "A-Mem: Agentic memory for LLM agents," in *Advances in Neural Information Processing Systems*, vol. 38, 2025.

[12] S. E. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.

[13] G. V. Cormack, C. L. A. Clarke, and S. Buttcher, "Reciprocal rank fusion outperforms condorcet and individual rank learning methods," in *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, Boston, MA, USA, 2009, pp. 758–759.

[14] SNAP Research, "LoCoMo: Evaluating very long-term conversational memory of LLM agents," GitHub repository, commit cbfbc1dba6bc53d00625212a0f22d55ffee7c1fc, file task_eval/evaluation.py, 2024, accessed: 2026-10-19. [Online]. Available: https://github.com/snap-research/locomo/blob/cbfbc1dba6bc53d00625212a0f22d55ffee7c1fc/task_eval/evaluation.py